Make or Buy the State? Consulting in the Public Sector*

Marta Morando† **[Job Market Paper]**

October 18, 2026

Click **here** for the latest version

Abstract

Governments increasingly rent professional services from the market rather than hiring permanently within the bureaucracy, yet public employment remains crucial in sustaining local economies. What shapes this make-or-buy decision? I examine whether it reflects bureaucratic productivity gains or responsiveness to citizens' policy preferences, focusing on consulting services in Italy. To estimate the impact of consulting on productivity, I exploit a national reform that lifted caps on municipal consulting expenditures, combined with quasi-exogenous variation in the intensity of prior restrictions. Access to consulting improves productivity: it enhances municipal public service provision while lowering budgetary needs. These effects are explained by a reorganisation of internal capacity: municipalities spend less on in-house employees, reducing the number of low-skilled administrative workers. Despite these productivity gains, a large-scale survey with embedded AI-led interviews reveals that most citizens want the state to rely more on its own employees. Interviews and information experiments show that this is partly motivated by beliefs that consulting is economically wasteful. Information on the economic benefits of consulting reduces citizens' willingness to pay to protect public employment by 40%. The remaining willingness to pay is consistent with perceived local economic benefits from public sector employment. Politicians and senior bureaucrats, surveyed separately, respond to citizens' demand for public production. Taken together, these results suggest that departures from productively efficient organisational forms in the public sector can be rationalised by understanding citizens' beliefs and preferences over how the state is organised.

*I am deeply indebted to my advisors, Oriana Bandiera, Tim Besley, Robin Burgess, and Noam Yuchtman for invaluable guidance and support. For very helpful feedback, I also thank Gharad Bryan. I also benefitted from comments by Lydia Assouad, Marco Bellifemine, Raffaele Blasone, Eduard Boehem, Maria Cedro, Adrien Couturier, Maarten De Ridder, Gaia Dossi, Sebastian Ernst, Bennet Feld, Fred Finan, Maitreesh Ghatak, Michael Gmeiner, Beyza Gulmezoglu, Jonathon Hazell, Xavier Jaravel, Camille Landais, Gilat Levy, Rocco Macchiavello, Isabela Manelici, Gad Nacamulli, Suresh Naidu, Arash Nekoei, Steve Pischke, Ronny Razin, Pol Simpson, Kate Smith, Silvia Vannutelli, John Van Reenen, Sarah Winton, Yoshiki Wiskamp, and Eddy Zou and numerous seminar participants at the London School of Economics and Barcelona Summer Forum. I thank STICERD, the Programme on Cohesive Capitalism, and Google Cloud Research for financial support and policymakers at the Institute for Finance and Local Economy for sharing contextual knowledge and helping with the survey on public officials. This project got ethics approval from the London School of Economics. Rodrigo Rocha provided excellent research assistance. The experiments have been pre-registered; registration ID: AEARCTR-0019084. All errors are mine.

†London School of Economics and STICERD, m.morando1@lse.ac.uk.

1. Introduction

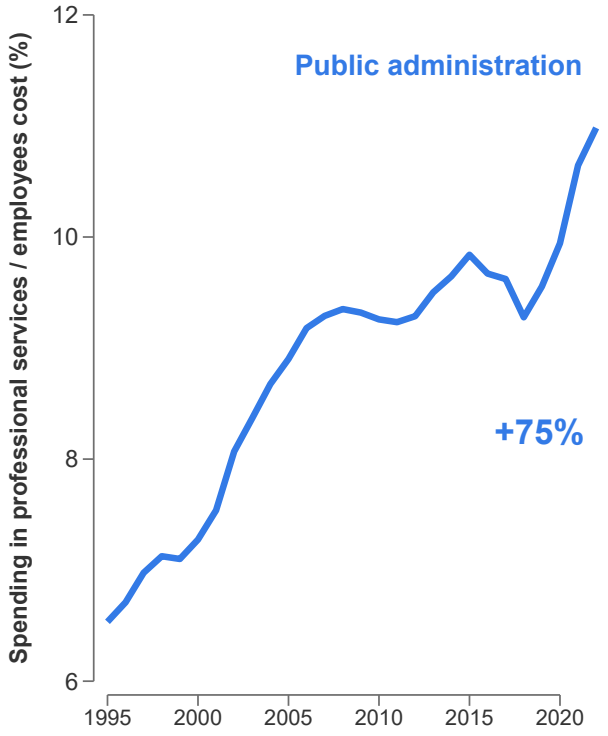
Effective states are central to economic growth. A key input into state effectiveness is personnel: governments need expertise to design and implement policies, provide public goods, and deliver public services. Traditionally, governments have acquired expertise by hiring public officials on permanent contracts. This model of public production has made governments important employers: public administration accounts for roughly 7% of total employment on average, across 189 countries. Over the past three decades, however, states have increasingly rented external expertise from the market: in 2022, spending on external professional services represented 11% of spending on in-house employees, a ratio 75% higher than in 1995 (Figure 1). What determines whether governments obtain skills through public employment or purchase it from external providers?

The decision to rely on external providers may be driven by productivity gains for the bureaucracy. If renting external personnel yields no such gains, then its use may instead reflect opportunities for politicians to extract rents through discretionary appointments. If renting external personnel improves productivity, however, the question is why bureaucracies continue to favour public employment. This paper identifies a new explanation: citizens may value public employment beyond the services they receive and the taxes they pay. Since states are ultimately accountable to citizens, these policy preferences may shape how public production is organised. Productivity gains alone may therefore be insufficient to explain why states adopt one organisational form over another.

I study this choice in the context of consulting services purchased by the Italian public sector, defined as external professional services contracted for a specific project from individuals or firms. I combine new administrative data, a natural experiment, large-scale surveys with embedded AI interviews and online experiments, and, lastly, a survey of local politicians and senior bureaucrats. I show that, when organising public sector production, departures from productive efficiency can be rationalised by understanding citizens' beliefs and preferences over how the state itself is organised. Consulting improves the performance of local governments, yet citizens would like the state to rely more on public officials, even when informed about the performance gains from consulting. Importantly, citizens' demand for public employment has political consequences: local policymakers respond to their policy preferences.

Studying the choice of external versus internal labour inputs requires tackling three steps. In the first step, I ask whether and how consulting shapes public sector performance. Answering these questions poses two main challenges. The first is measurement: the economic performance of public bodies is notoriously hard to observe (Besley, Burgess,

Figure 1: Professional Services as a Percent of Employee Costs in Public Administration



Notes. The figure reports expenditure on professional services as a ratio of labour costs in public administration, averaged across 80 countries with equal weights. The series is smoothed using a centred three-year moving average. The endpoint annotation reports the percentage change in the unsmoothed average between 1995 and 2022. Source: OECD input–output tables Referenced on page: 1.

Khan and Xu, 2022), and measuring consulting requires highly disaggregated spending data. The second is identification: whether to rely on consulting is an endogenous choice of public organisations.

I address the measurement challenge with new administrative data covering the universe of Italian municipalities over more than a decade. I combine municipal financial statements, which report disaggregated revenues and expenditures, with the census of public employees, which reports headcounts and characteristics by occupation. To these, I add an output index used by policymakers to monitor the municipal level of services, aggregating administrative tasks and frontline services. I validate it against citizens' satisfaction with their own municipality's services, measured through the survey I conducted, and define performance as output per euro of revenue or expenditure collected. I measure consulting using payment flows classified as such under centralised accounting guidelines. Evidence from a large-language-model classification of procurement contracts, service

spending by municipal function, and a survey of municipal officials shows that consulting spans a range of administrative tasks. In 2024, Italian municipalities spent, on average, 10% of their public employees wage bill on consulting.

I tackle the identification challenge by exploiting a national reform that lifted pre-existing caps on consulting expenditures, together with quasi-exogenous variation in the intensity of prior restrictions. Starting from 2011, in response to the eurozone debt crisis, all public bodies were required to cut consulting expenditures based on their own 2009 consulting spending level. The cap was removed in 2019. Because consulting spending is highly variable year to year, the 2009 baseline contains a large transitory component, generating quasi-exogenous variation in how tightly municipalities were bound. I isolate this component and interact it with year dummies in an event-study difference-in-differences design around the 2019 removal of restrictions. This identifies the effect of removing consulting restrictions. Instrumenting consulting spending with the exposure to restrictions interacted with an indicator for the post-removal period identifies the causal impact of consulting adoption and provides interpretable magnitudes. Event studies show no evidence of differential pre-trends across a wide range of outcomes. Placebo exposures constructed using alternative baseline years other than 2009 do not predict the post-reform increase in consulting spending. A large battery of robustness checks validates the empirical strategy.

I find that consulting improves municipal performance: it raises the quality and quantity of public services while lowering the revenues collected from citizens to finance them. A 10% increase in consulting expenditure per resident raises local government output by 3% and reduces revenues by 1%, relative to the pre-treatment mean. Because municipalities must balance their budgets, expenditures fall by a similar amount. This is not explained by greater reliance on debt or on transfers from other public bodies, but by a reduction in the wage bill of public employees. Output gains are concentrated in public goods and services closely tied to municipal administration.

These patterns match a conceptual framework in which politicians choose labour inputs to produce public services. How taxes and public employment respond to the removal of consulting restrictions depends on the substitutability between internal and external labour. Empirically, substitution is concentrated among clerical workers, rather than blue-collar staff or managers, consistent with descriptive evidence that consulting is an input into the administrative core of municipalities. It is also higher for low-education workers, suggesting that access to consulting raises the opportunity cost of low-skilled workers.

I find no evidence that politicians leverage consulting to obtain private benefits, including blame-shifting motives. Similarly, its take-up is uncorrelated with other forms of political misconduct, such as corruption or misuse of public funds, while it increases with

electoral accountability and with the skills of local politicians.

In the second step, I ask whether citizens' views and preferences on how the state is organised can help explain why governments restrict consulting use in the public sector, despite its performance gains. Citizens may prefer the state to rely on its own employees because they believe consulting has negative consequences for the public sector, or because they value public production beyond its effects on public sector performance and other implications it may have, such as loss of control over the public sphere. This cannot be recovered from observational data, so I conducted a nationally representative survey of around 5,000 Italian citizens. The survey embeds AI interviews to elicit unprompted reasoning about the use of external consultants as well as public employees, followed by information treatments and a discrete-choice experiment. Interviews led by a large language model allow me to elicit deeper reasoning through adaptive interactions with respondents (Geiecke and Jaravel, 2026). In my survey, 61% of respondents report preferring this format for a future study over an in-person interview or a questionnaire with open-ended questions. I measure attitudes toward the two modes of production through closed-ended questions on both stated and real-stakes choices, including signing a petition.

Citizens have limited knowledge of consulting in the public sector: 64% display little or no familiarity with the topic. Nevertheless, a majority would prefer the state to rely on public employees rather than consultants. This preference emerges in both stated policy views and a real-stakes choice: 51% sign a petition asking national policymakers to rely more on public employees relative to external providers. In the discrete-choice experiment, respondents choose between two policies previously implemented in Italy: cutting consulting expenditures and limiting public sector turnover. When the policies deliver equal savings to households, 68% choose to cut consulting. The implied willingness to pay to retain public employees rather than consultants is 124 euros per household per year. This magnitude represents 10% of municipal taxes, or, alternatively, the performance gains of increasing consulting per capita by 24%.

Citizens' demand for public production correlates with public sector ties, political ideology, and news consumption. Public employees and respondents with close relatives working in the public sector express stronger support for public officials. The gap in support between public employees and consultants is most pronounced on the far left, narrows towards the right, and widens again on the far right. Respondents who follow the news more closely also express greater support for public employees and less support for consultants. The association with news consumption suggests a possible role for information in shaping citizens' demand for public production.

Both the qualitative and experimental evidence point to the importance of beliefs about

the economic costs of consulting. In the AI interviews, economic arguments are used in favour of both sides, while non-economic concerns, such as fear of private interests, feature mainly in favour of public employees. The most common reason citizens give for opposing consulting is that it is economically wasteful. The information experiments support this interpretation. I randomly assign respondents to three information arms: the economic advantages of consulting, its non-economic disadvantages, and factual information on municipal consulting spending. Information about the economic advantages has the broadest effects: it shifts policy preferences and real-stakes behaviour, and is the only treatment with an economic and statistically significant effect on willingness to pay. Telling respondents that consulting can have economic benefits lowers willingness to pay for public employees by 40%. Citizens' remaining willingness to pay is higher in areas with greater unemployment and greater dependence on public employment. This pattern is driven by private sector workers, consistent with citizens valuing the local demand they expect public sector jobs to generate.

In the thid step, I examine whether public decision-makers respond to citizens' demand for public production. I survey municipal politicians and senior bureaucrats whose careers are closely tied to mayoral terms. Respondents report that citizens' views are an important consideration in decisions about how to organise public production. Informing them that most citizens prefer cuts to consulting over cuts to public employment significantly reduces their stated support for consultants relative to public employees. This highlights that elected officials take into account citizens' demand for public production.

Taken together, my results suggest that departures from productive efficiency can reflect governments' responsiveness to citizens' beliefs and preferences about how the state should be organised. These findings connect to a broader literature on the role of mission, social objectives, and values in shaping public and private organisations (e.g., Besley and Ghatak, 2005, Fioretti, 2022, Ashraf, Bandiera, Minni and Zingales, 2025, Colonnelli, McQuade, Ramos, Rauter and Xiong, 2026, Conway and Boxell, 2026).

Contribution to the literature. This paper contributes to four streams of literature. First, it speaks to the literature on the sources of poorly performing arrangements in the public sector. This has revolved around politicians benefiting from the organisation of the state, in the form of corruption (Shleifer and Vishny, 1993, Olken, 2007) or political control of personnel (Xu, 2018, Colonnelli, Prem and Teso, 2020, Akhtari, Moreira and Trucco, 2022, Aneja and Xu, 2024, Mastrorocco and Teso, 2026). The closest paper is Bandiera, Prat and Valletti (2009), which separates active and passive waste in government and finds that passive waste accounts for a sizable share of all waste. This paper provides an explanation

for passive waste. I suggest that states weigh performance gains against citizens' demands regarding how the state should be organised. Thus, departures from productive efficiency can stem from citizens' beliefs and preferences about how the state should be organised, rather than reflecting politicians' pursuit of private interests.

Second, it speaks to a related literature on the personnel economics of the state, which studies the importance of bureaucrats for government effectiveness (Best, Hjort and Szakonyi, 2023, Fenizia, 2022). A central question in this literature is how the selection and incentives of public officials affect performance (Deserranno, 2019, Bertrand, Burgess, Chawla and Xu, 2020, Ashraf, Bandiera and Jack, 2014, Besley and Ghatak, 2005, Khan, Khwaja and Olken, 2015, Deserranno, Kastrau and León-Ciliotta, 2025, Deserranno, Caria, Kastrau and León-Ciliotta, 2025), and whether granting bureaucrats more discretion improves or hinders service delivery (Muñoz and Otero, 2025, Bandiera, Best, Khan and Prat, 2021). This literature focuses on improving the bureaucracy within its existing boundaries. I ask a different question: whether shifting the boundary itself, by acquiring labour inputs externally through consulting services, affects public sector performance and how the organisation of the state is shaped by citizens' views and preferences.

Third, the paper relates to the literature on the make-or-buy decision of the state, both its theoretical foundations (Besley and Ghatak, 2001, Shleifer, 1998, Hart, Shleifer and Vishny, 1997) and a small empirical literature evaluating the consequences of privatisation and outsourcing (Duflo, Hanna, Olken and Tandon, 2025, Romero, Sandefur and Sandholtz, 2020, Banerjee, Hanna, Kyle, Olken and Sumarto, 2019, Galiani, Gertler and Schargrodsky, 2005). I bring new evidence on a form of outsourcing that has received little attention and is not a service directed to citizens, but rather a labour input into government production. Also, while the literature has focused on contractual frictions, I show that the effects of outsourcing interact with and depend on the *internal* organisational structure of bureaucracies, specifically the presence of skills in-house and the degree of decentralisation. I explain the state's make-or-buy decision through political economy lenses by looking at citizens' beliefs and preferences, and how elected officials react to citizens' demand for public production.

Fourth, the paper connects to recent work on the economics of consulting and its effects on client organisations. Existing evidence focuses mostly on the private sector, studying the impact of managerial consulting on firm productivity (Bijnens, Jäger and Schoefer, 2025) and the adoption of management practices (Anderson and McKenzie, 2022, Bloom, Eifert, Mahajan, McKenzie and Roberts, 2012, Bloom, Mahajan, McKenzie and Roberts, 2020, Bruhn, Karlan and Schoar, 2018, Bloom, Lemos, Sadun, Scur and Van Reenen,

2021).[1] I extend this work to the public sector and take a broader view of consulting that encompasses not only management services but also the range of specialised services that local governments procure externally. Additionally, this paper looks jointly at economic and political implications of consulting adoption.

Structure of the paper. The remainder of the paper is organised as follows. Section 2 provides information on the context and the administrative data used to understand the impact of consulting on public sector productivity. Section 3 lays out the identification and empirical strategy. Section 4 organises the empirical strategy and the results through a conceptual framework, reports the observational findings, and assesses the validity of the empirical design. Section 5 presents citizens' policy preferences over the organisation of the state through a large-scale survey of Italian citizens. Section 6 examines local policymakers' responses to citizens' demand for public production. Section 7 concludes.

2. Context and Data

This paper combines several administrative datasets on Italian municipalities to study how access to consulting affects public sector performance and the internal organization of public bodies.

Italy has just under 8,000 municipalities grouped in twenty regions. Municipalities are the smallest governmental units with decision-making authority, and, as such, the main point of contact between citizens and the state. They are responsible for a wide range of tasks, including land-use planning, social services, waste collection, public works and procurement, and the regulation of local economic activity through licenses and permits. They also handle substantial resources: municipal spending amounts to 4–5 percent of GDP, and municipalities finance nearly half of their revenues locally rather than through transfers. Locally-raised revenues include local taxes such as the waste tax, property tax, and income-tax surcharges, as well as non-tax revenues, such as fees or fines. Municipal finances are subject to national fiscal rules. Municipalities must adopt balanced budgets, with planned expenditures not exceeding planned revenues. Permanent hiring is also subject to spending limits, and recruitment occurs mainly through competitive examinations or transfers from other public bodies. Each municipality has a directly elected mayor and city council, and an executive body appointed by the mayor. The mayor is elected for a five-year term and makes all the key policy decisions, including how the annual budget is allocated.

[1] With the exception of Bijnens, Jäger and Schoefer (2025), who use data on the universe of Belgian firms, this literature consists of randomised interventions in small firms in developing countries.

2.1. Measuring Municipal Characteristics

Municipal budgets. I collect detailed municipal financial statements from AIDA PA (Bureau Van Dijk), which harmonizes data from official public accounts. It contains economic and financial information on the universe of Italian municipalities on a commitment basis, from 2000 to 2024. The data include comprehensive information on municipal government spending and revenues, each disaggregated by type, the number of residents, new borrowing, debt repayments, indicators used by policymakers to monitor municipalities' fiscal and financial sustainability, and unique municipal identifiers. Coverage varies across variables. For instance, spending on services is available between 2000 and 2015.

I measure the size of local government using locally-raised revenues and total expenditure. Locally-raised revenues are my preferred measure because they capture resources raised from local residents and businesses and are more directly under municipal control than revenues from transfers. The expenditure accounts do not allow an analogous distinction between discretionary and mandated spending. In 2024, the average municipality raised €1,047 per resident locally and recorded €2,740 per resident in total expenditure, with current expenditure accounting for 53% of the total.

Municipal workforces. Organisational data come from the annual census of Italian public sector employees (Conto Annuale), published by the State General Accounting Office. Municipality-level data are available from 2001 to 2024 and report personnel expenditure, employment, and separations, together with workforce composition by age, education, and occupational category.[2] Occupational categories distinguish managers, professionals, clerical workers, and blue-collar workers. The raw data are at the position–municipality–year level, covering 832,562 observations, which allows me observe the composition of the workforce within occupational categories.

These characteristics differ markedly across municipalities. In 2024, the average municipality employed 42 workers and spent €195 (SD €144) per resident on personnel. Employees were, on average, 52 years old, and 32% held at least a university degree. The highly educated employment share was 2 times larger among municipalities in the top quartile of the distribution than among those in the bottom quartile. The average number of job separations was 4, of which 39% were related to retirement.[3] I observe positions rather than specific jobs: 28.9% of workers are blue-collar, 45.9% clerical, 24.9% professional, and 0.4% managerial.[4] More generally, municipal employees include both frontline and

[2]These data refer to permanent employment.
[3]The remaining share consists of resignations, transfers, dismissals, and residual causes.
[4]See Appendix Table A.1 for the construction of these occupational categories.

back-office workers; while the literature devotes more attention to the former—for example, teachers, police, and healthcare professionals—this paper focuses mostly on administrative employees who operate in the background but are central to public sector production.

Other sources. I draw additional municipal characteristics from the 2001 population census, which provides cross-sectional data on demographics, labor market outcomes, education, and housing conditions. The Ministry of Economy and Finance provides annual municipality-level data on aggregate taxable and employment income, and the corresponding numbers of taxpayers, from 2000 to 2024. Variables related to corruption cases for the years 2004-2014 are gently shared by Daniele and Giommoni who aggregate at the municipal-year level corruption-related felonies from the Ministry of the Interior. I use data on municipal elections and elected official characteristics for each municipality, provided by the Italian Ministry of the Interior. Lastly, I assemble a new dataset of administrative rulings involving municipalities from the online archive of the Court of Auditors, an independent institution responsible for overseeing public finances. Using the archive's API, I retrieve more than 25,000 rulings issued between 1992 and 2025.

2.2. Measuring Public Services Output and Performance

Measuring public sector performance is notoriously difficult because government outputs are multidimensional and often not easily measurable (Besley, Burgess, Khan and Xu, 2022, Xu, Deserranno, Moreira and Teso, 2023). Existing work therefore typically focuses on public services whose output can be quantified, such as education or health care (e.g., Duflo, Hanna and Ryan, 2012, Bloom, Propper, Seiler and Van Reenen, 2015), or relies on subjective assessments (e.g., Rasul and Rogger, 2018, Limodio, 2021). I address this measurement challenge using an official score of municipal services provision that captures the multidimensionality of local government output.

I measure municipal output using OpenCivitas, a dataset covering approximately 6,600 municipalities in ordinary-statute regions. OpenCivitas reports public services output on a 1–10 scale, relative to municipalities in the same population class. Higher values indicate greater service provision relative to similarly sized municipalities. The measure is directly relevant to policymakers: it was designed to compare and monitor service provision across municipalities. It aggregates five components corresponding to education, social services and childcare, waste management, local police, and territory and roads.[5] Each component is based on underlying physical measures such as pupils transported

[5]The five function-specific components are combined using weights based on their shares of national standard expenditure needs. Waste collection has the largest weight.

and school meals provided; users of social services and childcare; the share of waste collected separately; fines, investigations, and authorizations issued by local police; and environmental authorizations, projects and road works. These physical output measures come from detailed questionnaires filled by municipalities. Appendix Figure A.1 shows a snippet of the questionnaire. Public services output is available in 2010, 2013, 2015–2019, and 2021–2022.[6] This aggregate, off-the-shelf measure is the main outcome variable in the empirical analysis.

The underlying physical measures clarify what public services output captures. Appendix Figure A.3 reports a lasso regression of the aggregate measure on these indicators. The model accounts for 77 percent of the cross-municipality variation; the largest coefficients are on separate waste collection, social-service coverage, school meal coverage, permits and authorisations, nursery coverage, and the maintenance of public lighting.[7] These predictors span frontline services and administrative tasks, so the measure is not dominated by a single municipal function.

The measure captures the level of services provided, but may reflect also their quality. The distinction matters because existing theory frames the choice between external and in-house provision as a tradeoff between cost efficiency and quality (Hart, Shleifer and Vishny, 1997, Besley and Ghatak, 2001, Shleifer, 1998). To assess this, I compare public services output with citizens' own assessments, drawn from the survey I fielded with 4,872 respondents across 1,472 municipalities. Citizens are asked how satisfied they are with the public goods and services provided by their municipality. Appendix Figure A.2 relates 2022 public services output to these 2026 assessments, controlling for population decile. A one-point higher citizen assessment –a fifth of the survey scale– is associated with 1.15 more points on the ten-point output scale (SE = 0.19).[8] The correlation suggests that the OpenCivitas measure tracks public provision as citizens experience it, not solely as municipalities report it. Since satisfaction responds to how services are delivered and not only to their quantity, I interpret public services output as capturing both the quantity and the quality of local provision.

This measure captures output rather than performance, as a municipality may deliver more services simply because it has more resources to spend. Combining it with municipal budget data lets me measure output relative to the revenues used to produce it, which maps directly onto the production function of local government.

[6] Some of the underlying physical measures are observed only in certain years.

[7] Variables in levels are normalized by 1,000 residents.

[8] The two measures are constructed differently: survey responses are absolute and on a 1–5 scale, whereas public services output runs from 1 to 10 and is benchmarked within population class. The regression therefore includes population decile fixed effects.

2.3. Measuring Consulting

In this paper, I define consulting as the purchase of external professional expertise, supplied either by an individual professional or by a firm, for a specific project with clear deliverables.[9] Consulting is a form of service outsourcing, but unlike contracts for waste collection or school transport, it does not directly deliver a service to citizens.[10] It instead supplies *labor input* into government production (e.g., financial accounting, IT design, strategic planning, legal).

Municipalities have discretion to hire external consultants through either individual professional appointments or service procurement contracts.[11] Procurement data cover only service procurement contracts and therefore disproportionately capture services supplied by firms and larger contracts.[12] Public expenditure data are typically too aggregated to identify consulting, reporting neither the type of transaction nor the counterparty.[13]

I overcome these obstacles by building a baseline measure from payment flows issued by public bodies and classified as consulting under centralised guidelines. Payments are recorded regardless of the legal channel, so the measure spans both legal regimes. Additionally, this measure does not depend on how a researcher chooses to define consulting. Given the limitation mentioned above, measures constructed from procurement data serve as validation and provide descriptive evidence on the nature of consulting engagements in the public sector.

I collect municipal payment flows from Italy's General Government Transactions Information System (SIOPE), which reports, for each municipality, the total amount paid under each invoice category, from 2010 to 2025. My baseline measure of consulting expenditure aggregates invoices coded as expert, professional, and consulting services. Appendix Table A.2 lists all the invoice codes used.

These data differ from municipal financial statements in two ways. First, they are

[9] It consists of both providing advice, which can or cannot be followed by local governments, as well as performing specific tasks. Neither the Italian legal framework nor the data can distinguish between the two.

[10] Consulting also differs from privatisation, which transfers residual claimant rights to a private entity and thereby aligns profit motives with performance (Hart, Shleifer and Vishny, 1997, Shleifer, 1998). Under a consulting contract, the municipality retains those rights, and the incentives derive from the expected probability of securing future contracts.

[11] Consulting provided by organisations falls under procurement law, whereas individual professionals can be hired under either legal regime (see Appendix A for more details).

[12] Coverage also varies over time; for example, legal services were not covered by procurement law prior to 2017.

[13] Bijnens, Jäger and Schoefer use VAT data to study consulting in the private sector, identifying consulting by counterparty type. In my setting, centralised guidelines classify the transaction rather than the counterparty, allowing consulting to be identified even when purchased from firms whose business activity lies elsewhere.

disaggregated at the invoice level, which is what makes consulting measurable. Second, they record bank disbursements rather than commitments. Italian accounting rules require municipalities to book revenues and expenditures when the legal obligation arises rather than when payment is made. This matters for the identification strategy in Section 3, which relies on the timing of consulting obligations rather than disbursements.

As a secondary source, I collect tender-level procurement data from the National Anti-Corruption Authority. These cover the quasi-universe of awarded contracts from 2012 to 2023, recording the object of each transaction, the tax code and name of the counterparty, along with the contract's value, procurement category, name and identifiers of the public body issuing the contract, and publication date.[14] The resulting sample contains 605,518 tenders. Classifying procurement contracts based on the awarded firm yields a measure highly correlated with the baseline from payment flows (Appendix Figure A.5).

Descriptive evidence. Table A.3 reports summary statistics. The average municipality spends roughly €12 per resident on consulting, compared to about €181 per resident on in-house personnel. Consulting thus represents approximately seven percent of personnel expenditures. Relative to locally-raised revenues, consulting accounts for two percent of the municipal budget. The distribution of consulting expenditure is right-skewed, with a long tail of high-spending municipalities (Figure A.4, Panel A). Consulting grew sharply after 2019: average consulting expenditures per resident increased by more than fifty percent, from about 10 to 16 euros. The timing coincides with the national reforms described in Section 3. Over the same period, in-house personnel expenditures declined, and consulting as a share of personnel costs rose from roughly six to nine percent. Figure A.4, Panel B maps consulting expenditures per resident by decile across Italian municipalities. The geographic distribution is dispersed: high- and low-spending municipalities coexist within the same regions, and there is no discernible North-South gradient.

Procurement records report the object of each contract, thus, they provide useful examples of what consulting services represent for Italian municipalities, for instance:

> ***Strategy.*** *"Strategic-plan guidelines for the redevelopment of Trieste's Old Port," Trieste, 2015, €170,000*
> ***Digital administration.*** *"Digitization of building and urban-planning archives," Messina, 2015, €205,000*

[14] As standard in the literature (Decarolis, Fisman, Pinotti and Vannutelli, 2025), I restrict the sample to contracts under "ordinary procedures," which are the majority of contracts, and contracts above 40,000 euros. Extraordinary procedure includes contracts that are in strategic sectors, as national security, which face different legal requirements. Public bodies are forced to acquire a tender identifier –and thereby be included in the dataset– only for payments above €40,000.

Procurement capacity. *"Specialist consulting and training on public procurement," Grosseto, 2019, €137,500*

Tax administration. *"Support to the municipal tax office for cadastral equalization and tax assessment," Mantova, 2020, €135,666.67*

Panel A of Figure 2 makes the same point quantitatively, classifying a random subset of consulting contracts into eight categories using a large language model. Spending is concentrated in technical and administrative services: IT and digital work account for 34 percent, followed by management (18%), and finance and insurance (12%), with the remainder split across safety, accounting, communication, legal, and training.[15] Local consulting thus covers a wide array of professional tasks.[16] These shares should be read with care: procurement records do not cleanly separate current from capital spending, and cover only the selected sample described above.[17]

Panels B and C of Figure 2 relate consulting spending to service spending across municipal functions. The coefficients come from a multivariate regression of standardised consulting spending per resident on standardised service spending per resident by function, with year fixed effects. In Panel B, a one-standard-deviation increase in administration service spending is associated with a 0.30-standard-deviation increase in consulting spending, which is more than three times the association for any other function. Panel C shows that, within administration, the largest coefficients are for the technical office and general secretariat, personnel and organisation. The technical office is responsible for managing and planning public works and urban planning, as well as issuing permits and authorisations. The general secretariat, personnel and organisation coordinates the offices, manages staff, and turns political decisions into administrative acts.

These patterns are consistent with evidence from a second survey I conducted among local politicians and bureaucrats. Municipal policymakers describe similar uses of consulting in response to the survey question "For which tasks and in which situations has your municipality used external consultants?" (Appendix Figure A.39). Taken together, the observational and survey evidence indicates that municipalities use consulting differently from other service outsourcing: it is a professional input into administration, rather than for direct delivery of frontline services.

[15] Safety contracts are common because public works require expert certification.

[16] Public debate (Mazzucato and Collington, 2023, Bogdanich and Forsythe, 2022) centres on management engagements, which account for under a fifth of municipal consulting spending recorded in procurement data. Additionally, consulting engagements involving large consultancy firms are relatively rare.

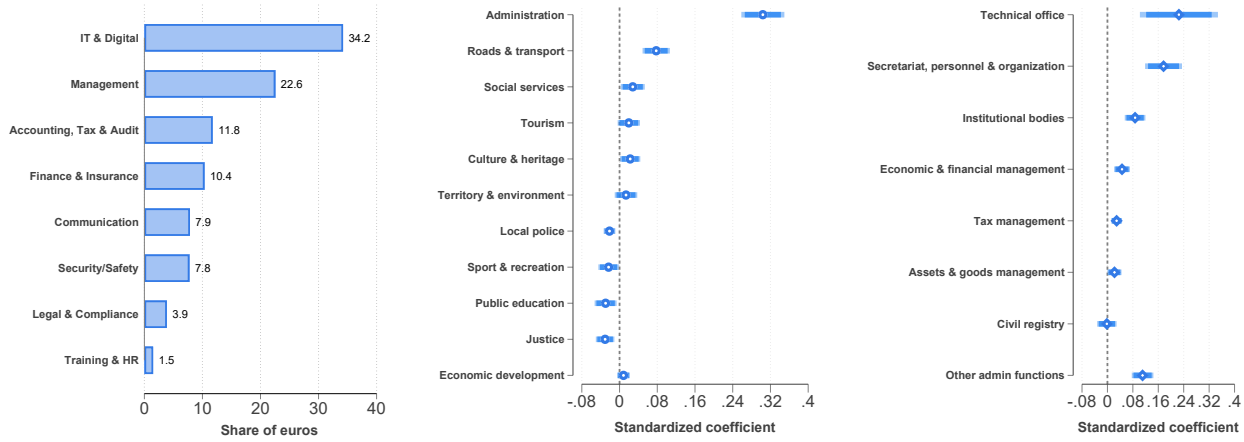
[17] For example, contracts coded as IT consulting may bundle hardware with installation services.

Figure 2: Consulting Across Public Sector Tasks and Functions

Panel A: LLM Classification of Procurement Contracts

Panel B: Broad Municipal Functions

Panel C: Within the Administration Function



Notes. Panel A plots the composition of consulting expenditures for a random subset of tenders. Contracts are first classified as consulting versus non-consulting using a large language model; consulting contracts are then assigned by the model to one of eight pre-trained subcategories. Bars report, for 2011–2023, each subcategory's share of total euros spent on consulting (shares sum to 100). The eight categories are: IT and Digital, Management, Finance and Insurance, Security/Safety, Accounting, Tax and Audit, Communication, Legal and Compliance, and Training and HR. Panels B and C report standardized coefficients from multivariate regressions of consulting spending per resident on service spending per resident by function. The sample is municipality-years between 2010–2015. Panel B includes broad service functions. Panel C decomposes administration into subfunctions and a residual category. Panel B and C specifications include year fixed effects and standard errors are clustered at the local labour market level. Referenced on page: 13.

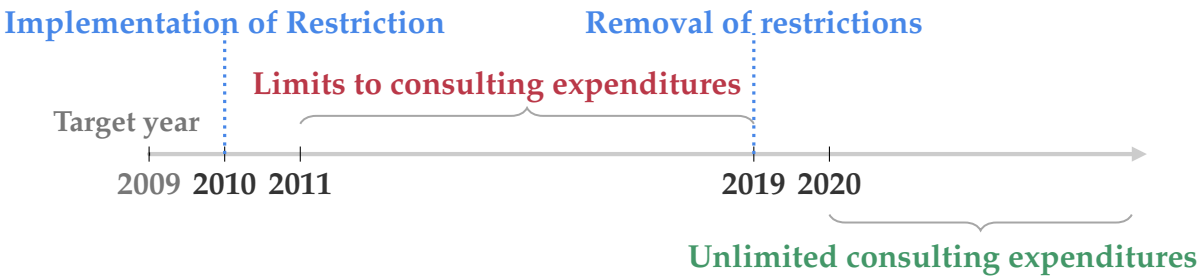
3. Identification and Empirical Strategy

This section explains the main empirical strategy to assess the effects of consulting in the public sector. It leverages a natural experiment based on a national reform that lifted pre-existing ceilings on consulting expenditures, combined with quasi-exogenous cross-municipality variation in the intensity of the prior restrictions.

3.1. Institutional Background

During the sovereign debt crisis, Italy aimed at reducing public debt, including measures that induced local public bodies to limit their spending. Among these, in 2010, the national government passed a decree law, D.L. 78/2010, which prescribed that all public bodies, except universities and research entities, cut their consulting expenditures. The law specified that from 2011 onward, public bodies had to reduce their consulting spending to 20% of their *own* 2009 consulting expenditure. The caps on consulting expenditures have

Figure 3: Timeline of the Reforms



Notes. The figure shows the timeline of the reforms, as well as the target year (in light gray). Dark blue indicates the year of the drafting of the laws, light red indicates the period in which restrictions are implemented and light green indicates the period in which restrictions were lifted. Referenced on page: 15.

been officially repealed with a law in 2019, effective from 2020 onward.[18] Thus, starting from 2020 municipalities' choices over consulting engagements were no longer restricted. Aggregate consulting expenditure is consistent with these legislative changes.[19] Consulting spending decreased during the restriction period and sharply increased when restrictions were removed.

Decree-Law 78/2010 was largely unanticipated, and there is no clear evidence explaining why policymakers selected 2009 as the reference year. Given that the law was drafted in 2010, anchoring the cap to the immediately preceding year was a natural administrative choice. Importantly, public bodies could *not* retroactively change their reported 2009 consulting expenditures.

The reform applied broadly to the public sector, including ministries, regions, provinces, healthcare facilities, and other public bodies that typically account for substantially more consulting expenditure than municipalities. Thus, it is unlikely that the choice of 2009 was tailored to any particular subset of municipalities. Similarly, the removal of restrictions in 2019 was largely unanticipated. Figure 3 shows the timeline of the events with all the relevant years.

3.2. Exposure to Consulting Caps

These institutional features generate cross-municipality variation in the intensity of the restrictions.

[18]The full text of Article 6, Decree-law No. 78/2010, N.7 and of Article 57, Decree-law 124/2019, N.2. can be found in Appendix A.

[19]Appendix Table A.3 reports consulting expenditure by period. My measure records cash payments, whereas the law applies on a commitment-basis, thus the timing may differ. The same pattern is visible using input-output tables for professional services purchased by the public administration (Appendix Figure A.7, Panel A).

Different municipalities had different levels of consulting spending in 2009. However, the naive comparison between high and low spenders would lead to biased estimates. The two groups differ in time-invariant characteristics (e.g., size and income), and in time-varying ones (e.g., growth rates).[20]

I address this concern by isolating the *transitory* part of 2009 consulting spending. Within a municipality, consulting expenditure moves substantially from year to year (Appendix Figure A.12). Part of this movement can be attributed to exogenous factors rather than local fundamentals, for instance, the proximity of municipal elections.[21]

Municipalities more exposed to the reform are those that spent little in consulting in 2009 compared to their *own* needs in that period. Those that spent relatively more faced a looser limit and did not have to change their behaviour much. Measuring how much municipalities deviated in 2009 from their own average consulting expenditure allows me to exploit the intensity of reform and, at the same time, remove differences due to municipalities being observationally different both in levels and trends.[22] An alternative way of interpreting this instrument is that it nets out the counterfactual exposure that may have been realised if the anchoring year chosen had been different from 2009.

The law anchors the cap to 2009 consulting expenditure as recorded in municipalities' financial statements. My consulting measure is based on cash disbursement, whose timing may not align with recorded spending.[23] Since the construction of the exposure measure relies on the exact timing, I proxy consulting with service spending.[24] This proxy is validated by the fact that across municipalities the raw correlation between consulting and services is around 70% (see Appendix Figure A.6), consistent with consulting being a type of service outsourcing.[25]

More precisely, I am defining a continuous exposure variable, as follows:

$$\text{Exposure}_m = \left(1 - \frac{\text{Services}_{m,2009}}{\frac{1}{4}\sum_{t=2006, t\neq 2009}^{2010} \text{Services}_{m,t}}\right) \times 100 \tag{1}$$

Municipalities with larger exposure are those who were spending relatively less in

[20] Thus, non-random exposure to the reform could bias estimates even if the reform is completely exogenous.

[21] See Appendix Figure A.21. For historical reasons, Italian municipalities are on different electoral cycles.

[22] Balance tests and placebo exercises validate this intuition. See Appendix Figures A.31 and A.32.

[23] Italian municipalities are on average six months late to the first payment and two years on the final payment (Bellifemine, Couturier and Tozzo, 2025).

[24] Consulting data are also mostly missing before 2010.

[25] Additionally, if services and consulting spending were not related, I would not be able to detect any effect.

services in 2009 compared to their own mean.[26] This means that they would have faced tighter restrictions during the restriction period and thereby benefited more from the removal of restrictions in 2019. Panel A of Figure A.13 shows that this exposure measure has substantial variation and is well-distributed. Panel B shows that it is not clustered geographically.

Variables and sample definition. All the monetary outcomes, such as locally-raised revenues or total expenditures, are expressed in euros per resident and winsorized at 1st and 99th percentiles. Consulting expenditure per resident is instead winsorized at the 2nd and 98th percentiles.[27] The main analysis covers municipality-years between 2014-2024.[28] I begin in 2014 because compliance with the restrictions may not have been immediate.[29] For the first stage, the effect of removing the consulting restrictions on consulting spending itself, I use consulting data through 2025. Public services output has more limited coverage, since it is observed only in selected years and only for municipalities in ordinary-statute regions.

3.3. Event-Study Specification

To understand how the removal of consulting restrictions affected public sector performance and the internal organization of public bodies, I estimate the following event-study difference-in-differences specification, which interacts the exposure measure from Equation (1) with year indicators:

$$Y_{m,t} = \alpha_m + \delta_{r(m),t} + \gamma_{d(m),t} + \sum_{j \in \mathcal{T} \setminus \{2019\}} \beta_j \Big(\text{Exposure}_m \times \mathbb{1}\{t = j\} \Big) + \varepsilon_{m,t} \tag{2}$$

Here m indexes municipalities and t years, $r(m)$ denotes the region of municipality m, and $d(m)$ its population decile.[30] I omit 2019, the year Parliament lifted the limits on consulting expenditure, so all coefficients are relative to that year. The coefficients of interest are the β_j, which trace the differential change in Y_{mt} associated with a one-percentage-point increase in Exposure_m. Higher values of Exposure_m mean that municipality m faced tighter restrictions before 2019, and therefore gained more from their removal.

[26] The window includes 2010, the year the reform was drafted. Using only pre-2009 years would load more heavily on municipality-specific trends.

[27] In Appendix Figure A.35 I show that my results are unchanged with different winsorizations or taking the log of the outcome variables.

[28] I remove observations with missing or extreme personnel expenditure growth.

[29] I validate this intuition in Appendix Figure A.33.

[30] Population deciles are based on the 2001 census.

The identifying assumption is a conditional (strong) parallel-trends assumption: absent the reform, outcomes in municipalities with different values of Exposure_m would have evolved similarly, conditional on municipality fixed effects, region×year fixed effects, and population-decile×year fixed effects.[31] Under this assumption, β_j for $j > 2019$ captures the dynamic effect of the reform, while coefficients for $j < 2019$ provide a test for differential pre-trends. The term $\delta_{r(m),t}$ absorbs region-specific shocks. Similarly, $\gamma_{d(m),t}$ controls for shocks common to municipalities in the same population decile. Consequently, violations of the identifying assumption would require time-varying confounders that (i) generate differential trajectories aligned with the reform timing, and (ii) correlate with the exposure measure even after conditioning on region-by-year and population-decile-by-year fixed effects. Standard errors are clustered at the local labour market level to allow for arbitrary correlation in the error structure across municipalities within local labour markets over time.[32]

3.4. Instrumental Variable Strategy

Under the conditional (strong) parallel trends assumption, the event-study design described above identifies the reduced-form (intent-to-treat) effect of removing consulting restrictions, as well as the first stage impact on consulting expenditure. To isolate the impact of greater access to consulting and therefore interpret magnitudes, I use an instrumental variable strategy that leverages cross-sectional variation in exposure to the reform and the timing of the removal of restrictions.[33]

I define the instrument as the interaction between a municipality's exposure to the intensity of restrictions and an indicator for the post-removal of restriction period:

$$z_{m,t} \equiv \text{Exposure}_m \times \text{Post}_t, \qquad \text{Post}_t \equiv \mathbb{1}[t > 2019]. \tag{3}$$

The first stage and the reduced form are the static versions of Equation 2, with consulting expenditure per resident, $\text{Consulting}_{m,t}$, and other outcomes, $Y_{m,t}$, as the dependent variables, respectively.

[31] This is a *strong* parallel trends assumption (Callaway, Goodman-Bacon and Sant'Anna, 2026) because it requires that the outcome path municipalities would have followed at exposure $\bar{e}$ coincides with the path realised by those actually at exposure $\bar{e}$, that is, that municipalities with different exposures would have responded identically to the same exposure intensity.

[32] There are around 600 local labour markets in Italy.

[33] Because exposure is defined so that the first stage is positive, the reduced form and the IV estimates have the same sign.

The 2SLS second-stage equation is:

$$Y_{m,t} = \alpha_m + \delta_{r(m),t} + \gamma_{d(m),t} + \pi \widehat{\text{Consulting}}_{m,t} + u_{m,t}, \tag{4}$$

where $\widehat{\text{Consulting}}_{m,t}$ denotes consulting expenditure per resident predicted from the first stage. The 2SLS coefficient π equals the reduced-form coefficient scaled by the first stage.

Identification requires two conditions. First, instrument relevance: differential exposure to the reform must generate meaningful variation in consulting intensity. Second, the exclusion restriction: conditional on the fixed effects, $\text{Exposure}_m \times \text{Post}_t$ must affect $Y_{m,t}$ only through its effect via consulting spending. Under these conditions, $\hat{\pi}$ recovers the local average treatment effect of consulting expenditures per resident for municipalities whose consulting decisions are shifted by the removal of consulting restrictions.

4. Results

This section first introduces a conceptual framework to organise the empirical results. I then present the empirical results from the observational data, starting with the first stage and the reduced-form results. To interpret the magnitudes, I also show the static IV results. After this, I provide evidence on the validity of my empirical design.

4.1. Conceptual framework.

Public employees and consultants are two inputs into public production. A politician chooses in-house labour, B, and consultants, C, to provide public goods and services $\mathsf{g}(\mathsf{B},\mathsf{C})$, financed by tax revenue T. At input prices $w_\mathsf{B}, w_\mathsf{C} > 0$, the politician's problem is

$$\begin{aligned} \max_{\mathsf{B},\mathsf{C},\mathsf{T}\geq 0} \quad & u(\mathsf{g}(\mathsf{B},\mathsf{C})) - \mathsf{T} \\ \text{s.t.} \quad & w_\mathsf{B}\mathsf{B} + w_\mathsf{C}\mathsf{C} = \mathsf{T}, \qquad w_\mathsf{C}\mathsf{C} \leq K \end{aligned} \tag{5}$$

The value of public services is increasing and strictly concave, $u' > 0$ and $u'' < 0$, while taxes enter the objective linearly. The spending cap K captures the consulting restrictions. In the baseline, $\gamma(\mathsf{C}) = 0$. Production has constant returns to scale and takes the CES form

$$\mathsf{g}(\mathsf{B},\mathsf{C}) = \left[\alpha \mathsf{B}^\rho + (1-\alpha)\mathsf{C}^\rho\right]^{1/\rho}, \qquad 0 < \alpha < 1, \quad \rho < 1,$$

The elasticity of substitution is $\sigma = 1/(1-\rho)$. Appendix F provides the derivations.

To map into the natural experiment, I consider the case in which the binding consulting

cap gets relaxed. Relaxing a binding cap makes an additional euro spent on consultants produce more services than an additional euro spent on public employees. Relaxing the cap thus raises consulting spending, public services output $g(\cdot)$, and output per euro of revenue, $g(\cdot)/T$, as the input mix moves towards consultants.

The responses of public employment and revenue are ambiguous. Consultants raise the marginal product of public employees, but the resulting increase in services lowers their marginal value. Employment falls when the latter effect dominates, that is, when $-gu''(g)/u'(g) > 1-\rho$. Greater substitutability between the two labour inputs therefore makes a reduction in the number of public employees more likely. Revenue falls only if the saving on public employees exceeds the additional consulting bill. Conversely, sufficiently flat marginal utility and low substitutability lead to increases in both public employment and revenue. Thus, removing consulting restrictions has an ambiguous effect on in-house bureaucrats or on the municipal budget.

Politicians may also value consulting for reasons unrelated to service production. This can be captured by adding a term, $\gamma(C)$, to their objective, representing, for example, private benefits from discretionary appointments.[34] In the presence of such private benefits, relaxing consulting restrictions need not improve output per euro spent.[35] The effects of relaxing a binding consulting constraint, therefore, depend on whether consulting is used primarily as a productive input or whether there are political economy reasons behind its use.

First stage. Figure 4 presents the event-study estimates from Equation 2, with consulting expenditures per resident as the outcome variable.[36] Municipalities that faced tighter restrictions increased their consulting spending more after the limits were removed. The effect begins to appear in 2020 and persists throughout the post period. The magnitude is economically meaningful: the static specification implies that a one standard deviation increase in exposure (corresponding to 19.27 percentage points) raises consulting expenditures by 50 cents per resident.[37] Since average consulting spending over 2014 to 2019 was approximately 10.3 € per resident, this represents an increase of 4.9% relative to the

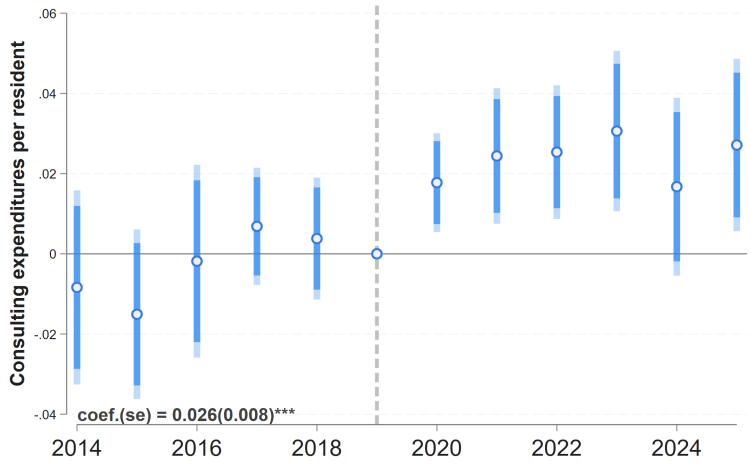
[34] Conversely, a policy preference for public employment can be represented by $\gamma'(C) < 0$.

[35] This mechanism is consistent with evidence that patronage in public appointments undermines bureaucratic selection and performance (Colonnelli, Prem and Teso, 2020, Xu, 2018). Appendix F derives the conditions under which relaxing consulting restrictions improves performance in this case.

[36] Appendix Table A.12 shows the results for the static difference-in-differences in table format in the full sample as well as restricting the time window around 2019.

[37] A one percentage point increase in exposure raises consulting expenditures by 2.60 cents per resident (p-value=0.00113). A joint F-test that the exposure-by-year coefficients for 2014–2018 are zero yields $F(5, 605) = 1.234$ ($p = 0.2914$); 2019 is the omitted year.

Figure 4: First-Stage Effect of the Removal of Consulting Restrictions



Notes. The figure shows first-stage event-study estimates from Equation 2, with municipal consulting expenditures per resident as the outcome. The main regressor is the exposure measure defined in Equation (1) interacted with year dummies. The omitted year is 2019, the year in which the law removing consulting restrictions is drafted. The specification controls for region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. The coefficient and standard error reported in the bottom left quadrant refer to the static specification. Referenced on pages: 20, 21, 32.

pre-removal mean. Most post-2019 coefficients are statistically significant at the 5% level. Prior to the removal of restrictions, the coefficients are close to zero, with 90 and 95% confidence intervals that consistently include zero. There is no evidence of differential pre-trends, in line with the identifying assumption. The estimates in Figure 4 serve as the first stage of the IV strategy described in Equation 3.

Effects of the removal of consulting restrictions. Figures 5 shows the reduced-form effects of removing consulting restrictions. Panel A, B, C, and D, reports the event-study estimates from Equation 2 for public services output, locally-raised revenues, total expenditures, and public employees wage bill. The bottom-left corner of each panel reports the coefficient and its standard error from the static difference-in-differences specification.

Panel A of Figure 5 shows that municipalities facing tighter consulting restrictions experienced larger increases in public services output after the limits were removed. Before removal, the coefficients are statistically indistinguishable from zero.[38] The post-reform coefficients are positive and statistically significant at the 5% level in both 2021 and 2022, the two post-reform years observed. The static specification implies that municipalities with one standard deviation greater exposure experienced a larger increase in public services

[38] A joint F-test that the exposure-by-year coefficients for 2015–2018 are zero yields $p = 0.1380$.

output, equivalent to 1.2% of the pre-removal mean ($p = 0.0253$). Under the identifying assumption, this differential response captures the effect of removing consulting restrictions on public services output.

As discussed in Section 2, the output index covers both frontline services and administrative tasks and is positively associated with citizens' assessments of service quality. I therefore interpret this result as an improvement in a broad measure of local service provision, capturing aspects of both quantity and quality. This finding is also consistent with public officials' expectations from the survey I fielded: 55% of elected officials expect greater use of external consultants to improve the quality of public services (Appendix Figure A.41).

If consulting served exclusively non-productive purposes, such as shifting responsibility or serve as a tool to extract private rents for politicians, captured by positive $\gamma(C)$ in the conceptual framework, we would not expect greater consulting use to improve public sector output.

Panels B and C investigate whether this output improvement comes at a higher cost for municipal budgets and, ultimately, for citizens. Panel B reports reduced-form event-study estimates from Equation 2 for locally-raised revenues per resident, while Panel C reports the estimates for total expenditures.[39] Since municipalities face balanced-budget requirements, savings in expenditure allow them to raise less revenue from residents and businesses. The estimates point in this direction.

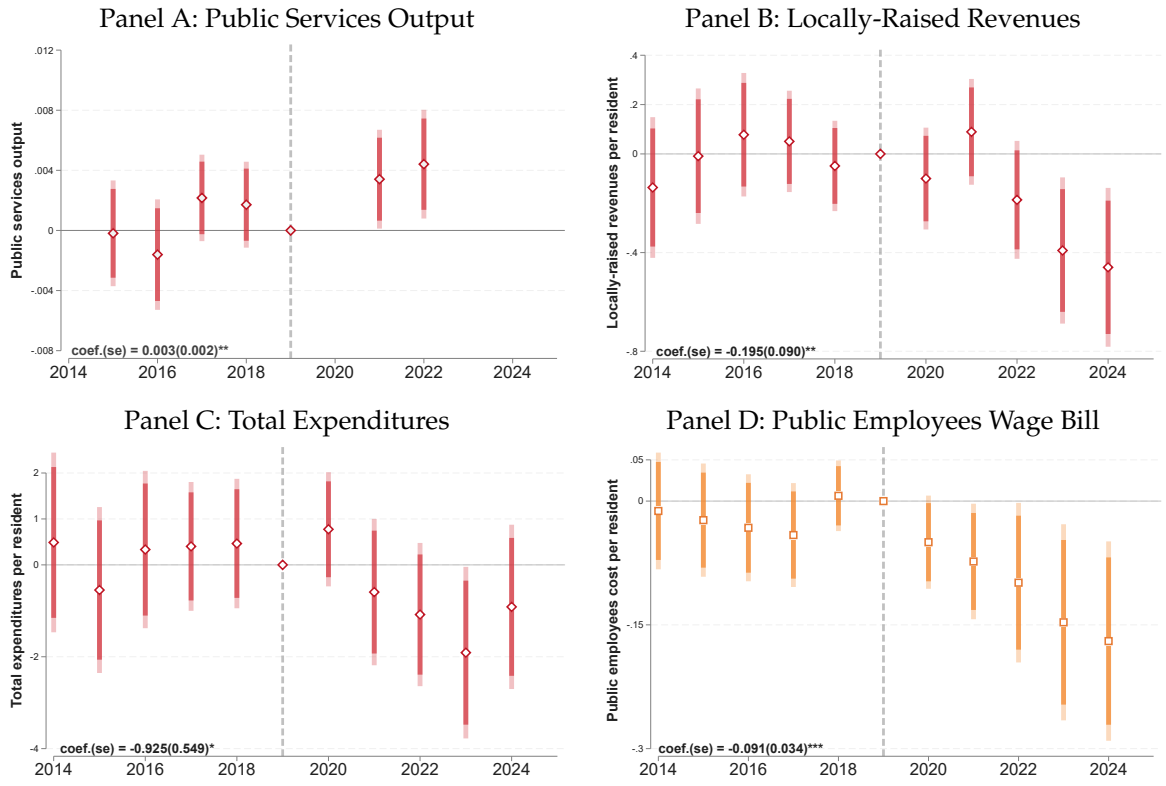
Panel B shows that before the removal of consulting restrictions, municipalities with different levels of exposure follow parallel paths.[40] After the removal, more exposed municipalities lower their revenues gradually over time. The coefficients become statistically significant at the 5% level in both 2023 and 2024. The static specification implies that a one standard deviation increase in exposure lowers locally-raised revenues by 3.7 € per resident, or 0.4% of the pre-removal mean ($p = 0.0307$).

Panel C suggests that this reduction in locally-raised revenues is accompanied by lower total expenditures. The static specification implies a reduction of 17.7€ per resident for a one standard deviation increase in exposure, equivalent to 1.0% of the pre-removal mean ($p = 0.0929$). The annual estimates are negative from 2021 onward, although only the 2023 coefficient is statistically significant at the 5% level.

[39] As explained in Section 2, these two measures differ. Locally-raised revenues are my preferred measure because they capture resources raised locally and are more directly under municipal control. Total expenditures include both current and capital expenditures, as well as discretionary and mandated spending.

[40] Joint tests of the exposure-by-year coefficients for 2014–2018 yield $p = 0.3869$ for locally-raised revenues and $p = 0.6945$ for total expenditures.

Figure 5: Reduced-Form Effects of the Removal of Consulting Restrictions



Notes. The figure shows reduced-form event-study estimates from Equation 2. Panels A, B, C, and D report estimates for public services output, locally-raised revenues per resident, total expenditures per resident, and in-house personnel expenditures per resident, respectively. In all panels, the main regressor is the exposure measure defined in Equation 1 interacted with year dummies. The omitted year is 2019, the year in which the law removing consulting restrictions is drafted. All panels control for region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. The coefficient and standard error reported in the bottom left quadrant of each panel refer to the static specification. Referenced on page: 21.

Together with the increase in output, these findings suggest an improvement in local government performance: more public services are provided while fewer resources are raised from residents and businesses. I show this more directly by constructing a measure of output per unit of revenues in Appendix Figure A.14.

The conceptual framework provides a channel through which greater consulting use can reduce budgetary needs: municipalities may substitute consultants for internal personnel, with savings on the wage bill exceeding the additional consulting expenditure. Panel D provides evidence consistent with this channel. After removal, more exposed municipalities reduce their public employees' wage bill, pointing to substitution between consulting

and in-house labour.[41] Before removal, there is no statistically detectable differential trend in personnel spending. After removal, more exposed municipalities reduce their public employees' wage bill. The decline grows over time, with the annual coefficients statistically significant at the 5% level from 2021 onward. The static specification implies that a one standard deviation increase in exposure lowers personnel expenditures by 1.8 € per resident, or 1.0% of the pre-removal mean of 173.4 € ($p = 0.00809$). The gradual decline is consistent with municipalities adjusting their reliance on in-house labour over time. Together with the increase in consulting expenditure, this pattern supports substitution between consulting and internal personnel as one channel through which the reform can reduce budgetary needs.

Taken together, the results show that the removal of consulting restrictions had a meaningful first-stage effect, increasing consulting expenditures among the municipalities most constrained by the former caps. The reform also affected the broader performance of local government. On the output side, municipalities that expanded consulting increased public services output. On the input side, they substituted away from in-house public officials, effectively shrinking the size of the local bureaucracy and the amount of resources that citizens transfer to the state. The net effect on short-run performance, measured as public services delivered per unit of locally-raised revenue, is positive. I explore the mechanisms behind these findings in Section 4.1.

Effects of consulting on public sector performance. Table 1 reports OLS, reduced-form (RF), and 2SLS estimates for four outcomes: public services output, locally-raised revenues per resident, total expenditures per resident, and public employees' wage bill per resident. All three specifications use a common sample for each outcome.[42] Panel A reports the preferred specification; Panel B adds baseline covariates interacted with year indicators. The instrument is the interaction of pre-reform exposure and the post-reform indicator, as defined in Equation 3, with a first-stage F-statistic of around 10. The IV estimates translate the effects of removing consulting restrictions into effects per additional euro of consulting expenditure per resident. The table also reports the effect of a one standard deviation increase in consulting expenditure for OLS and IV, and in the instrument for RF, expressed as a percentage of the outcome's pre-removal mean. In the discussion below, I report the effects of an increase in consulting equal to 10% of its pre-removal mean, again expressed as a percentage of the outcome's pre-removal mean. Both means are calculated in the

[41] A joint test that the exposure-by-year coefficients for 2014–2018 are zero yields $p = 0.5771$.

[42] For each outcome, all three specifications use the corresponding 2SLS estimation sample, which additionally requires non-missing consulting expenditure. This sample restriction can produce small differences relative to the RF estimates in Figure 5 and the OLS estimates in Appendix Figure A.11.

corresponding IV sample.

For public services output (columns 1–3), the OLS coefficient is close to zero ($\hat{\beta} = 0.001$, $p = 0.3974$). The IV estimate implies that an additional euro of consulting per resident raises the output index by 0.135 points on its 1–10 scale ($p = 0.0541$). A 10% increase in consulting implies a 2.6% increase in output.[43] The larger IV estimate is consistent with "negative" selection of municipalities into consulting: municipalities hire consultants precisely because of poor output provision. Such selection would bias OLS downward, making consulting appear uncorrelated with state capacity.

For locally-raised revenues (columns 4–6), the OLS coefficient is positive ($\hat{\beta} = 0.574$, $p < 0.001$), whereas the IV estimate is negative. An additional euro of consulting per resident lowers locally-raised revenues by 7.612 € per resident ($p = 0.0652$). An increase in consulting equal to 10% implies a reduction in revenues of 0.9%.[44] Again, the sign reversal between the OLS and the IV is consistent with "negative" selection of municipalities into consulting: local governments purchasing more consulting services are those that extract more resources from citizens. This would bias OLS upward: consulting would be associated with higher revenues even if it reduces the amount that municipalities need to raise from residents and businesses.

For total expenditures (columns 7–9), the OLS coefficient is positive ($\hat{\beta} = 1.581$, $p = 0.0012$). The IV point estimate implies a reduction of 36.168 € per resident for each additional euro of consulting per resident, but is not statistically distinguishable from zero ($p = 0.1488$). An increase in consulting equal to 10% of its pre-removal mean corresponds to a reduction of 2.1% of mean pre-removal expenditures. The direction is consistent with lower budgetary needs, albeit more imprecisely estimated.

For personnel expenditures (columns 10–12), the OLS coefficient is small and negative ($\hat{\beta} = -0.045$, $p = 0.0759$). The IV estimate implies that an additional euro of consulting per resident lowers personnel expenditure by 3.687 € per resident ($p = 0.0360$). An increase in consulting equal to 10% of its pre-removal mean reduces personnel expenditure by 2.2% of its pre-removal mean.[45] In magnitude, the estimated in-house personnel saving is roughly half the estimated reduction in locally-raised revenues, pointing to wage-bill savings as an important source of lower budgetary needs. As with revenues and expenditures, selection could bias OLS upward: municipalities with higher public employees' expenditure tend

[43] A one standard deviation increase in consulting expenditure implies an increase of 2.28 output-index points, or 41.02% of the pre-removal mean.

[44] A one standard deviation increase in consulting expenditure implies a reduction of 134.87 € in locally-raised revenues per resident, or 15.98% of the pre-removal mean.

[45] A one standard deviation increase in consulting expenditure implies a reduction of 65.40 € in personnel expenditure per resident, or 37.73% of the pre-removal mean.

Table 1: Effect of Consulting on Public Sector Performance

	Public Services Output			Locally-Raised Revenues			Total Expenditures			Public Employees Wage Bill		
	OLS (1)	RF (2)	IV (3)	OLS (4)	RF (5)	IV (6)	OLS (7)	RF (8)	IV (9)	OLS (10)	RF (11)	IV (12)
Panel A: Preferred Specification												
Consulting	0.001		0.135*	0.574***		-7.612*	1.581***		-36.168	-0.045*		-3.687**
	(0.001)		(0.070)	(0.087)		(4.120)	(0.485)		(25.019)	(0.025)		(1.754)
Exposure × Post		0.003**			-0.195**			-0.924*			-0.091***	
		(0.002)			(0.090)			(0.549)			(0.034)	
N	44,097	44,097	44,097	81,648	81,648	81,648	81,692	81,692	81,692	82,759	82,759	82,759
𝔼(Y)	5.56	5.56	5.56	844.24	844.24	844.24	1774.86	1774.86	1774.86	173.35	173.35	173.35
Effect size (%)	0.31	0.67	41.02	1.20	-0.31	-15.98	1.58	-0.69	-36.12	-0.46	-0.71	-37.73
KP F-Stat			8.452			10.806			10.736			10.387
Panel B: Specification With Baseline Controls × Year												
Consulting	0.001		0.124*	0.548***		-6.487*	1.533***		-29.238	-0.047*		-3.359**
	(0.001)		(0.065)	(0.086)		(3.825)	(0.484)		(22.482)	(0.025)		(1.624)
Exposure × Post		0.003**			-0.173*			-0.775			-0.087**	
		(0.002)			(0.090)			(0.527)			(0.034)	
N	44,097	44,097	44,097	81,648	81,648	81,648	81,692	81,692	81,692	82,759	82,759	82,759
𝔼(Y)	5.56	5.56	5.56	844.24	844.24	844.24	1774.86	1774.86	1774.86	173.35	173.35	173.35
Effect size (%)	0.34	0.65	37.73	1.15	-0.27	-13.61	1.53	-0.58	-29.20	-0.48	-0.67	-34.37
KP F-Stat			9.156			11.490			11.428			11.199

Notes. The table reports OLS, static reduced-form, and IV estimates for public services output, locally-raised revenues per resident, total expenditures per resident, and public employees wage bill per resident. The IV columns instrument consulting expenditures per resident with $\text{Exposure}_m \times \text{Post}_t$, and the reduced-form columns report estimates from regressions of each outcome on $\text{Exposure}_m \times \text{Post}_t$. All specifications control for municipality fixed effects, region×year fixed effects, and population decile×year fixed effects. Panel B additionally controls for baseline municipal controls interacted with year fixed effects. Standard errors are clustered at the local labour market level. Effect size (%) is defined as the estimated coefficient multiplied by one standard deviation in the right-hand side variable (consulting per capita for OLS and IV, and $\text{Exposure}_m \times \text{Post}_t$ for RF) divided by the mean of the dependent variable before the restrictions have been removed. Referenced on pages: 24, 32.

to hire more consultants. This would make OLS understate the personnel savings from consulting and the degree of substitution between the two inputs.

Fiscal sustainability. A remaining question is whether lower revenues come at the expense of fiscal sustainability. Municipalities could reduce the burden on current residents by accumulating debt or relying more heavily on transfers. This matters for the national objective of maintaining sound municipal finances (Vannutelli, 2022). In this period, Italian municipalities are required to have a balanced planned budget, and indeed, the estimates above suggest they comply to this rule: both revenues and expenditures decline, although the expenditure effect is slightly imprecise. I examine the fiscal sustainability of municipalities directly in Appendix Figure A.15. The overall sustainability of municipal finances can be synthesised by their ability to repay debt and their tendency to issue new debt, as well as the presence of persistent deficit ("structural deficit"). Another object that national

policy-makers target to monitor municipal finances is their reliance on locally-raised revenues over total revenues ("fiscal autonomy"). The estimates show no clear increase in cumulative net borrowing or in the probability of structural deficit, and no clear decline in debt repayments or in the share of revenues raised locally. These results thus suggest that municipalities raise less revenue locally without becoming more indebted or more reliant on external resources over the period studied.

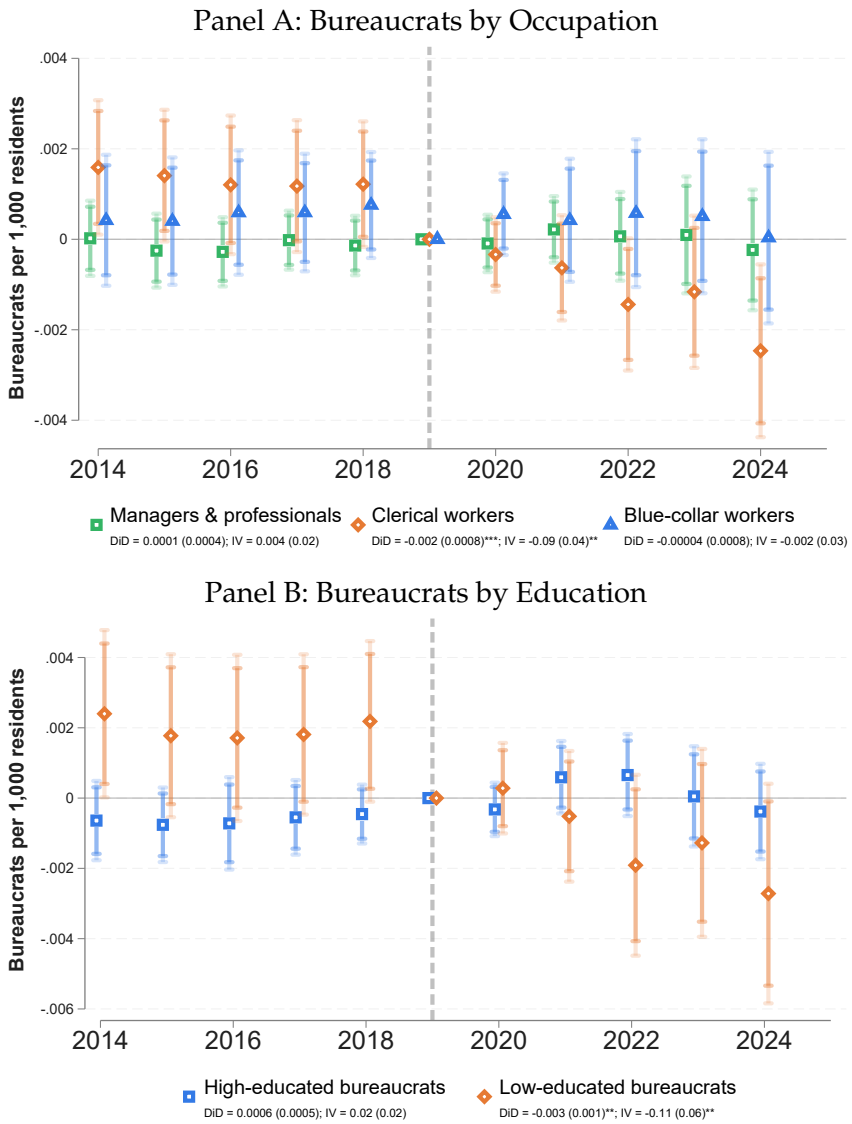
Decomposing public sector output. Table A.4 shows that the clearest evidence of improvement is in waste and territory management, which are two of the most important functions of Italian municipalities (see Appendix Figure A.10). As described in Section 2, these measures cover activities such as separate waste collection, environmental authorisations and inspections, urban permits, and public works. The pattern is consistent with consulting being a labour input into the administration of local governments. I find no statistically significant effects on social services (kindergartens and social assistance), local policing, or education-related services (school meals and transportation).

Consulting reshapes the internal organisation of the state. The wage bill can fall because municipalities employ fewer people or spend less per employee. Appendix Figure A.16 points to the former: employment declines, while the estimated change in average personnel expenditure per employee is small and statistically insignificant.

Figure 6 shows which employees account for this reduction. Panel A shows a decline in clerical employment, with no statistically significant change among managers and professionals or blue-collar workers. Panel B gives a separate breakdown by education: employment falls among less-educated workers, while the estimate for highly educated workers is small and statistically insignificant. This pattern suggests that consulting replaces administrative labour, consistent with the descriptive evidence on the nature of consulting services. Panel B shows that splitting the sample by education level, the number of low-skilled bureaucrats decreases, while the number of high-skilled bureaucrats is unchanged. Public officials and consultants are highly substitutable, but highly-educated bureaucrats less so. These findings are consistent with external expertise improving the efficiency of administrative production and thereby raising the opportunity cost of retaining low-skilled in-house staff.

Politicians' private benefits from consulting. Viewed through the lens of the conceptual framework, the gains in performance suggest a limited role for politicians' private benefits in explaining the response to the reform. Additional findings support this interpretation.

Figure 6: The Effects of Consulting on the Skills of Bureaucrats



Notes. The figure shows reduced-form event-study estimates from Equation 2. Panel A splits municipal employees into managers and professionals, clerical workers, and blue-collar workers. Panel B splits municipal employees into high- and low-educated workers. Outcomes are measured per 1,000 residents using 2001 population and winsorized at the 1st and 99th percentiles. The estimates reported below each outcome are from the pooled difference-in-differences specification and the static IV specification, which instruments consulting expenditures per resident with post-reform exposure. All specifications control for municipality, region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. Referenced on page: 27.

First, officials might hire consultants to shift responsibility for their decisions, *blame-shifting*. If this were an important motive, demand for consulting should rise when personal liability becomes more salient. Using the timing of Court of Auditors liability judgments, I find no evidence of an increase in consulting relative to personnel expenditure (Appendix Section C.3, Figure A.23). This is consistent with interviews in which local politicians describe remaining responsible for authorisation and oversight when tasks are contracted out.

Second, if consulting were used as a means for politicians to achieve personal objectives, this may distort the allocation of resources. I study this explanation using municipal expenditure data, which classifies expenditure across approximately twenty functions, or "missions," including administration, education, public safety, transport, environmental protection, and social services. Changes in expenditure shares associated with these functions are informative about shifts in local-government priorities. I do not find that access to consulting leads municipalities to distort their policy-making. Appendix Figure A.19 shows that access to consulting does not change the share of expenditure directed to each municipal function.

If consulting were primarily a means of extracting rents, one might expect its use to be associated with other forms of political misconduct. I find no statistically significant relationship between consulting expenditure and recorded local corruption (Appendix Figure A.20). If anything, the consulting response to the reform is smaller in municipalities with prior corruption or Court of Auditors cases (Appendix Table A.6). Consulting expenditure increases as elections approach, that is, when electoral accountability is higher and politicians objectives are plausibly more in line with citizens' ones (see Appendix Figure A.21). The same pattern does not apply to public employees' spending. Consulting expenditure is also higher under mayors with more education or with a background in high-skilled occupations (Appendix Figure A.22).

Finally, politicians might use consultants to secure immediate gains while neglecting the bureaucracy's future capacity. Over the four years following the reform, I find no statistically significant reduction in administrative capacity (Appendix Table A.5). The evidence, therefore, does not support for the interpretation of consulting being used prevalently as a tool to extract resources.

4.2. Validity of the Design

In this subsection, I assess the validity of the empirical design.

Parallel trends. The key identifying assumption underlying both the reduced-form and first-stage specifications is that, absent the removal of consulting restrictions, municipalities with different levels of exposure would have followed similar paths. A potential concern is that, even after conditioning on the full set of fixed effects, unobserved factors correlated with the cross-sectional variation in exposure to consulting restrictions may coincide with the lifting of the restrictions and thereby confound the estimates. Although the identifying assumption cannot be tested directly, its plausibility can be assessed using the pre-reform event-study estimates. Throughout the paper, I report event-study estimates for many different outcomes spanning consulting expenditure, municipal finances, public services, workforce composition, and placebo measures.[46] These estimates provide no evidence of differential pre-trends. This evidence is consistent with the institutional features of the national reforms and the use of transitory spending variation to construct municipal exposure, as described in Section 3.

Balance tests. Even in the absence of detectable pre-trends, exposure may be correlated with local characteristics that shape municipalities' responses to aggregate shocks. Such differences could have affected spending during the financial and sovereign debt crises, when exposure was determined, and outcomes during the pandemic, when restrictions were lifted. I first perform balance tests to understand whether different levels of exposure to consulting restrictions correlate with pre-restrictions observable characteristics. This exercise helps support the idea that the exposure measure can be considered quasi-exogenous in nature.

Appendix Figure A.31 shows balance tests on around thirty observable characteristics, measured either in 2001, the most recent census year prior to the restrictions, or in 2000. It reports the standardized coefficient from the cross-sectional regression of Exposure_m on each characteristic X_m, controlling for population decile and region fixed effects, as in the main specification. Standard errors are clustered at the local labour market level. Municipality level characteristics are grouped into four categories: public sector, local economy, culture, education, and demographics. Being exposed to the reform is not systematically correlated with the reported characteristics. Most coefficients are close to zero and not statistically significant at 5 or 10% level.

I also examine whether exposure is associated with differential income growth.[47] Appendix Figure A.34 shows no evidence of a systematic relationship between exposure and

[46] Besides the results shown in this section, see Section 4.1 and Appendix Section C.

[47] One may think that municipalities with different service spending in 2009 – relative to their 2006-2010 average – may have been affected differently by the crisis.

changes in taxable income across municipalities. These exercises help support the idea that being exposed to tighter restrictions is not systematically correlated with underlying observable local fundamentals.

Falsification exercises. I conduct two falsification exercises. The first tests whether the results are specific to 2009, the year used to determine the consulting spending cap, by constructing exposure measures anchored to alternative years. The second examines outcomes that should not respond to the removal of consulting restrictions.

For the first exercise, I replace the 2009 service expenditure with expenditure in each alternative anchor year and adjust the comparison-period average accordingly. Specifically, I define:

$$\text{Exposure}_m^y = \left(1 - \frac{\text{Services}_{m,y}}{\frac{1}{4}\sum_{t=2006, t\neq y}^{2010} \text{Services}_{m,t}}\right) \times 100, \quad y \in \{2006, \ldots, 2009\} \tag{6}$$

Equation (6) mirrors the baseline construction in Equation (1), changing only the anchoring year.[48]

These exercises assess whether the first-stage relationship is specific to the statutory anchor year. If exposure primarily captured persistent municipal characteristics associated with subsequent consulting expenditure, similar relationships might emerge when exposure is constructed using other years. Panels A and B of Appendix Figure A.32 report estimates from the static version of Equation 2, with consulting expenditure per capita as the outcome and exposure constructed as in Equations 6 and 7, respectively. The placebo estimates are uniformly small, generally negative, and statistically indistinguishable from zero. Only exposure anchored to 2009, the year used to determine the statutory cap, yields a positive and statistically significant estimate. This pattern supports the interpretation that the first stage reflects exposure to the consulting restrictions.

Second, I examine population and average income as placebo outcomes. These variables are unlikely to respond materially to the removal of consulting restrictions over the period studied, but may reveal broader demographic or economic changes correlated with exposure. Panels C and D of Appendix Figure A.32 report estimates from Equation 2 using,

[48] To extend the exercise to earlier anchor years, I construct a second set of placebo exposure measures using a common denominator: average service expenditure over 2006–2010. This allows the anchor year to vary from 2000 to 2009 while holding the average consulting spending fixed:

$$\text{Exposure}_m^y = \left(1 - \frac{\text{Services}_{m,y}}{\frac{1}{5}\sum_{t=2006}^{2010} \text{Services}_{m,t}}\right) \times 100, \quad y \in \{2000, \ldots, 2009\} \tag{7}$$

respectively, the logarithm of the number of residents and the logarithm of the average income as outcome variables. There is little reason to expect either variable to respond to the lifting of consulting restrictions. Consistent with this, both panels show no differential post-reform change across municipalities with different levels of exposure. The estimated coefficients are close to zero and statistically insignificant throughout. Importantly, the pre-treatment coefficients also display no evidence of differential trends.

External validity. The estimates may be driven by municipalities for which the consulting restrictions were most binding. To assess this concern, I re-estimate the first-stage and reduced-form specifications separately for municipalities below and above median exposure. Appendix Figure A.36 shows similar point estimates across the two groups for each outcome. This pattern suggests that the results are not driven solely by the most exposed municipalities.

Instrument validity. Figure 4 and Table 1 confirm the instrument is relevant – the removal of restrictions, combined with differential exposure to pre-reform consulting caps, produces a strong first stage. The IV interpretation additionally requires that, conditional on the fixed effects, exposure to the removal of restrictions affects municipal outcomes only through consulting expenditure. The exclusion restriction is not directly testable. However, the baseline balance, absence of detectable pre-trends, and falsification exercises lend support to the exclusion restriction.

Imperfect compliance. A further concern is imperfect compliance with the consulting restrictions, specifically municipalities facing more restrictions may have fewer incentives to comply with the law. These municipalities would not react to the removal of restrictions, and this could lead to underestimating the first-stage effect. Although I cannot directly measure compliance, rulings by the Court of Auditors provide evidence of active judicial scrutiny of consulting. During the restriction period, 15% of the Court's rulings concerned consulting.[49] Non-compliance does not correlate with exposure (Appendix Table A.13). Also, aggregate patterns (see Appendix Table A.3 and Figure A.7, Panel A) are consistent with the law having a bite.

Implementation of restrictions. I do not use the introduction of restrictions to identify their effects, for two reasons. First, several key outcomes, including consulting expenditures and public service delivery, have insufficient coverage before 2010 to assess whether

[49]This figure excludes references to consulting services used in the judicial proceedings themselves.

municipalities with different exposure were following parallel trends. Second, the exposure measure embeds a mechanical relationship with pre-reform spending. Because it is defined using 2009 service spending relative to surrounding years, municipalities with higher exposure are, by construction, those with lower spending in 2009 relative to 2010. In an event-study specification that omits 2010, this mechanically generates a negative coefficient on the 2009 lead. Post-reform estimates are therefore measured relative to a negative pre-reform estimate, which biases the estimated effects toward zero. Third, because consulting expenditures have high yearly variation, the resulting estimates are likely to be contaminated by mean reversion: municipalities with elevated spending in 2009 were both less exposed to the reform and more likely to experience subsequent declines, regardless of the policy change. Together, these features invalidate any event-study design around the implementation of restrictions.

I nevertheless report descriptive evidence on spending during the restriction period. Appendix Table A.14 shows that more-exposed municipalities spent less on consulting during this period, conditional on region-by-year and population-decile-by-year fixed effects. This association is consistent with the restrictions having been binding, but does not identify their causal effect.

Other robustness. Appendix Figure A.35 shows that the main event-study patterns are robust to clustering by municipality or province, adding baseline characteristics interacted with year indicators or time-varying log population and income, and omitting region×year or population-decile×year fixed effects. Similar patterns emerge with alternative exposure definitions, winsorization thresholds, and log monetary outcomes. A quadratic exposure-response specification (Callaway, Goodman-Bacon and Sant'Anna, 2026) also yields similar results.

5. Citizens' Demand for Public Production

The findings shown so far create a puzzle. Access to consulting improved public sector performance and lowered the resources citizens transferred to the state. Why, then, was it restricted to begin with? When the sovereign debt crisis forced Italy to consolidate its budget, policymakers could have reduced spending on internal and external inputs proportionally while preserving the input mix. Instead, they shifted away from external toward internal resources (see Appendix Figure A.7, Panel A). Restricting consulting use in the public sector is a common national policy adopted by many countries in the world (see Appendix Figure A.8).

This section develops an explanation that departs from standard arguments for restricting external appointments, such as limiting corruption or patronage in the public sector. States are ultimately accountable to citizens, and citizens value not only what the state delivers and how much it costs, but also *how* production is organised. Productively efficient organisational arrangements may therefore not be adopted because of the way citizens want the state to be organised. I refer to this broad concept as *citizens' demand for public production*, encompassing both their beliefs and their preferences over the modes of production.

Citizens' beliefs and preferences about how the state is organised cannot be recovered from administrative data. I thus designed and administered a large-scale survey with embedded AI interviews on a representative sample of Italian citizens. The survey instrument combines AI-led interviews, closed-ended questions, information treatments, and a discrete choice experiment to quantify citizens' policy preferences and understand what shapes them. Citizens exhibit substantial demand for public production. Part of this demand reflects misperceptions about the economic consequences of consulting. The remainder can be explained by ideology and the value citizens attach to public employment for its perceived benefits to the local economy.

5.1. Large-Scale Survey with Embedded AI-Interviews

Data collection and sample. My sample comprises 4,828 respondents surveyed in July and August 2026. The survey was administered online, with participants recruited through the survey company, *Bilendi*, which is widely used in economics (e.g., Chinoy, Nunn, Sequeira and Stantcheva, 2026, Alesina, Miano and Stantcheva, 2023, Alesina, Stantcheva and Teso, 2018). Bilendi maintains online panels recruited through opinion platforms and telephone, and sends survey links to panelists with targeted socioeconomic characteristics. Participation is restricted to Italian residents aged 18–74. Respondents who complete the survey are rewarded with points redeemable for a variety of perks, including cash, hotel points, frequent flyer miles, shopper rewards, and charitable donations. This allows the company to recruit a diverse sample, including higher-income respondents who might be indifferent to small cash incentives but who value other rewards.

Respondents are drawn by quota sampling on gender, age, income, and macro-region of residence, so that the sample matches the national distribution along each dimension.[50]

[50] Income quotas are computed combining the share of individuals aged 18–74 with no personal income, estimated from the 2022 Bank of Italy Survey on Household Income and Wealth, with the distribution of gross personal income among taxpayers with positive income in 2024 Ministry of Economy and Finance tax records.

Stantcheva (2023) shows that the pool of respondents available to commercial survey companies is broadly representative of national populations, but tends to include more educated and younger individuals. Appendix Table A.15 compares the final sample with population benchmarks on the four quota dimensions, and with education and employment, which quotas do not target. The sample tracks the national population closely on all dimensions except education, where respondents are positively selected.

Median completion time was around 15 minutes, and the survey design automatically excludes respondents who complete the survey implausibly quickly.[51] Appendix Table A.16 shows that 22% of respondents who begin the survey do not complete it, within the 15–30% range Stantcheva (2023) reports for online surveys. Dropouts are concentrated at the AI-interview stage, which precedes the randomized interventions. Attritors are less familiar with the use of artificial intelligence than completers, but the differences in all other observable characteristics remain small. Thus, I interpret my estimates as applying to a population somewhat more comfortable with the use of artificial intelligence than the Italian adult population.

I designed the survey in-house and developed a dedicated web application to administer the AI interviews. The application is embedded directly in the survey: respondents launch the interview from within the questionnaire and return to it upon completion, so the interview is integrated seamlessly into the survey flow. More details can be found in Appendix E.

Consent and AI-interview framing. Before deciding whether to participate, respondents are told that the study is academic and nonpartisan, that their responses are anonymous, and that they may withdraw at any time. The same screen also discloses that the questionnaire includes a short interview developed by the research team and conducted with the support of an artificial-intelligence tool. Respondents are told that the tool cannot modify their answers and is programmed to ask neutral, non-leading questions. Disclosing the AI component before consent aligns respondents' expectations with the interview they will encounter and reduces surprise-induced attrition when the interview begins. Additionally, using neutral language and making clear that AI tool is programmed by the research team is a way to limit differential selection and reduce skepticism. This is particularly important given the context and given that attitudes toward artificial intelligence vary systematically across the population. For instance, women and older adults are generally more concerned about artificial intelligence. More broadly, Italian citizens appear to be

[51] See the survey duration distribution in Appendix Figure A.25.

skeptical by international standards.[52]

Survey structure. Panel A of Figure 7 presents a block diagram of the survey flow; Appendix E reproduces the full questionnaire and the interview prompt, translated into English. Respondents first provide consent, then answer a block of demographic questions and a short module on broader societal views, and then proceed directly to the AI-led interview. Placing the interview early limits fatigue and keeps engagement high. More importantly, it ensures that closed-ended questions are asked after the interview to avoid priming respondents with predetermined answer categories.

After the interview, respondents complete an attention check, a randomized elicitation of prior quantitative beliefs, and a randomized information module. They then report their support for consultants and for public employees—with the order of the two questions randomized—and whether they would prefer the government to employ more or less public employees relative to consultants. The former two questions capture citizens' attitudes toward each mode of production in absolute terms, the latter, instead, forces citizens to trade off one versus the other. These three questions constitute my main outcome variables. Respondents then complete, in order: a discrete choice experiment; a short set of closed-ended items that both validate the AI-interview evidence and serve as a manipulation check; a real-stakes behavioral question; and, for respondents not asked prior quantitative beliefs, posterior beliefs about the extent of consulting use.[53] The survey closes with a feedback module.

AI-led interviews. AI-led interviews are a novel tool that allows me to understand citizens' mental models of how the state is organized at scale (Geiecke and Jaravel, 2026, Haaland, Roth, Stantcheva and Wohlfart, 2025, Chopra and Haaland, 2024). Interviews conducted through a large language model (LLM) are a cost-effective way to collect thousands of qualitative responses within hours. Well-designed prompts can induce the LLM to carry out interviews of comparable quality to expert human interviewers, extracting richer information than open-ended text boxes while preserving the benefit of not priming respondents toward particular answers (Geiecke and Jaravel, 2026).

Before clicking on the AI-interview link, respondents are encouraged to answer in their

[52] A 2025 Pew Research Center survey reports that roughly half of Italian respondents are more concerned than excited about artificial intelligence, placing Italy among the most skeptical countries in the sample. See https://www.pewresearch.org/global/2025/10/15/how-people-around-the-world-view-ai/.

[53] The prior-belief randomization assigns 50% of respondents to the consulting prior, 10% to the personnel prior, and 40% to no prior elicitation. The last group reports the posterior consulting belief, avoiding anchoring from an earlier elicitation.

own words and are told that there are no right or wrong answers. These guidelines are in line with the best-practices to motivate effort for open-ended questions (Haaland, Roth, Stantcheva and Wohlfart, 2025).[54] It also invites respondents to ask for clarification, a feature that open-ended text questions do not offer.

Respondents are interviewed by an AI agent that follows a prompt I wrote (reproduced in Appendix E). Panel B of Figure 7 shows the interface they see when the interview opens. The AI agent introduces the topic: "As you may know, public administrations can sometimes carry out a task with their own employees, or hand it over to consultants or outside firms, even though the task remains paid for and overseen by the state." Respondents can then answer by typing or by speaking. Interviews are carried out in a non-directive manner through follow-ups, while keeping the overall duration short.[55] The aim is to reduce respondent burden in an online setting and limit differential attrition while still preserving room for unprompted views and deepening respondents answers. At the same time, the short format imposes tighter discipline on follow-ups and ensures coverage of a small set of issues: whether they are familiar and what comes to their mind when thinking about consulting in the public sector, the perceived advantages and disadvantages of consulting and external firms, the perceived advantages and disadvantages of public officials, and lastly, why public administrations rely on external resources. This application of AI-led interviews shows how this tool can flexibly adapt the best practices from sociology (Small and Calarco, 2022) to relatively short online interviews preserving a representative sample throughout.

Response quality and biases. To ensure data are of high quality, an attention check follows the interview block. Attention checks are useful to screen out inattentive respondents whose answers may be inaccurate and increase noise in the data. Only 3% of respondents fail the attention check.

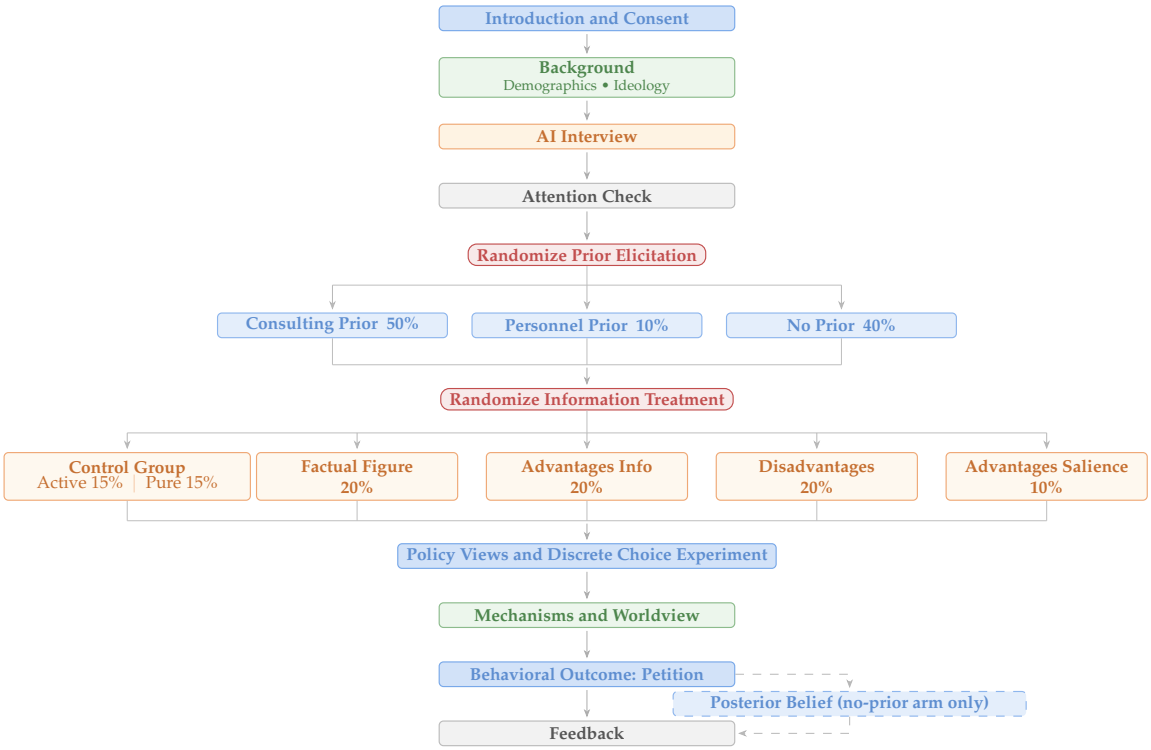
As a quality metric for AI interviews, following Geiecke and Jaravel (2026), I ask which format respondents would prefer in a future study: an automated interview with AI – like the one they just took, an interview with a human, a survey with open text fields, or no preference. 61% of respondents would rather do an AI-interview compared to an interview with a human enumerator or traditional open-ended questions, and 24% has no

[54] Respondents are reminded then to return to the questionnaire afterward to complete the study and receive compensation. To discourage low-effort responses, respondents are also informed that answers that are excessively short or rushed may be flagged and may not qualify for compensation.

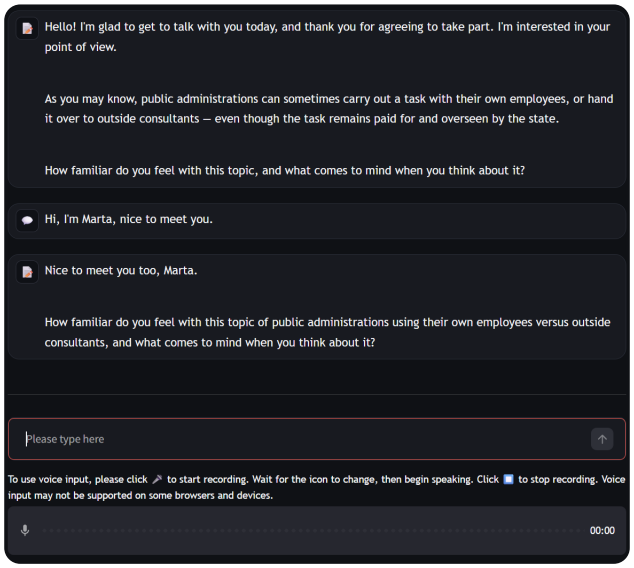
[55] At the end of the interview the LLM summarizes the points raised by the respondent, who are asked to evaluate the summary. My interview is built to reach the summary after about 6 to 7 interviewer questions, with a hard cap of 8.

Figure 7: Survey with Embedded AI-Led Interview

Panel A: Survey Flow



Panel B: Interview Example



Notes. Panel A shows the flow of the online survey with embedded AI-led interview. The dashed branch denotes the posterior-belief module shown only to respondents assigned to the no-prior arm. Panel B shows a snippet of the respondent interface during the text interview. Referenced on pages: 36, 37.

preferences. Additionally, respondents gave an excellent rating to the interview summary which amounts to 3.5 out of 4.

One relevant concern with online surveys is that stated preferences are hypothetical. I address these concerns by measuring outcome variables –support for each mode of production and one vs the other– in a more realistic way. First, in the discrete choice experiment, respondents have to choose their preferred policy option, whether to cut consultants or bureaucrats. These policy options closely mimic policies implemented in Italy during the sovereign debt crisis. Additionally, discrete choice experiments (or conjoint designs) have been shown to closely match real-life outcomes (Hainmueller, Hangartner and Yamamoto, 2015). Second, I design a "real-stakes" behavioral question. Immediately before the feedback module, all respondents are asked whether they would like to support a petition. They can select a petition asking Italian municipalities to rely more on public employees, or an exactly parallel petition asking them to rely more on consultants and external firms, or neither.[56] Respondents are told that aggregate support will be communicated to the National Association of Italian Municipalities, my policy partner on the bureaucrats survey. The results hold for this behavioural measure as well as for stated policy preferences, and the two are strongly correlated (see Appendix Figure A.37).

Three features of the setting limit experimenter demand and social desirability bias. Consultants versus public employees is not a domain with an obvious socially desirable answer or a well-defined norm.[57] The whole survey and the interview questions are worded neutrally, and the AI is instructed to "let the respondent bring up the substance of their views in their own terms." Indeed, 73% of respondents report that the survey was not biased or didn't express a view.[58]. Lastly, responses are fully anonymous. Anonymity is an important feature to limit experimenter demand effect (Haaland, Roth and Wohlfart, 2023).

5.2. Citizens' policy preferences over how the state is organized

This subsection describes respondents' knowledge and views of consulting in the public sector. Are citizens familiar with how the state is organized? What are their policy preferences on how the public sector recruits its labour force?

[56]The order of the two opposing petitions is randomized, while the opt-out remains last.

[57]Bursztyn, Haaland, Röver and Roth (forthcoming) list domains where social desirability bias is severe, such as health and moral or criminal behavior.

[58]Additionally, treatment assignment does not explain whether respondents report report no directional survey bias, see Appendix Table A.17

Knowledge and familiarity. Citizens know little about consulting in the public sector, yet substantially overstate how much municipalities spend on it. The AI interview opens by asking respondents how familiar they are with the topic. Panel A of Figure 8 classifies their answers: 64% display no or only low familiarity, and just 6% display high familiarity.

Immediately after the interview, and before any information treatment, I elicit quantitative beliefs. A random half of respondents are asked how many euros municipalities spend on external consulting for every €100 spent on their own employees. The true value is about €10.[59] As Panel B shows, 95% report more, 61% report above €50, and only 3% answer correctly. The misperception is thus both widespread and large.

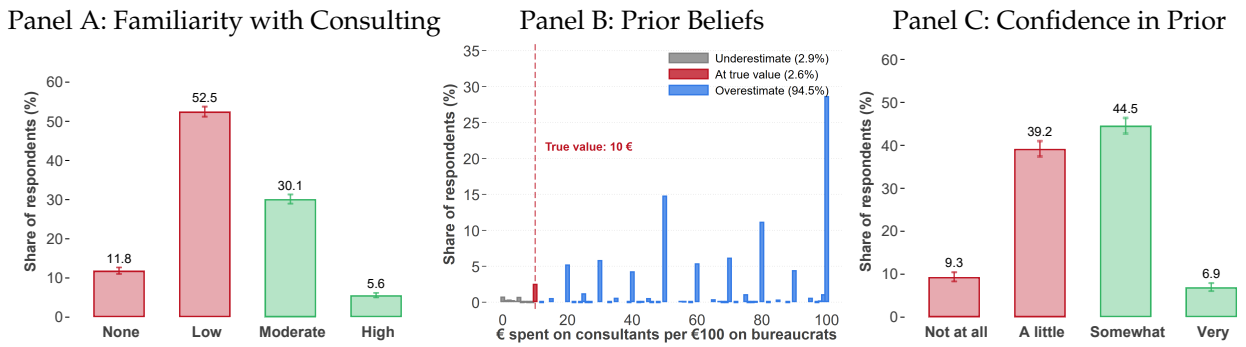
To assess whether this simply reflects the difficulty of estimating municipal budgets, I benchmark it against a parallel question on personnel spending, asked at the same point in the survey to the separate randomized group: how many euros municipalities spend on their own employees for every €100 of locally-raised revenues. Beliefs are dispersed here too, but errors are smaller: 60% exceed the true value of €30 and 12% identify it exactly (Appendix Figure A.26). Overestimation is therefore specific to consulting rather than a general tendency to overstate what municipalities spend.

After each point estimate, I ask respondents how confident they are in their answer. Panel C shows that 49% report little or no confidence. Overall, Figure 8 highlights that citizens lack knowledge and familiarity, yet this does not translate into random guesses: citizens systematically believe the public sector relies more on consultant than the reality.

Support for Consultants, Bureaucrats, and Relative Reliance. Despite their limited knowledge of consulting in the public sector, most respondents take a position on how the state should organize production. Panel A of Figure 9 shows that 45.1% prefer the state to rely less on consultants and external firms, while only 29.5% prefer no change and 25.4% favor greater reliance. Panel B shows a more balanced distribution for bureaucrats: 33.5% prefer less reliance, 28.0% prefer no change, and 38.5% prefer greater reliance. The share of respondents expressing a neutral position is identical across the two inputs. The main difference is that respondents are more likely to favor reducing reliance on consultants than reducing reliance on bureaucrats, and more likely to favor greater reliance on bureaucrats than greater reliance on consultants. Panel C makes the tradeoff between the two inputs explicit: 58.1% favor relatively greater reliance on bureaucrats, compared with 21.4% who favor relatively greater reliance on consultants; 20.5% prefer neither. Thus, when it comes to choose one versus the other, respondents put even more emphasis on in-house production.

[59] Approximately the average ratio of consulting to personnel expenditures across municipalities in 2024.

Figure 8: Knowledge and Beliefs About Consulting Use

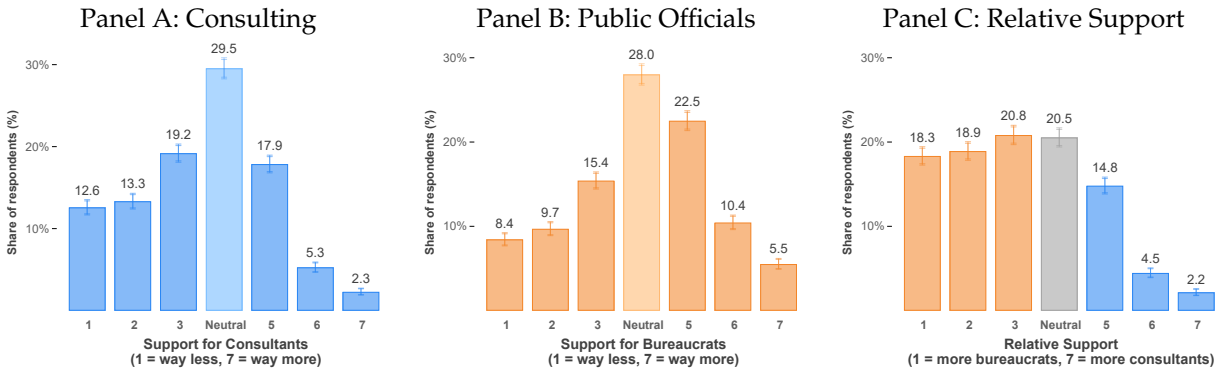


Notes. Panel A reports the share of respondents by familiarity with the topic, coded from Part I of the AI-led interview on a scale from 0 (no knowledge) to 3 (detailed or professional knowledge); the sample is restricted to respondents with a codable interview. Panel B reports the distribution of prior quantitative beliefs about how many euro municipalities spend on external consulting for every 100 euro spent on their own employees. The dashed red line marks the true value of 10 euro. Gray bars denote respondents who underestimate consulting spending and blue bars respondents who overestimate it; the legend reports the share of each group. Panel C reports the distribution of respondents' confidence in this belief, elicited right after the belief question. All panels use the analysis sample of respondents who submitted the final survey page. In Panel A and C, the capped lines correspond to the 90 and 95 percent confidence bands. Referenced on page: 40.

Similarly, 51.2% of respondents would support a petition asking to expand the number of public officials, while only 24.9% would support one asking to expand the number of consultants. Taken together, these responses point to a preference for in-house labour force, which become even more prominent when the tradeoff between the two is made explicit.

These policy views vary widely across demographic groups, and the variation tracks both direct economic interests and political leanings. Appendix Figure A.27 plots average support for consultants, support for bureaucrats, and preferred relative reliance by demographic group. Public employees report the highest support for bureaucrats of any employment group, and respondents with a close family member working in the public sector show the same tilt: more support for bureaucrats, less support for consultants. Support for bureaucrats is also higher among the unemployed and students, and in the South and Islands, where local economies depend more heavily on public employment. There is a stark political divide. At the extreme left, strong support for bureaucrats coexists with strong opposition to consultants, a gap of almost two points on the seven-point scale. The gap narrows steadily moving rightward and closes among respondents on the right, who rate the two inputs similarly; it reopens at the extreme right. News consumption reinforces the tilt toward in-house production: respondents who follow the news more closely are less favorable to consultants and more favorable to bureaucrats. Support for consultants falls by about one point on the seven-point scale between the youngest and

Figure 9: Distribution of Support for Consulting and Bureaucrats



Notes. The figure shows the distribution of survey respondents' preferred level of use of consultants and public employees in the public sector, and their preferred relative allocation of the two inputs. Each bar reports the share of respondents in the analysis sample selecting a given response category on a seven-point scale. In Panels A and B, higher values correspond to preferring more use of the relevant input. In Panel C, lower values correspond to preferring greater reliance on bureaucrats and higher values to preferring greater reliance on consultants. The neutral option ('no more nor less') is represented by the choice number 4. In all panels, the capped lines correspond to the 90 and 95 percent confidence bands. Referenced on page: 40.

oldest cohorts, while support for bureaucrats stays roughly flat, so preferences tilt toward in-house production over the life cycle. Men and women hold similar views on consultants, but women are more favorable to bureaucrats. Education plays almost no role, and income matters only at the very top: respondents earning more than €50,000 are the only group favoring fewer bureaucrats and marginally more consultants.

Lastly, preferences over how the state is organized track respondents' broader beliefs about public and private institutions. Appendix Figure A.28 shows these conditional associations.[60] Favorable views of private firms produce a clean substitution pattern: respondents who believe that firms and profits contribute positively to society support consultants more, bureaucrats less, and prefer a more consultant-oriented mix. Favorable views of the public sector are not the mirror image. Trust in public institutions predicts stronger support for consultants, both in levels and relative to bureaucrats, and is only modestly related to support for bureaucrats. One interpretation is that institutional trust captures confidence in the state's capacity not only to rely on its own workforce, but also to select, monitor, and coordinate private agents. By contrast, respondents who want public

[60] Each point is obtained from a separate OLS regression of one of the three preference measures (coded from 1 to 7) on one of the four belief measures (coded from 1 to 5). All specifications include fixed effects for birth year, gender, education, employment status, public sector ties, income category, and macroregion, and report heteroskedasticity-robust standard errors. The corresponding unconditional specifications yield nearly identical estimates.

institutions to do more, and the state to be more present in daily life, favor bureaucrats in both absolute and relative terms. Confidence in state capacity is therefore distinct from a preference for the *size* of government: trusting the public sector does not translate into demand for a larger public workforce, while wanting a larger state does.

Citizens' reasoning on the modes of production. AI interviews are helpful to get qualitative evidence on citizens' mental models about consulting and public employees, and their relative advantages. To convert qualitative into quantitative data, I use a large language model to translate open-ended answers into short concepts, which are grouped by the mode of production they favor.[61] These are shown in Figure 10.

Panels A and B present word-clouds of perceived benefits of consulting and public employees, respectively. Bigger fonts indicate more respondents mentioning each benefit. Three main considerations come out. First, economic considerations appear prominently in both panels. Respondents associate consultants with specialized expertise, speed, and professionalism. At the same time, arguments in favor of public employees often refer to the higher cost of consulting and to the fact that public employees are already paid by the state. This suggests that some respondents view consulting mainly as an additional expense, rather than as an input that may improve productivity. These perceptions contrast with the main empirical results shown in Section 4, which show that greater access to consulting improves public sector performance. Second, the lower-cost rationale for public employees is the most prevalent rationale, even when compared with non-monetary aspects. Third, non-monetary considerations are still relevant especially in support for public employees, less so in support for consulting. Corruption risk, conflict of interests, lack of control and transparency are indeed prominent concerns.

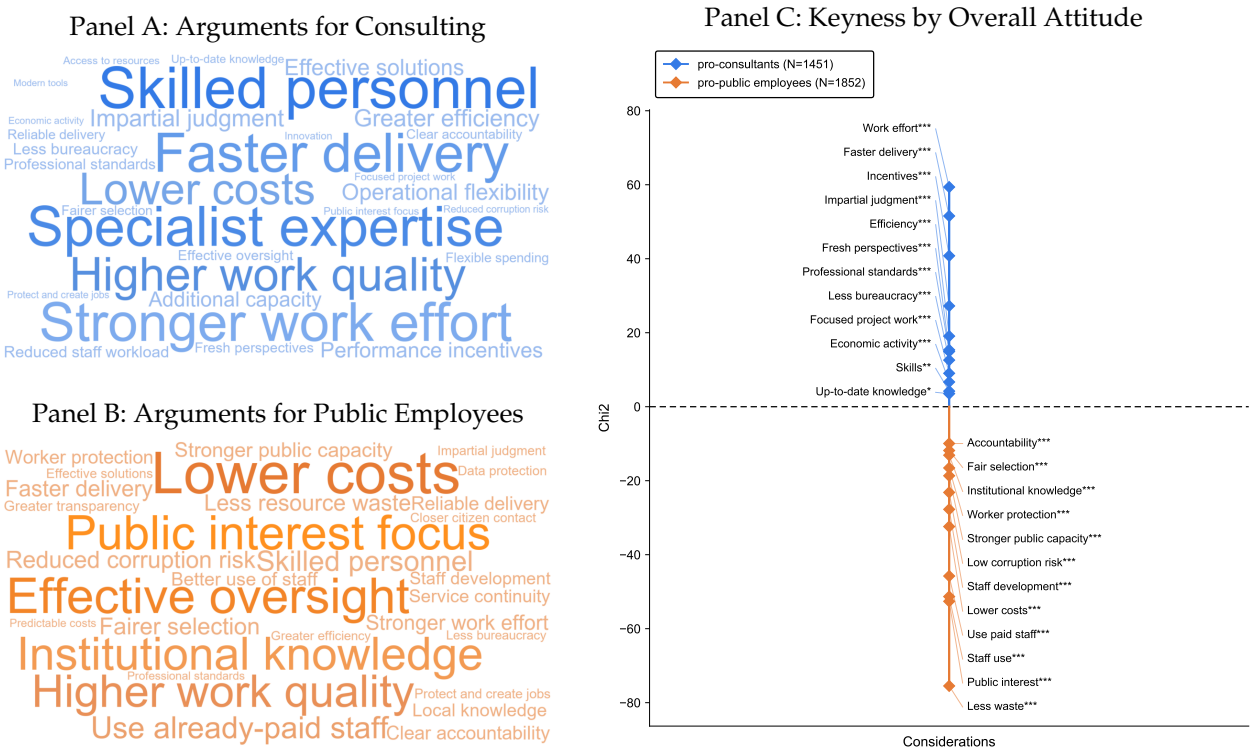
Panel C compares how often each reviewed consideration is mentioned by respondents with different overall attitudes. For each person and consideration, mentions are combined across the target-option panels and counted once.[62] The signed Yates-corrected chi-squared statistic compares the share mentioning each feature in the two attitude groups. Pro-consultant respondents particularly mention work effort (22.2% versus 12.1%) and faster delivery (21.2% versus 11.9%). Pro-public-employee respondents particularly mention less resource waste (1.2% versus 7.9%) and public interest (5.2% versus 12.6%).

Taken together, AI interviews suggest that citizens think about the organization of

[61] More details on this procedure in Appendix E.

[62] I measure overall net position towards consulting through an LLM which is instructed to consider the interview as a whole. This measure is highly correlated with close-ended measures. In Panel C of Figure 10, I remove those with neutral, mixed or not classifiable positions, which amount to 31.6% of the full analysis sample.

Figure 10: Arguments for Consultants and for Public Employees from AI Interview



Notes. The figure uses the full survey analysis sample of 4,829 respondents. A large language model maps respondents' answers about favoring consultants or public employees into short English concepts. Panels A and B present word clouds of perceived benefits of the favoured option, combining that option's advantages with disadvantages of the alternative. Panel C uses source-reviewed considerations, pooling each consideration across target options and counting each respondent once. It shows the signed Yates-χ^2 for each consideration and tests equal mention rates across the overall-attitude groups. Respondents without a retained directional concept contribute zero concept mentions rather than being dropped. In Panels A and B, each extracted claim is checked against the respondent's original answer and its question context before grouping; unsupported or unresolved assignments are excluded. Each panel shows the 30 most frequently mentioned benefit groups; font size increases with the number of respondents mentioning a group, counting each respondent once per group within a panel. Font sizes are normalized separately within each cloud and are not comparable across panels. Panel C classifies the respondent's overall position from the full interview and compares pro-consultant respondents (in blue; $N = 1{,}451$) with pro-public-employee respondents (in orange; $N = 1{,}852$); neutral, unclassified or mixed respondents are excluded. It plots the chi-squared statistics for considerations mentioned by at least three respondents; positive (negative) values indicate greater prevalence in the blue (orange) group. The 12 largest values in each direction are shown; stars use unadjusted p-values. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 43.

the state largely in economic terms. But economic reasoning does not uniformly favor consultants. High public sector cost is often the leading argument against their use, in contrast with the observational evidence in Section 4.

5.3. Experimental Results: The Causal Effects of Information

This subsection presents the results from the experimental part of the survey, which showed respondents information about the use of consulting in the public sector. This serves two purposes. First, it validates the descriptive findings: people have very negative views and strongly oppose the use of consulting in the public sector, and these negative views are mostly linked to citizens' perceptions of the economic consequences. Second, it shows that information is effective in shifting policy views and real-stakes decisions.

Design. Treatments were designed to be as neutral and simple as possible to reduce priming and experimenter demand effect, as well as cognitive fatigue. The exact text used in the survey is reported in Appendix E.

The first information treatment ("factual information") is a short text that reports factual information on how much Italian municipalities spend on consulting, which amounts to roughly €10 per €100 euros municipalities spend on in-house personnel. The objective of this treatment is to understand whether adjusting citizens' factual beliefs on the extent of consulting use may shift their attitudes towards how the state is organized.

The second information treatment ("advantages information") reports the potential economic benefits of hiring consultants rather than relying on in-house public officials. Its content mirrors the estimates in Section 4: respondents are told that access to consultants can reduce costs and improve the quality of public services. To distinguish this information treatment from pure priming, I design an alternative treatment ("advantages-salience") where respondents are asked to think about the advantages from consulting adoption, without providing any information.

The third information treatment ("disadvantages information") is symmetric to the second: respondents read a text describing the potential drawbacks of relying on consultants rather than on in-house officials. These drawbacks are non-economic, and concern conflicts of interest, favoritism, and weakened accountability. They mirror the concerns respondents themselves raise during AI interviews.

To understand the effects of these randomized information interventions on signing the petition and stated policy preferences, I estimate the following ordinary least squares (OLS) specification:

$$y_i = \beta_0 \text{factual}_i + \beta_1 \text{advantages}_i + \beta_2 \text{disadvantages}_i + \beta_3 \text{salience}_i + \varepsilon_i \tag{8}$$

where y_i denotes one of these outcomes variables for respondent i, and each treatment is

an indicator equal to one if the respondent is assigned to that treatment and zero otherwise. The omitted category is the pooled control group. Standard errors are heteroskedasticity-robust.

Results. Table 2 reports estimates of Equation 8 for five different outcome variables. Columns 1 and 2 report the two real-stakes behavioral measures: whether respondents sign a petition supporting the use of public employees, and whether they sign one supporting the use of consultants. Columns 3–5 report three measures of attitudes toward how the state should be organized, each on a 1–7 scale: support for external consultants (column 3), support for in-house public employees (column 4), and relative support for consultants over public employees (column 5).

The effects largely confirm the descriptive patterns outlined in subsection 5.2. First, because most respondents overestimate how much municipalities rely on consultants, the factual treatment amounts, for most respondents, to a downward correction (see Appendix Table A.11). And because respondents hold largely negative perceptions of consulting, such a correction should lead to improved support towards relying on external resources. Indeed, when provided the correct information on the extent of consulting use by municipalities, citizens increase their support for consulting in both absolute ($\hat{\beta} = 0.197$, $p = 0.002$), and relative terms ($\hat{\beta} = 0.136$, $p = 0.038$). However, in both cases, the economic magnitudes remain modest: 5.7% and 4.4% of the respective control means. I cannot reject the null of no effect for signing any petition, neither on attitudes towards public officials.

Information about the economic benefits of consultants relative to public employees produces larger and broader effects. It raises support for consultants in absolute terms ($\hat{\beta} = 0.380$, $p < 0.001$) and relative to public employees ($\hat{\beta} = 0.377$, $p < 0.001$), corresponding to 11.1% and 12.3% of the respective control means. The treatment also shifts real-stakes behavior: respondents who read about the economic benefits are 24.1% ($p = 0.002$) more likely to sign the petition favoring the use of consultants in the public sector. The treatment also lowers support for public employees ($\hat{\beta} = -0.157$, $p = 0.012$) and reduces the probability of signing the petition favouring public employees by 10.0% ($p = 0.015$).

The effects are not driven by priming alone. First, asking respondents to think about the advantages of consulting without supplying any information leaves four of the five outcomes unchanged, with small point estimates throughout (Appendix Table A.9). Second, I reject the equality of the two treatment effects (see Appendix Table A.10). Information about the economic consequences of consulting therefore shifts stated preferences, real-stakes behavior, and attitudes toward public employees alike, and it does so through its content rather than by directing attention to the advantages. This is consistent with the

descriptive evidence from the AI interviews, in which citizens characterize consulting as a waste of public money—a perception at odds with the estimates in Section 4 which constitute the main takeaway from this information intervention.

Lastly, information about the non-economic drawbacks of consulting relative to public officials does not simply mirror the effects of the economic-benefits treatment. It moves two outcomes in the expected direction: it raises support for the public-employee petition ($\hat{\beta} = 0.065$, $p = 0.002$) and lowers support for consultants ($\hat{\beta} = -0.197$, $p = 0.001$. The effect sizes are more muted: citizens are 12.8% more likely to support the public-employees petition, while those assigned to the advantages treatment are 24.1% more likely to support the consulting one. Similarly, the magnitude is comparatively small for their stated policy preferences towards consultants: only 5.7% of the control group mean. Additionally, the effects on support for public employees, on relative support, and on the probability of signing the pro-consultant petition are not significant, and the magnitudes are much smaller than those of the advantages treatment.

Two interpretations are consistent with these patterns. First, respondents' priors may already align with the content of the disadvantages information: conflicts of interest, favouritism, and accountability are concerns they raise unprompted in the interviews, leaving little room to update. Second, respondents may weight non-economic considerations less heavily than economic ones when forming preferences over how the state is organized, in which case updating on this dimension translates into smaller attitudinal changes. The descriptive evidence is consistent with both: citizens reason primarily in economic terms and misperceive the costs to the public sector, while non-economic concerns are both less prominent and in line with the information provided.

Overall, the experimental results line up with the descriptive evidence. Citizens hold negative views on consulting, and in the interviews they articulate these views primarily in economic terms. Consistent with this, information on the economic benefits of consulting shifts preferences over how the state is organized, while information on the non-economic drawbacks has more muted effects.

5.4. Willingness-to-Pay for Public Employees vs Consultants

This subsection quantifies citizens' demand for public production: how much they value retaining public officials rather than external consultants. Building on the labour economics literature that uses discrete-choice experiments to value non-wage amenities (Folke and Rickne, 2022, Mas and Pallais, 2017, Maestas, Mullen, Powell, von Wachter and Wenger, 2023), I use an experiment embedded in the nationally representative survey described

Table 2: Effects of the Information Treatments

	Support petition for		Support use of		
	Public employees	Consultants	Consultants	Public employees	Consultants vs. employees
	(1)	(2)	(3)	(4)	(5)
Factual information	0.010	0.007	0.197***	-0.083	0.136**
	(0.021)	(0.018)	(0.062)	(0.066)	(0.065)
Advantages information	-0.051**	0.057***	0.380***	-0.157**	0.377***
	(0.021)	(0.019)	(0.060)	(0.063)	(0.065)
Disadvantages information	0.065***	-0.012	-0.197***	0.029	-0.071
	(0.021)	(0.017)	(0.061)	(0.065)	(0.064)
Control group mean	0.506	0.238	3.436	4.043	3.063
Effect size, percent of control group mean					
Factual information	1.9%	2.8%	5.7%	-2.1%	4.4%
Advantages information	-10.0%	24.1%	11.1%	-3.9%	12.3%
Disadvantages information	12.8%	-5.0%	-5.7%	0.7%	-2.3%
Respondents	4,828	4,828	4,828	4,828	4,828

Notes. The table reports effects of the three information treatments relative to the omitted category, which pools the pure and active control groups. Every regression additionally controls for the advantages-salience assignment. For columns (1) and (2) the outcome is a dummy equal to one if a respondent signed up for the petition supporting public employees or consultants, respectively. In columns (3)–(5) the outcome variables are the policy preferences for the use of consultants, public employees, and consultants relative to public employees, measured on the seven-point scales. The control group mean is the outcome mean among respondents in the pooled control groups. Effect sizes divide each treatment coefficient by the corresponding control group mean and express the result in percent. Heteroskedasticity-robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 46.

above to express citizens' policy preferences in monetary terms.

Design. Respondents evaluate two proposals to reduce Italian public spending, side by side, each with a short description and an estimated annual saving. The first mirrors the consulting cap studied in Section 3: reduce the use of consulting, for example by not renewing or by scaling back existing contracts. Because this option leaves production relatively more reliant on in-house public employees, denote its saving by s_B euros per household per year. The second is a hiring freeze: reduce new hiring in the public sector, for example by temporarily leaving vacant part of the posts that open up.[63] Because this option leaves production relatively more reliant on consultants, denote its saving by s_C euros per household per year. Respondents choose the measure they prefer, twice. Each task is indexed by j with $j \in 1, 2$. The two descriptions stay fixed and only the savings

[63]These policies were quite common in Italy, see Rizzica (2020) for a thorough description of laws limiting turnover in the Italian public sector.

change between the two tasks (see Appendix E for more details).[64]

Savings are the experimental variation. Each proposal's saving takes one of five values, 120, 135, 175, 215, or 240 euros per household per year, a grid calibrated to official spending totals.[65] The first task draws the ordered pair (s_B, s_C) uniformly from the 25 combinations; the second redraws it uniformly among the unequal pairs, excluding the one just seen. The savings gap $s_B - s_C$ therefore ranges from -120 to $+120$ euros.

Estimation. To express citizens' demand for public production in monetary terms, I estimate how their choices respond to differences in the household savings offered by the two policies. Savings are expressed in euros per household per year. I denote the marginal utility from private consumption as α, while μ is a reduced-form parameter capturing citizens' policy preference for retaining public employees rather than external consultants, holding household savings constant. In task j, with $j \in 1, 2$ cutting consultants leaves production relatively more in-house and delivers saving s_{Bj}, whereas restricting public hiring leaves production relatively more reliant on consultants and delivers saving s_{Cj}. The two option-specific utilities are therefore:

$$\begin{aligned} &\text{Choose to keep bureaucrats and cut consultants:} \quad U_{iB,j} = \alpha s_{Bj} + \mu + \varepsilon_{iB,j}, \\ &\text{Choose to keep consultants and cut bureaucrats:} \quad U_{iC,j} = \alpha s_{Cj} + \varepsilon_{iC,j}. \end{aligned}$$

The respondent chooses to cut consultants when:

$$U_{iB,j} - U_{iC,j} = \mu + \alpha \Delta s_j + (\varepsilon_{iB,j} - \varepsilon_{iC,j}) > 0, \qquad \Delta s_j \equiv s_{Bj} - s_{Cj}. \tag{9}$$

Assuming the errors are i.i.d. type-I extreme value yields the binary logit:

$$\Pr\left(\text{cut consultants} \mid \Delta s\right) = \frac{\exp(\mu + \alpha \Delta s)}{1 + \exp(\mu + \alpha \Delta s)}. \tag{10}$$

I estimate Equation (10) at the respondent-task level, with standard errors clustered by respondent. The logit constant estimates μ, while the slope estimates the marginal utility of

[64] In the discrete-choice experiment, I randomized whether respondents were told that the two policies would have the same effects on public-good provision and service delivery. 37% of respondents considered this equivalence implausible; among those reading the equivalence description, that share was 42%. I thus restrict the following analysis to respondents who were not given that description ("baseline sample").

[65] Reapplying the 2010 rule today—capping consulting spending by the Italian public administration at 20% of its current level—would save about 5.6 billion euros a year, or 215 euros per household. Caps at 35% and 50% save 4.5 and 3.5 billion (175 and 135 euros). The endpoints, 120 and 240 euros (3.1 and 6.2 billion), widen the support of the savings gap.

household savings, α. At equal savings, the predicted share cutting consultants is $\frac{\exp(\hat{\mu})}{1+\exp(\hat{\mu})}$, so $\widehat{\mu} > 0$ means that the population leans toward keeping public employees when the two policies have the same economic benefits. The money-equivalent for μ is the following willingness-to-pay:

$$\text{WTP}^{\text{bureaucrats}} = \frac{\mu}{\alpha}, \tag{11}$$

with inference by the delta method. It is the additional annual household saving that restricting public employees hiring must offer, relative to cutting consultants, for the two policies to be equally attractive. I also estimate the linear probability model $\Pr(\text{cut consultants} \mid \Delta s) = \beta + \gamma \Delta s$. Its intercept β estimates the share cutting consultants at equal savings, and the indifference point solves $\beta + \gamma \Delta s = 1/2$, so the OLS analogue of Equation (11) is $(\beta - 1/2)/\gamma$.

Results. Table 3 reports willingness-to-pay estimates from the discrete choice experiment. Column (1) presents my preferred specification, a binary logit estimated on the pooled control group—respondents not exposed to any information treatment. Column (2) adds demographic controls, and column (3) reports the corresponding linear probability model. Column (4) estimates the logit on the baseline sample, pooling all information arms.

Choices respond strongly to the saving gaps. In the pooled control group, a 100 euro increase in the savings gap raises the probability of choosing to cut consulting contracts by 13.5 percentage points in the logit and by 13.3 percentage points in the linear probability model. At equal savings, 68.5% of respondents choose to cut consultants rather than freeze public hiring. I reject the null hypothesis that this fraction equals one half ($p < 0.001$). The share is stable across specifications and statistically above one half throughout.

The implied willingness to pay to keep public employees over consultants is also stable across specifications. In the control-group logit, it amounts to 124.0 euros per household per year (s.e. 19.7). Residualizing by demographic characteristics yields 134.1 euros (s.e. 21.5), while the linear probability model yields 133.8 euros (s.e. 22.9). The baseline-sample logit produces a smaller estimate of 99.5 euros (s.e. 8.0).

Citizens thus have a sizeable demand for public in-house provision, and a clear majority would choose to cut consulting contracts rather than freeze public sector hiring. This demand reflects both citizens' intrinsic preferences and their beliefs about the consequences of each mode of production.

To assess the role of beliefs, Figure 11 compares willingness to pay across information arms. Information about the economic benefits of relying on external rather than internal resources substantially reduces demand for public production. Relative to pooled controls,

Table 3: Willingness to Pay to Keep In-House Provision Relative to External

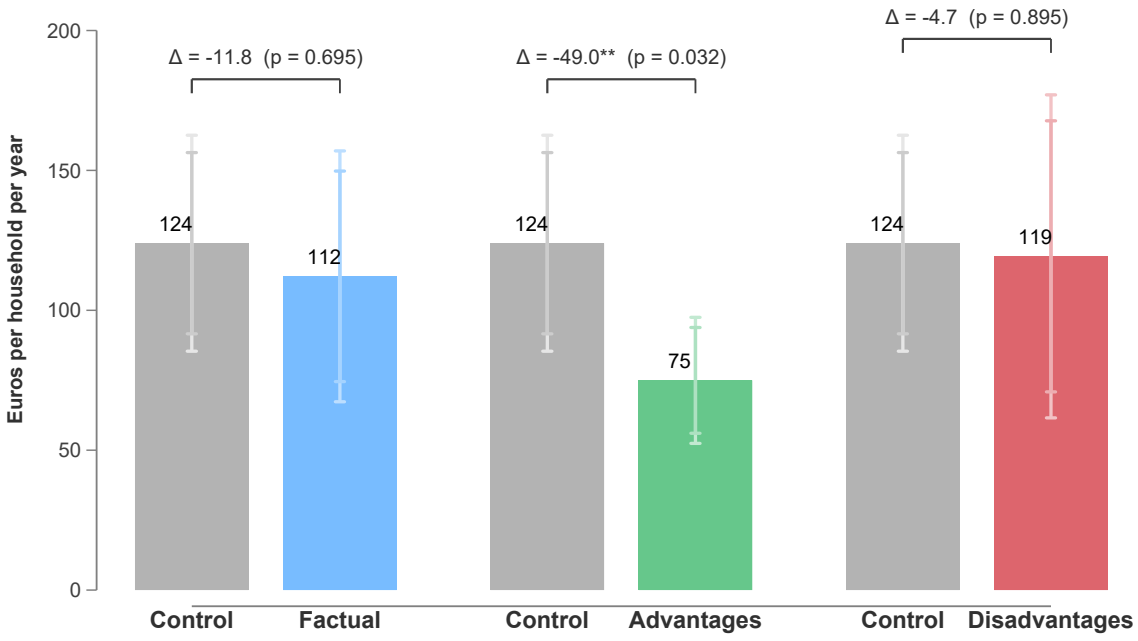
	Control Groups			Baseline sample
	Logit (1)	Logit (2)	OLS (3)	Logit (4)
Panel A. Choice model				
In-house premium (μ; β)	0.778***	0.879***	0.678***	0.630***
	(0.067)	(0.070)	(0.014)	(0.032)
Savings gap, 100 euros (α; γ)	0.628***	0.656***	0.133***	0.633***
	(0.092)	(0.099)	(0.019)	(0.043)
Panel B. Estimated quantities				
Share cutting consultants at equal savings	0.685***	0.686***	0.678***	0.652***
p-value, test share $= 0.5$	[0.000]	[0.000]	[0.000]	[0.000]
WTP to keep bureaucrats, euros per year	124.0***	134.1***	133.8***	99.5***
	(19.7)	(21.5)	(22.9)	(8.0)
Demographic controls	–	✓	–	–
Observations	1,446	1,444	1,446	6,236
Respondents	723	722	723	3,118

Notes. The table reports estimates from the discrete choice experiment. The dependent variable equals one if the respondent chooses to cut consultants rather than restrict public hiring, which corresponds to $U_{i\mathsf{B},j} > U_{i\mathsf{C},j}$. All columns restrict the sample to respondents who were not told that the two policies would have the same effects on public-good and service delivery. Columns (1)–(3) additionally restrict the sample to respondents in the pooled pure and active control groups; column (4) includes all information arms. Columns (1), (2), and (4) report logits, while column (3) reports a linear probability model. Column (2) controls for birth year, gender, education, work situation, public sector ties, income, and macroregion fixeed effects. The savings-gap coefficient is reported for a 100-euro change. In the uncontrolled logits, willingness to pay is $\widehat{\mu}/\widehat{\alpha}$; in the controlled logit it is averaged over demographics; and in OLS it is $(\widehat{\beta} - 1/2)/\widehat{\gamma}$. Standard errors are clustered by respondent, and nonlinear quantities use delta-method inference. Square brackets report the p-value of the two-sided test that the equal-savings share equals one half; stars on that row refer to this test, while all other stars test the null of zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 50.

the advantages treatment reduces willingness to pay by 49.0 euros ($p = 0.032$). The corresponding differences for the factual and disadvantages treatments are not statistically and economically significant.

Providing information about the potential economic benefits of consulting shifts not only stated policy preferences and real-stakes behaviour, but also citizens' willingness to pay higher taxes to retain public officials. The effect is sizeable: willingness to pay falls from 124 to 75 euros, a decline of 40%. Demand for in-house public production therefore does not reflect intrinsic preferences alone: a sizeable share reflects beliefs about the adverse economic consequences of consulting. This interpretation is consistent with the AI-led interviews, in which citizens identify lower costs relative to external consultants as one of the main advantages of public employment.

Figure 11: Willingness to Pay to Keep Public Employees, by Information Treatment



Notes. The figure reports willingness to pay, in euros per household per year, to keep public employees for the pooled control (grey bars), factual-information (light blue bars), advantages-information (green bars), and disadvantages-information (red bars) treatment arms. The sample is restricted to respondents who were not told that the two policies would have the same effects on public-good and service delivery. Estimates come from a logit model that interacts the treatment arm with the savings difference. The pooled control combines the pure and active control groups. Numbers above the bars report willingness to pay. Capped lines report the 90 and 95% confidence bands. Brackets report the treatment-minus-control difference, Δ, and the corresponding two-sided p-value. Standard errors are clustered by respondent; willingness-to-pay estimates and differences use delta-method inference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 50.

Even after adjusting perceptions, willingness to pay remains substantial: respondents are willing to forgo 75 euros per household to retain public employees rather than consultants. The information intervention therefore narrows, but neither closes nor reverses, the gap.

Magnitudes: back-of-the-envelope calculation. To put citizens' willingness to pay in perspective, I calculate how much consulting expenditure would need to increase to generate performance gains of equivalent monetary value. I combine the IV estimates of consulting's effects on service output and revenues with hedonic estimates that value service improvements using house prices or rents (Appendix Table A.7). I express these service gains in annual euros per household, annualising the house-price valuation, and

add the household savings from lower revenue requirements. Appendix Table A.8 details these calculations. Applying these estimates linearly, I then calculate the increase in consulting expenditure that would generate combined annual gains equal to citizens' willingness to pay to retain public employees. This amounts to approximately 24% of consulting expenditure per capita at 2025 levels.

Citizens value public sector jobs and their impact on the economy. Substantial demand for public employment remains. In the AI interviews, citizens often cite protecting and creating jobs as an advantage of public employment. I therefore test whether citizens value public employment for the local economic benefits they expect it to generate. Figure 12 examines this possibility among respondents informed of consulting's potential economic advantages. Within each employment group, it compares respondents residing in municipalities where unemployment and public sector employment are both above their respective municipal medians with all other areas.

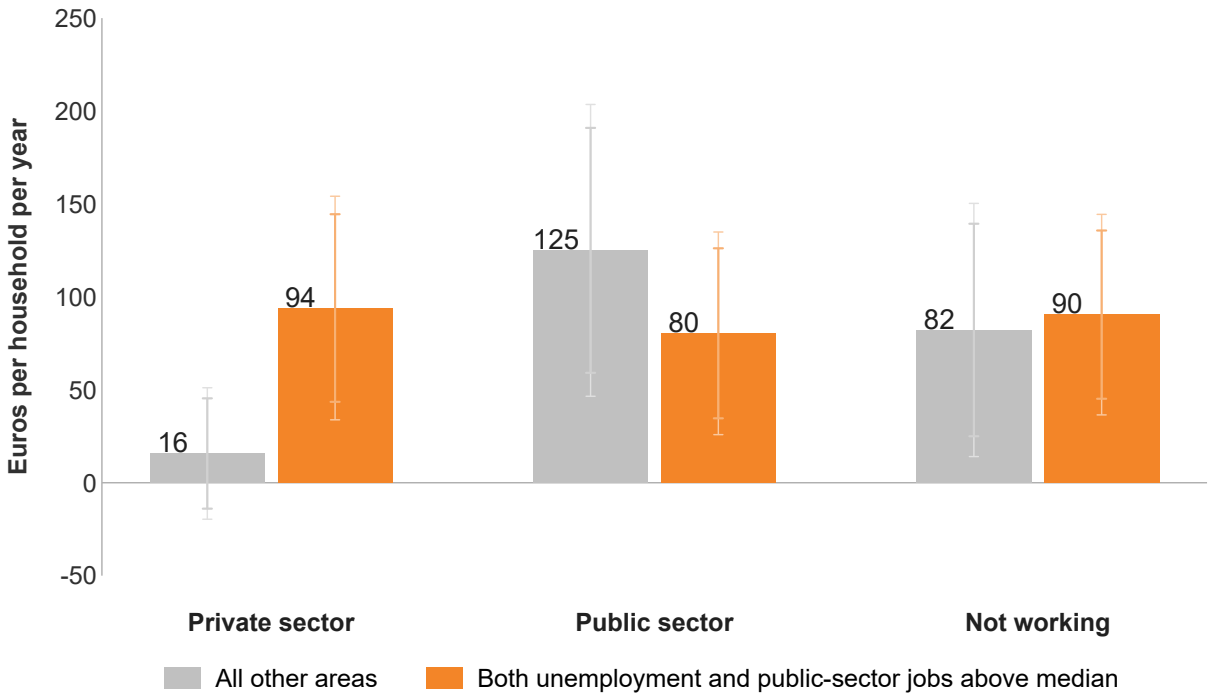
Willingness to pay varies little across local economic conditions among public sector workers and respondents who are not working. Among private-sector workers, however, demand for public production is substantially higher in areas where the local economy relies on public employment. In these municipalities, residual willingness to pay to retain public employees is €94 per household per year, compared with €16 elsewhere. Appendix Figure A.24 shows a related pattern: support for consultants relative to public employees declines with local unemployment among private-sector workers, but varies little among the other groups.

These patterns are consistent with private-sector workers valuing public employment for the demand it sustains for local goods and services, particularly where employment opportunities are limited. Public employment may therefore be valued both for its role in delivering public services and for supporting the local economy.

6. Policymakers react to citizens' demand

Elected officials are accountable to citizens and may therefore respond to their views and preferences over how the state organises production. Even when consulting raises output and lowers fiscal costs, elected officials may have little incentive to use it if citizens favour in-house public production. This section directly tests whether policymakers respond to citizens' demand for public production. I design and field a separate online survey of local policymakers, embedding a randomised information intervention.

Figure 12: Residual Willingness to Pay for Public Officials by Local Economy



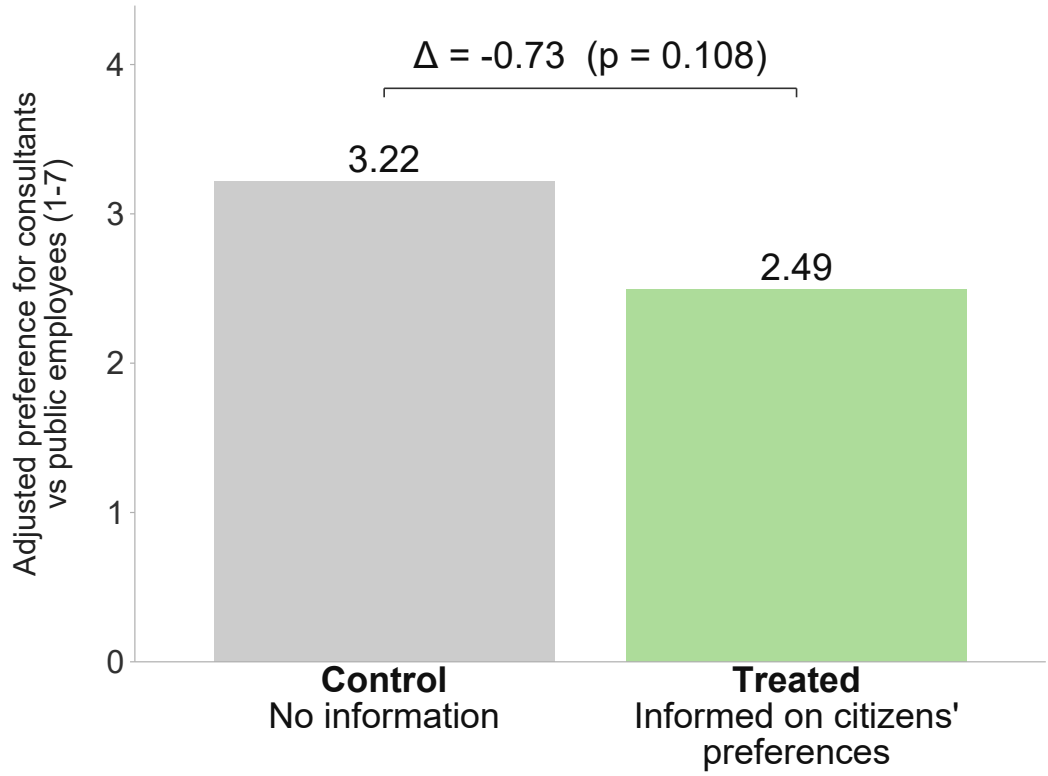
Notes. The figure reports residual willingness to pay to retain public employees over consultants estimated through the logit model in Equation 10, in euros per household per year, among respondents assigned to the advantages-information arm. Residual willingness to pay is the premium remaining after information on the economic advantages of consulting. Within each employment group, orange bars identify municipalities where both the 2024 unemployment rate and the 2020 public sector employment measure are above their respective municipal medians; grey bars identify all other combinations. Private-sector workers include the self-employed; the not-working group includes unemployed, student, and retired respondents. Standard errors are clustered by respondent, with delta-method inference for willingness to pay. Darker and lighter capped lines indicate 90% and 95% confidence intervals, respectively. Referenced on page: 54.

Survey design. The survey sample comprises 151 respondents.[66] The sample includes 45 mayors (29.8%). The remaining respondents include directors-general, municipal secretaries, other municipal staff, and members of the municipal executive, appointed by the mayor, and the municipal council, directly elected by citizens. Many public officials' positions are tied to the mayor's tenure. Directors-general, for example, are appointed by the mayor and cannot serve beyond the mayor's term.

Before the intervention, respondents report their views about consulting and estimate

[66]I recruited respondents by email using a contact list of municipal secretaries and directors-general provided by the Institute for Finance and Local Economy (IFEL), supplemented by a systematic online search for municipal email addresses. I restricted the mailing list to elected officials, senior municipal administrators, and their offices or staff.

Figure 13: Informing Politicians and Bureaucrats of Citizens' Policy Views & Preferences



Notes. Bars report average predicted outcomes from an OLS regression of preference for consultants versus public employees on treatment assignment and categorical indicators for municipal role, private-sector experience, and consulting-contract involvement. Predictions are averaged over the estimation sample with treatment set to zero or one. The sample includes 86 elected politicians and senior municipal officials. Treated respondents are informed that, at equal savings for the state, 69% of Italian citizens prefer reducing expenditure on consultants and external firms to reducing public sector hiring. The control group receives no information. The outcome ranges from 1 (much greater reliance on public employees) to 7 (much greater reliance on consultants and external firms), with 4 denoting the current mix. The bracket reports the treatment coefficient, Δ, and its two-sided p-value using heteroskedasticity-robust standard errors. Referenced on page: 56.

the share of Italian citizens who, at equal savings for the state, would prefer cutting consulting expenditure to reducing public sector hiring. Their mean estimate is 50.3%. Respondents are then randomly assigned to receive information about citizens' preferences or to a control group receiving no information. The treatment reports the results from the citizens' survey, i.e., that 69% of citizens prefer consulting cuts at equal savings. It thus makes public opposition to consulting explicit. Afterward, both groups indicate whether the state should rely more on public employees or on consultants and external firms. Responses range from 1 (much more on public employees) to 7 (much more on

external providers), with 4 denoting the current mix.

Results. Figure 13 reports regression-adjusted mean responses by treatment arm for 86 local politicians and senior officials whose career is tied to the one of the mayor.[67] Officials informed about citizens' preferences report lower support for consulting: the adjusted mean is 2.49 in the treatment group and 3.22 in the control group. The adjusted treatment effect is -0.73 points, with a heteroskedasticity-robust standard error of 0.45 ($p = 0.108$; 95% confidence interval $[-1.62, 0.16]$). The direction is consistent with responsiveness to citizens' demand for public production.

Despite being slightly imprecise, these findings suggest local policymakers are responsive to citizens' demand for public production and that they are largely misinformed about citizens' policy preferences. The experiment measures stated preferences, rather than subsequent hiring or contracting decisions.

7. Conclusion

This article introduces the choice between consulting services and public employment to the literature on the organisation of the state. Using administrative and original survey data in Italy, this paper documents that, when organising public sector production, departures from productive efficiency can be rationalised by understanding citizens' policy preferences over how the state itself is organised. While consulting improves public sector productivity, citizens have a strong demand for public in-house production. Productivity gains come from accessing specialised labour in a flexible way. Demand for public production can be attributed—to a large extent—to beliefs about the negative economic consequences of consulting. Citizens' willingness-to-pay for public production is large, but providing citizens with information on the economic advantages of consulting reduces it by 40%. The remaining demand for public production reflects the value citizens attach to stable public employment and its role in sustaining weak local economies. Citizens' demand for public production matters, as local policymakers react to citizens' views and preferences over the organisation of the state. Together, these findings rationalise why many countries around the world cut consulting services in the public sector, despite their positive productivity effects.

Important questions remain for future research. First, how does acquiring expertise externally on the market affect long-run state capacity relative to retaining low-skilled public officials? This paper shows that decisions to outsource labour input have an impact

[67] I restrict the sample to respondents with survey duration below 1.31 hours and observed controls.

on the internal organisational structure of bureaucracies, as accessing consulting reduces employment among low-skilled public workers. However, this paper does not analyse the long-run consequences of one mode of production versus the other. Second, how do these gains depend on the institutional environment? In contexts with weak state capacity, monitoring the use of flexible work arrangements is harder and may create opportunities to appropriate public resources.

My findings also contribute to a broader literature on the importance of non-monetary objectives for organisations (e.g., Besley and Ghatak, 2005, Fioretti, 2022, Colonnelli, Prem and Teso, 2020, Conway and Boxell, 2026, Ashraf, Bandiera, Minni and Zingales, 2025). States are ultimately accountable to citizens, who value not only the cost-effectiveness of the services they receive but also how those services are produced. Citizens' welfare therefore makes the state maximise an objective that goes beyond productive efficiency.

References

Akhtari, Mitra, Diana Moreira, and Laura Trucco. 2022. "Political Turnover, Bureaucratic Turnover, and the Quality of Public Services." *American Economic Review*, 112(2): 442–493. ◁ p. 5

Alesina, Alberto, Armando Miano, and Stefanie Stantcheva. 2023. "Immigration and Redistribution." *The Review of Economic Studies*, 90(1): 1–39. ◁ p. 34

Alesina, Alberto, Stefanie Stantcheva, and Edoardo Teso. 2018. "Intergenerational Mobility and Preferences for Redistribution." *American Economic Review*, 108(2): 521–54. ◁ p. 34

Anderson, Stephen J., and David McKenzie. 2022. "Improving Business Practices and the Boundary of the Entrepreneur: A Randomized Experiment Comparing Training, Consulting, Insourcing, and Outsourcing." *Journal of Political Economy*, 130(1): 157–209. ◁ p. 6

Aneja, Abhay, and Guo Xu. 2024. "Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age." *American Economic Review*, 114(8): 2352–2387. ◁ p. 5

Ashraf, Nava, Oriana Bandiera, and B. Kelsey Jack. 2014. "No Margin, No Mission? A Field Experiment on Incentives for Public Service Delivery." *Journal of Public Economics*, 120: 1–17. ◁ p. 6

Ashraf, Nava, Oriana Bandiera, Virginia Minni, and Luigi Zingales. 2025. "Meaning at Work." National Bureau of Economic Research Working Paper 33843. ◁ pp. 5 and 57

Bandiera, Oriana, Andrea Prat, and Tommaso Valletti. 2009. "Active and Passive Waste in Government Spending: Evidence from a Policy Experiment." *American Economic Review*, 99(4): 1278–1308. ◁ p. 5

Bandiera, Oriana, Michael Carlos Best, Adnan Qadir Khan, and Andrea Prat. 2021. "The Allocation of Authority in Organizations: A Field Experiment with Bureaucrats." *The Quarterly Journal of Economics*, 136(4): 2195–2242. ◁ p. 6

Banerjee, Abhijit, Rema Hanna, Jordan Kyle, Benjamin A. Olken, and Sudarno Sumarto. 2019. "Private Outsourcing and Competition: Subsidized Food Distribution in Indonesia." *Journal of Political Economy*, 127(1): 101–137. ◁ p. 6

Bellifemine, Marco, Adrien Couturier, and Jacopo Tozzo. 2025. "The Sentiment Channel of Fiscal Policy." *Mimeo*. ◁ p. 16

Bertrand, Marianne, Robin Burgess, Arunish Chawla, and Guo Xu. 2020. "The Glittering Prizes: Career Incentives and Bureaucrat Performance." *The Review of Economic Studies*, 87(2): 626–655. ◁ p. 6

Besley, Timothy, and Maitreesh Ghatak. 2001. "Government Versus Private Ownership of Public Goods." *The Quarterly Journal of Economics*, 116(4): 1343–1372. ◁ pp. 6 and 10

Besley, Timothy, and Maitreesh Ghatak. 2005. "Competition and Incentives with Motivated Agents." *American Economic Review*, 95(3): 616–636. ◁ pp. 5, 6, and 57

Besley, Timothy, Robin Burgess, Adnan Khan, and Guo Xu. 2022. "Bureaucracy and Development." *Annual Review of Economics*, 14: 397–424. ◁ pp. 1 and 9

Best, Michael Carlos, Jonas Hjort, and David Szakonyi. 2023. "Individuals and Organizations as Sources of State Effectiveness." *American Economic Review*, 113(8): 2121–2167. ◁ p. 6

Bijnens, Gert, Simon Jäger, and Benjamin Schoefer. 2025. "What Does Consulting Do?" National Bureau of Economic Research Working Paper 34072. ◁ pp. 6, 7, and 11

Bloom, Nicholas, Aprajit Mahajan, David McKenzie, and John Roberts. 2020. "Do Management Interventions Last? Evidence from India." *American Economic Journal: Applied Economics*, 12(2): 198–219. ◁ p. 6

Bloom, Nicholas, Benn Eifert, Aprajit Mahajan, David McKenzie, and John Roberts. 2012. "Does Management Matter? Evidence from India." *The Quarterly Journal of Economics*, 128(1): 1–51. ◁ p. 6

Bloom, Nicholas, Carol Propper, Stephan Seiler, and John Van Reenen. 2015. "The Impact of Competition on Management Quality: Evidence from Public Hospitals." *The Review of Economic*

Studies, 82(2): 457–489. ◁ p. 9

Bloom, Nicholas, Renata Lemos, Raffaella Sadun, Daniela Scur, and John Van Reenen. 2021. "The World Management Survey at 18: Lessons and the Way Forward." *Oxford Review of Economic Policy*, 37(2): 231–258. ◁ p. 6

Bogdanich, Walt, and Michael Forsythe. 2022. *When McKinsey Comes to Town: The Hidden Influence of the World's Most Powerful Consulting Firm.* New York:Doubleday. ◁ p. 13

Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. "Revisiting Event-Study Designs: Robust and Efficient Estimation." *The Review of Economic Studies*, 91(6): 3253–3285. ◁ p. A.29

Bruhn, Miriam, Dean Karlan, and Antoinette Schoar. 2018. "The Impact of Consulting Services on Small and Medium Enterprises: Evidence from a Randomized Trial in Mexico." *Journal of Political Economy*, 126(2): 635–687. ◁ p. 6

Bursztyn, Leonardo, Ingar K. Haaland, Nicolas Röver, and Christopher Roth. forthcoming. "The Social Desirability Atlas." *Journal of Political Economy: Microeconomics*. NBER Working Paper No. 33920. ◁ p. 39

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225(2): 200–230. ◁ p. A.29

Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. 2026. "Difference-in-Differences with a Continuous Treatment." *American Economic Review*. Forthcoming. ◁ pp. 18, 33, and A.45

Chinoy, Sahil, Nathan Nunn, Sandra Sequeira, and Stefanie Stantcheva. 2026. "Zero-Sum Thinking and the Roots of US Political Differences." *American Economic Review*, 116(3): 1052–1096.

Chopra, Felix, and Ingar Haaland. 2024. "Conducting Qualitative Interviews with AI." CESifo Working Paper No. 10666. ◁ p. 36

Colonnelli, Emanuele, Mounu Prem, and Edoardo Teso. 2020. "Patronage and Selection in Public Sector Organizations." *American Economic Review*, 110(10): 3071–3099. ◁ pp. 5, 20, and 57

Colonnelli, Emanuele, Tim McQuade, Gabriel Ramos, Thomas Rauter, and Olivia Xiong. 2026. "Polarizing Corporations: Does Talent Flow to "Good" Firms?" Working paper. ◁ p. 5

Conway, Jacob, and Levi Boxell. 2026. "Consuming Values." Working paper. Revise and resubmit at Econometrica. ◁ pp. 5 and 57

Daniele, Gianmarco, and Tommaso Giommoni. 2025. "Fiscal Rules, Corruption and Electoral Accountability." *The Journal of Politics*. Forthcoming. ◁ pp. 9 and A.25

Decarolis, Francesco, Raymond Fisman, Paolo Pinotti, and Silvia Vannutelli. 2025. "Rules, Discretion, and Corruption in Procurement: Evidence from Italian Government Contracting." *Journal of Political Economy Microeconomics*, 3(2): 213–254. ◁ p. 12

Deserranno, Erika. 2019. "Financial Incentives as Signals: Experimental Evidence from the Recruitment of Village Promoters in Uganda." *American Economic Journal: Applied Economics*, 11(1): 277–317. ◁ p. 6

Deserranno, Erika, Philipp Kastrau, and Gianmarco León-Ciliotta. 2025. "Promotions and Productivity: The Role of Meritocracy and Pay Progression in the Public Sector." *American Economic Review: Insights*, 7(1): 71–89. ◁ p. 6

Deserranno, Erika, Stefano Caria, Philipp Kastrau, and Gianmarco León-Ciliotta. 2025. "The Allocation of Incentives in Multilayered Organizations: Evidence from a Community Health Program in Sierra Leone." *Journal of Political Economy*, 133(8). ◁ p. 6

Duflo, Esther, Rema Hanna, and Stephen P. Ryan. 2012. "Incentives Work: Getting Teachers to Come to School." *American Economic Review*, 102(4): 1241–78. ◁ p. 9

Duflo, Esther, Rema Hanna, Benjamin A. Olken, and Shreya Tandon. 2025. "The Limits of Government Outsourcing: Property Tax Re-Assessments in India." Working Paper. ◁ p. 6

Fenizia, Alessandra. 2022. "Managers and Productivity in the Public Sector." *Econometrica*,

90(3): 1063–1084. ◁ p. 6
Fioretti, Michele. 2022. "Caring or Pretending to Care? Social Impact, Firms' Objectives, and Welfare." *Journal of Political Economy*, 130(11): 2898–2942. ◁ pp. 5 and 57
Folke, Olle, and Johanna Rickne. 2022. "Sexual Harassment and Gender Inequality in the Labor Market." *Quarterly Journal of Economics*, 137(4): 2163–2212. ◁ p. 47
Galiani, Sebastian, Paul Gertler, and Ernesto Schargrodsky. 2005. "Water for Life: The Impact of the Privatization of Water Services on Child Mortality." *Journal of Political Economy*, 113(1): 83–120. ◁ p. 6
Geiecke, Friedrich, and Xavier Jaravel. 2026. "Conversations at Scale: Robust AI-led Interviews." *Review of Economic Studies*. ◁ pp. 4, 36, 37, and A.52
Haaland, Ingar, Christopher Roth, and Johannes Wohlfart. 2023. "Designing Information Provision Experiments." *Journal of Economic Literature*, 61(1): 3–40. ◁ p. 39
Haaland, Ingar K., Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart. 2025. "Understanding Economic Behavior Using Open-Ended Survey Data." *Journal of Economic Literature*, 63(4): 1244–1280. ◁ pp. 36 and 37
Hainmueller, Jens, Dominik Hangartner, and Teppei Yamamoto. 2015. "Validating vignette and conjoint survey experiments against real-world behavior." *Proceedings of the National Academy of Sciences*, 112(8): 2395–2400. ◁ p. 39
Hart, Oliver, Andrei Shleifer, and Robert W. Vishny. 1997. "The Proper Scope of Government: Theory and an Application to Prisons." *The Quarterly Journal of Economics*, 112(4): 1127–1161. ◁ pp. 6, 10, and 11
Khan, Adnan Q., Asim I. Khwaja, and Benjamin A. Olken. 2015. "Tax Farming Redux: Experimental Evidence on Performance Pay for Tax Collectors." *The Quarterly Journal of Economics*, 131(1): 219–271. ◁ p. 6
Limodio, Nicola. 2021. "Bureaucrat Allocation in the Public Sector: Evidence from the World Bank." *The Economic Journal*, 131(639): 3012–3040. ◁ p. 9
Maestas, Nicole, Kathleen J. Mullen, David Powell, Till von Wachter, and Jeffrey B. Wenger. 2023. "The Value of Working Conditions in the United States and the Implications for the Structure of Wages." *American Economic Review*, 113(7): 2007–2047. ◁ p. 47
Mas, Alexandre, and Amanda Pallais. 2017. "Valuing Alternative Work Arrangements." *American Economic Review*, 107(12): 3722–3759. ◁ p. 47
Mastrorocco, Nicola, and Edoardo Teso. 2026. "State Capacity as an Organizational Problem." *Review of Economic Studies*. Published online June 17, 2026. ◁ p. 5
Mazzucato, Mariana, and Rosie Collington. 2023. *The Big Con: How the Consulting Industry Weakens Our Businesses, Infantilizes Our Governments, and Warps Our Economies.* New York:Penguin Press. ◁ p. 13
Muñoz, Pablo, and Cristóbal Otero. 2025. "Managers and Public Hospital Performance." *American Economic Review*, 115(11): 4040–4074. ◁ p. 6
Olken, Benjamin A. 2007. "Monitoring Corruption: Evidence from a Field Experiment in Indonesia." *Journal of Political Economy*, 115(2): 200–249. ◁ p. 5
Oster, Emily. 2019. "Unobservable Selection and Coefficient Stability: Theory and Evidence." *Journal of Business & Economic Statistics*, 37(2): 187–204. ◁ p. A.27
Rasul, Imran, and Daniel Rogger. 2018. "Management of Bureaucrats and Public Service Delivery: Evidence from the Nigerian Civil Service." *The Economic Journal*, 128(608): 413–446. ◁ p. 9
Rizzica, Lucia. 2020. "The Italian Public Sector Workforce: Recent Evolution in the Light of the Rules on Turnover." Bank of Italy Questioni di Economia e Finanza (Occasional Papers) 560. ◁ p. 48
Romero, Mauricio, Justin Sandefur, and Wayne Aaron Sandholtz. 2020. "Outsourcing Education:

Experimental Evidence from Liberia." *American Economic Review*, 110(2): 364–400. ◁ p. 6
Shleifer, Andrei. 1998. "State versus Private Ownership." *Journal of Economic Perspectives*, 12(4): 133–150. ◁ pp. 6, 10, and 11
Shleifer, Andrei, and Robert W. Vishny. 1993. "Corruption." *The Quarterly Journal of Economics*, 108(3): 599–617. ◁ p. 5
Small, Mario Luis, and Jessica McCrory Calarco. 2022. *Qualitative Literacy: A Guide to Evaluating Ethnographic and Interview Research.* University of California Press. ◁ p. 37
Stantcheva, Stefanie. 2023. "How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible." *Annual Review of Economics*, 15: 205–234. ◁ p. 35
Sun, Liyang, and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics*, 225(2): 175–199. ◁ p. A.29
Vannutelli, Silvia. 2022. "From Lapdogs to Watchdogs: Random Auditor Assignment and Municipal Fiscal Performance." National Bureau of Economic Research Working Paper 30644. ◁ p. 26
Xu, Guo. 2018. "The Costs of Patronage: Evidence from the British Empire." *American Economic Review*, 108(11): 3170–3198. ◁ pp. 5 and 20
Xu, Guo, Erika Deserranno, Diana Moreira, and Edoardo Teso. 2023. "Bureaucracy." *VoxDevLit*, 8(1). ◁ p. 9

Make or Buy the State?
Consulting in the Public Sector
Online Appendix

Marta Morando

Contents

A. Additional Background Information **A.2**

A.1. Legal Frameworks Regulating Consulting Adoption A.2

A.2. Laws Shaping Consulting Expenditures A.2

B. Additional Data Information **A.4**

B.1. Measuring Personnel Positions A.4

B.2. Measuring Public Services Output A.5

B.3. Measuring Consulting A.7

C. Additional Empirical Evidence **A.11**

C.1. From Global Data A.11

C.2. From Administrative Data A.13

C.3. From Court of Auditors Rulings A.28

C.4. From Survey Data A.31

D. Additional Validation and Robustness Checks **A.39**

E. Details on Surveys **A.52**

E.1. Citizen Survey with Embedded AI-Led Interview A.52

E.2. Politicians and High-Level Bureaucrats Survey A.70

F. Derivations Conceptual Framework **A.78**

A. Additional Background Information

A.1. Legal Frameworks Regulating Consulting Adoption

Municipalities acquire external expertise either through individual professional appointments or through procurement contracts for services. Individual professional appointments are ordinarily awarded through a public comparative procedure. Contracts above €5,000 must be transmitted to the relevant chamber of the Court of Auditors for ex post review. Consulting contracts governed by procurement law are instead assigned through direct awards, or competitive tenders (negotiated or open tenders), depending on the contract value and the rules in force at the time. In both cases, the municipality must identify the specific public need served and record why the expertise is bought rather than produced in house. For individual appointments, public bodies must also prove that the required expertise is unavailable internally and that the assignment is temporary, highly qualified, and directed toward a specific predefined objective. These requirements are meant to keep consulting from duplicating internal capacity or substituting for restricted public hiring, while transparent and competitive awarding and ex post review by the Court of Auditors are intended to limit favouritism in how contracts are assigned.

A.2. Laws Shaping Consulting Expenditures

I report the translated paragraphs of the two laws that limited and then removed the limits to consulting use. I highlight in bold the text relevant to the identification strategy.

Article 6, D.L. 78/2010, N.7 To enhance the professional skills within public administrations, starting from 2011 **the annual expenditure on studies and consultancy assignments** –including those awarded to public employees– incurred by the public administrations referred to in Article 1, paragraph 3, of Law No. 196 of 31 December 2009, including independent authorities, but excluding universities, research bodies and foundations and equivalent organisations, as well as study and consultancy assignments connected with privatisation processes and the regulation of the financial sector, **may not exceed** 20 per cent of **the amount incurred in 2009**. Awarding assignments without meeting the conditions set out in this paragraph constitutes a **disciplinary offence and gives rise to liability for loss to the public finances**. The provisions of this paragraph do not apply to health/medical activities connected with the recruitment, advancement, and employment of personnel of the Armed Forces, the Police Forces, and the National Fire Brigade.

Article 57, D.L. 124/2019, N.2 Starting from the year 2020, the following provisions regarding expenditure containment and reduction, as well as training obligations, **shall no longer apply to** the regions, the Autonomous Provinces of Trento and Bolzano, **local authorities**, and their instrumental bodies and entities, as defined by Article 1, paragraph 2, of Legislative Decree No. 118 of June 23, 2011, as well as to their instrumental entities in corporate form: a) Article 27, paragraph 1, of Decree-Law No. 112 of June 25, 2008, converted, with amendments, by Law No. 133 of August 6, 2008; b) **Article 6, paragraphs 7**, 8, 9, 12 and 13, **of Decree-Law No. 78 of May 31, 2010, converted, with amendments, by Law No. 122 of July 30, 2010**; c) Article 5, paragraph 2, of Decree-Law No. 95 of July 6, 2012, converted, with amendments, by Law No. 135 of August 7, 2012; d) Article 5, paragraphs 4 and 5, of Law No. 67 of February 25, 1987;e) Article 2, paragraph 594, of Law No. 244 of December 24, 2007;f) Article 12, paragraph 1-ter, of Decree-Law No. 98 of July 6, 2011, converted, with amendments, by Law No. 111 of July 15, 2011;g) Article 24 of Decree-Law No. 66 of April 24, 2014, converted, with amendments, by Law No. 89 of June 23, 2014.

B. Additional Data Information

B.1. Measuring Personnel Positions

I group municipal employees using the broad occupational categories reported in Conto Annuale. The national category labels changed in 2023, so Table A.1 shows how both the earlier and current labels map into the four underlying occupational groups.

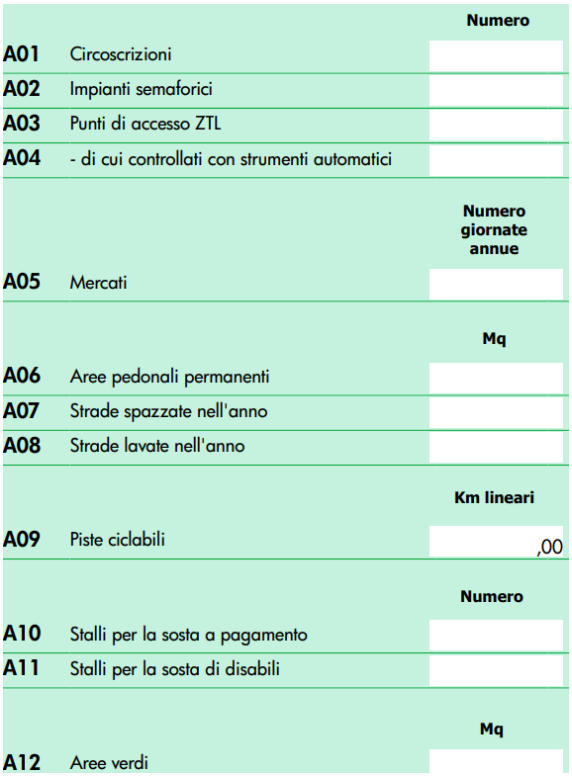
Table A.1: Occupational Classification of Municipal Employees

Paper category	Conto Annuale category (Italian)	English translation
Blue-collar workers	Through 2022: *Categoria A*; *Categoria B* From 2023: *Operatori*; *Operatori esperti*	Through 2022: Categories A and B From 2023: Operators; skilled operators
Clerical workers	Through 2022: *Categoria C* From 2023: *Istruttori*	Through 2022: Category C From 2023: Administrative and technical officers
Professionals	Through 2022: *Categoria D*; *Alte specializzazioni in dotazione organica*	Through 2022: Category D; highly specialized staff within the staffing establishment
	From 2023: *Funzionari ed elevata qualificazione*; *Alte specializzazioni e resp. servizi e uffici in d.o.*	From 2023: Officials and highly qualified staff; highly specialized staff and heads of services or offices within the staffing establishment
Managers	*Dirigenti*; *Direttivi*; *Direttori generali*	Managers; general directors

Notes. The classification follows the occupational hierarchy in Conto Annuale and the official transposition table in the 2019–2021 CCNL for Local Government Functions. Municipal secretaries and residual categories are omitted. Referenced on pages: 8, A.4.

B.2. Measuring Public Services Output

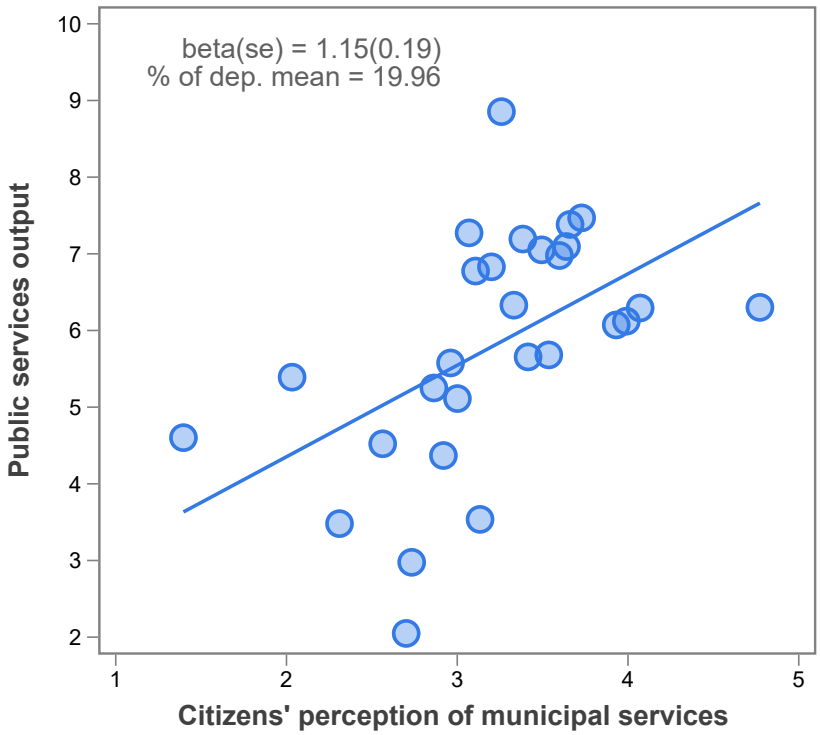
Figure A.1: Examples of Physical Measures Underlying Public Services Output



		Numero
A01	Circoscrizioni	
A02	Impianti semaforici	
A03	Punti di accesso ZTL	
A04	- di cui controllati con strumenti automatici	
		Numero giornate annue
A05	Mercati	
		Mq
A06	Aree pedonali permanenti	
A07	Strade spazzate nell'anno	
A08	Strade lavate nell'anno	
		Km lineari
A09	Piste ciclabili	,00
		Numero
A10	Stalli per la sosta a pagamento	
A11	Stalli per la sosta di disabili	
		Mq
A12	Aree verdi	

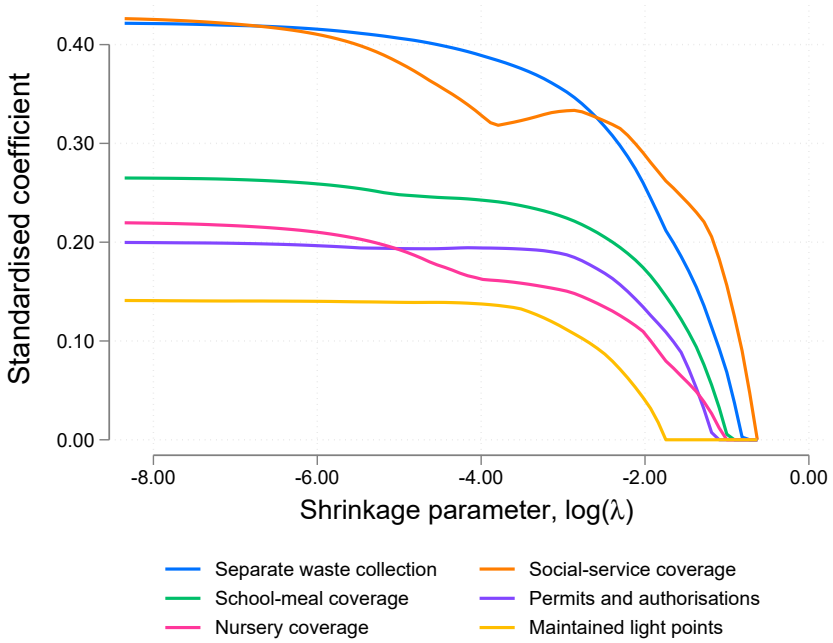
Notes. The figure shows a snippet of the OpenCivitas questionnaire in which municipalities report physical measures of local public goods services. OpenCivitas combines a subset of the reported measures, together with other administrative data, to construct public services output. Referenced on page: 10.

Figure A.2: Public Services Output and Citizens' Assessments of Public Services



Notes. The figure relates the average service-quality assessment among respondents to my survey (fielded July–August 2026) with the municipality's 2022 OpenCivitas measure of public services output. The matched municipality is the unit of observation, and both variables are shown in their original units. Citizen assessments are first averaged within municipality. The binscatter residualizes both measures with respect to population-decile fixed effects and weights each municipality by the number of survey respondents. Markers report binned means and the solid line reports the corresponding linear fit. The slope and standard error come from the same weighted municipality-level regression, with heteroskedasticity-robust standard errors. Referenced on page: 10.

Figure A.3: Lasso Decomposition of Public Services Output



Notes. The figure plots standardized lasso coefficient paths for the six predictors with the largest absolute coefficients in a municipality-level model of average OpenCivitas public services output. Raw measures are averaged over 2015–2019 and 2021–2022. Count variables are then expressed per 1,000 residents, and then winsorized at the 1st and 99th percentiles. Larger values represent greater public services output. Both the outcome and predictors are standardized across municipalities. The vertical axis reports standardized coefficients; the horizontal axis reports $\log(\lambda)$, where larger values imply stronger shrinkage. The model selects $\lambda = 0.0359$ and retains 28 of 79 predictors, with $R^2 = 0.773$ and $N = 3{,}802$. Referenced on page: 10.

B.3. Measuring Consulting

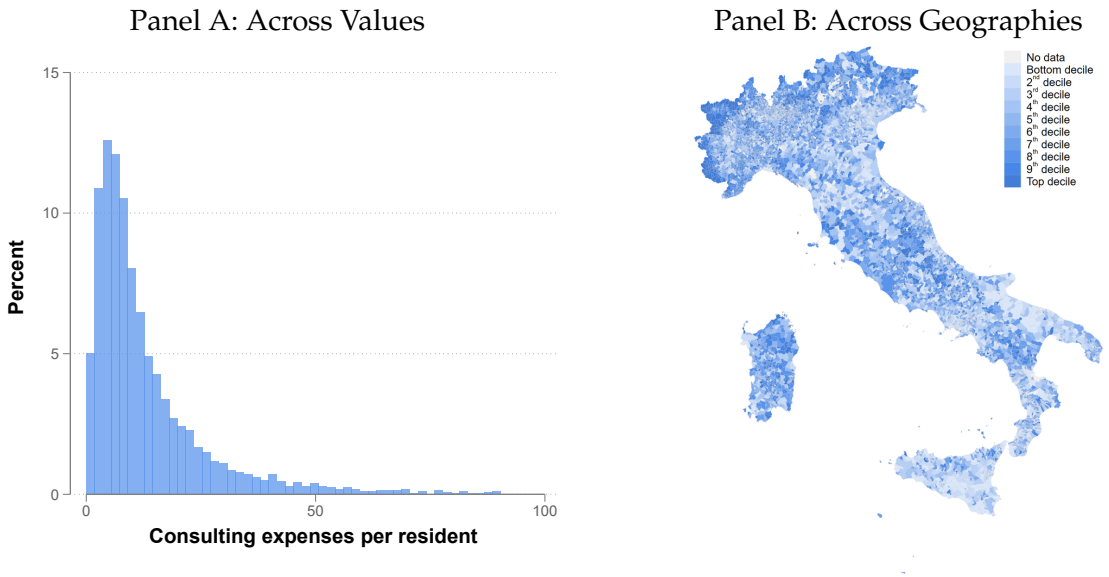
Main measure construction. The main measurement of consulting engagements come from municipal cash flows data. Table A.2 shows the description of the invoices used to form the baseline measure. In 2017 municipalities adopted a more granular coding system to report their budgets. Prior to 2017, cash-flow data used relatively coarse expenditure codes; after the harmonization, public bodies adopted a much more granular coding system, expanding the number of expenditure categories from roughly 250 to roughly 750. Because no official crosswalk exists to harmonize codes over time, I use cash-flow data only for items not available in the financial statements, including those related to consulting engagements. Note I consider only invoice codes referring to current expenditures, as capital expenditures refer primarily to investments. All personnel expenditures fall within current expenditures.

Table A.2: Consulting Expenditure Definitions

Invoice Code	Description	Year
1307	Consulting and professional engagements	Before 2017
1.03.02.10.001	Freelance engagements for studies, research and consulting	After 2017
1.03.02.10.002	Experts for commissions, committees and councils	After 2017
1.03.02.10.003	Engagements to studies, research and consulting firms	After 2017
1.03.02.11.999	Other professional and specialist services n.e.c.	After 2017
1.03.02.19.010	ICT consulting and professional services	After 2017
1110	Compensation for coordinated and continuous collaborations	Before 2017
1.03.02.12.003	Coordinated and project-based collaborations	After 2017
1.03.02.11.004	Appraisals and expert assessments	After 2017
1.03.02.11.008	Accounting, tax and labor professional services	After 2017

Notes. The table describes the specific invoice categories used to define consulting expenditures from municipal cash-flow data (SIOPE). In 2017 new accounting rules required municipalities to report expenditures using more disaggregated invoice categories. Referenced on pages: 11, A.7.

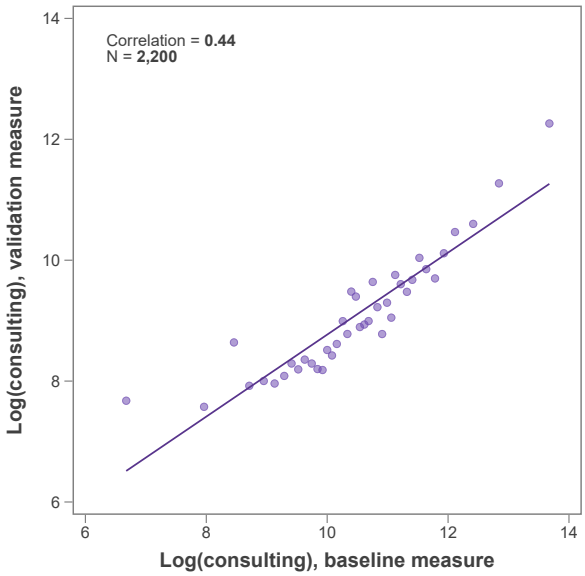
Figure A.4: Distribution of Consulting Expenditures



Notes. This figure shows the distribution of consulting expenditures, across Italian municipalities. Panel A reports the histogram of consulting expenditures per capita, with municipalities grouped into 50 equally spaced bins over [0€,100€]. The y-axis indicates the percent of municipalities falling within each bin. Panel B maps consulting expenditure per resident across municipalities, where each municipality is assigned a decile. Darker shading indicates a higher decile of consulting expenditures per capita. Gray areas indicate municipalities with no data. Referenced on page: 12.

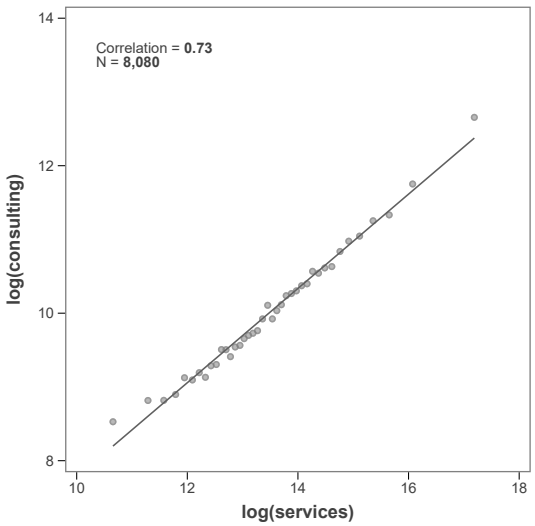
Validation using procurement suppliers. Because the LLM-based classification covers only a subset of procurement contracts, I construct an alternative measure of consulting expenditure based on the identities of winning firms. I searched on Orbis Italian firms which have consulting-related terms in their business purposes, as well as sector and industry descriptions. This leaves around 200 thousand candidate firms. I then retain firms that Orbis assigns to Business Services and use their industry and sector descriptions to exclude activities clearly unrelated to consulting, including bank holding companies, translation, photography, copy shops, specialised design, logistics, and brokerage. I also exclude sole proprietorships, leaving 135,645 unique firm identifiers. I classify a service contract as consulting if at least one reported winning firm belongs to this set. In the 2012–2023 sample, this procedure classifies 7,495 of the 439,208 municipal tenders with matched winner information as consulting. Figure A.5 compares municipality-level average expenditure on these contracts with the baseline measure over the same period.

Figure A.5: Validation of Consulting Expenditure Using Procurement Data



Notes. The figure compares municipality-level consulting expenditure measured from payment flows (baseline measure) with an alternative measure based on procurement contracts awarded to consulting firms identified in Orbis. For each municipality, both measures are averaged across matched municipality-years from 2012 to 2023 and then transformed into logs. Markers report means within 40 bins, and the solid violet line is the linear fit. Referenced on pages: 12, A.9.

Figure A.6: Correlation Between Consulting and Service Expenditure

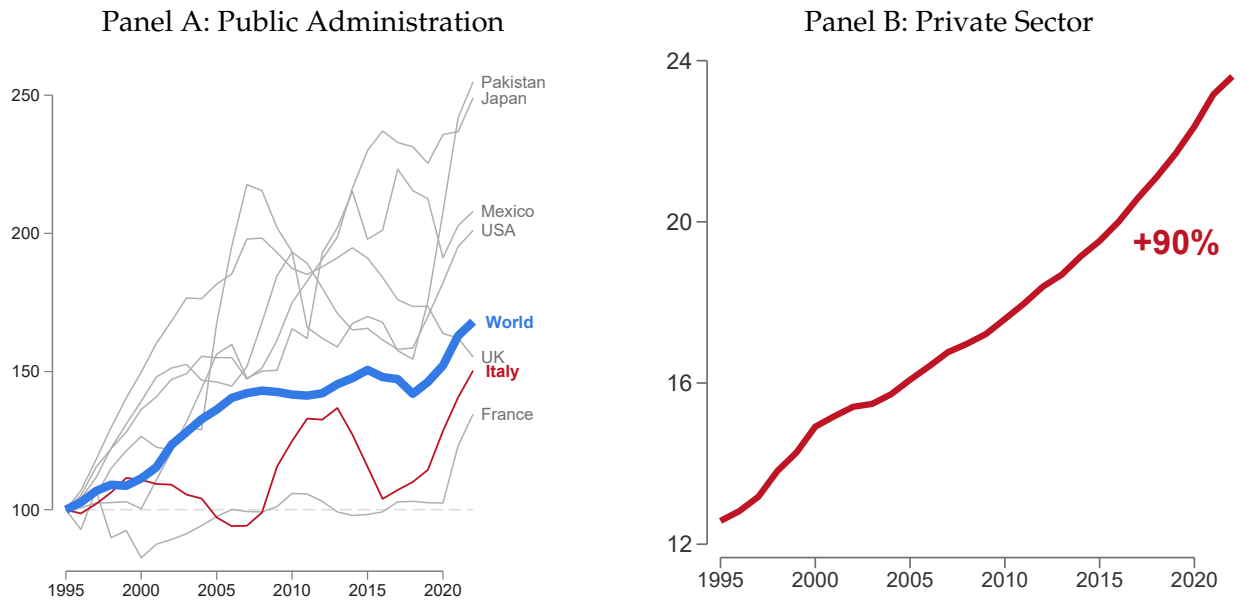


Notes. The figure compares municipality-level consulting expenditure from payment flows with total service expenditure from financial statements. For each municipality, each measure is averaged over its available years and then transformed into logs. Markers show means for 40 bins. The gray line is the linear fit, and the reported correlation coefficient is calculated from the underlying municipality-level observations before binning. N is the number of municipalities. Referenced on page: 16.

C. Additional Empirical Evidence

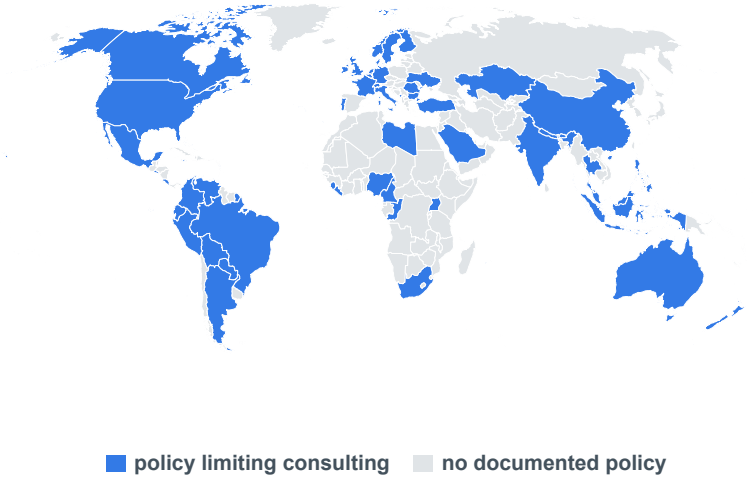
C.1. From Global Data

Figure A.7: Professional Services Across Countries and in the Private Sector



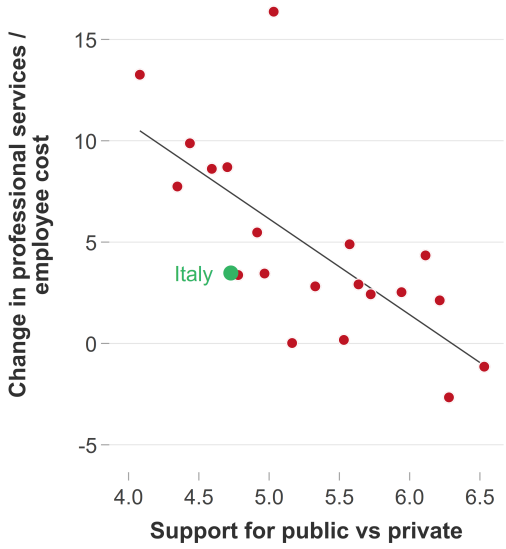
Notes. Panel A reports professional-services inputs used by public administration, expressed relative to labour compensation and indexed to 1995 = 100. The blue line is the world average; the red line is Italy; gray lines show selected comparison countries, including the United States, United Kingdom, France, Japan, Mexico, and Pakistan. Panel B reports world averages of expenses for professional-services input relative to labour cost, in percent, for a private-sector proxy (red). The private-sector proxy covers selected market industries, including agriculture, manufacturing, utilities, construction, trade, transport, information and communications, finance, professional services, education, health, arts, other services, and household activities. Series are smoothed using centered three-year moving averages. The endpoint annotation in Panel B reports the percentage change in the unsmoothed averages from 1995 to 2022. Source: OECD inter-country input-output tables. Referenced on pages: 15, 32, 33.

Figure A.8: Policies Limiting public sector Consulting



Notes. Blue marks countries with at least one documented policy restricting public sector consulting, including implemented measures and publicly discussed proposals, from 2000 onwards. Source: author's compilation, assembled through AI-assisted multilingual searches under a common coding protocol. Referenced on page: 33.

Figure A.9: Growth in Professional Services and Support for Public Ownership



Notes. The figure plots the change in professional-services spending relative to employee costs against support for public versus private ownership. Each point represents a country; Italy is highlighted in green. The solid line shows the linear fit.

C.2. From Administrative Data

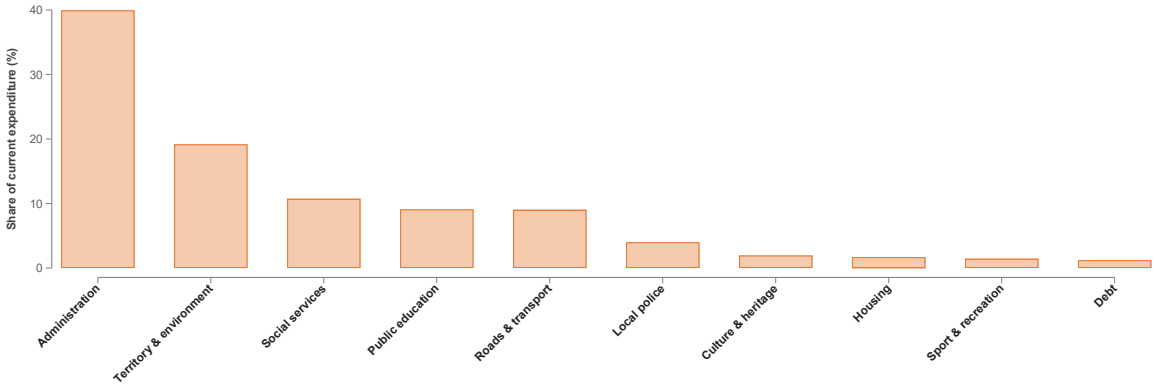
Table A.3: Summary Statistics

	All years (1)	Pre-2011 (2)	2011–2019 (3)	2020–onward (4)
Revenues per capita				
Median	1448.97	1279.93	1391.97	1803.04
Mean	1921.24	1678.39	1848.17	2601.23
Standard Deviation	1690.51	1471.42	1730.77	2967.93
Locally-raised revenues per capita				
Median	612.68	451.50	709.22	786.60
Mean	734.81	538.74	852.14	960.11
Standard Deviation	561.18	742.53	602.26	655.77
Expenses per capita				
Median	1415.92	1296.47	1359.99	1668.78
Mean	1883.28	1697.08	1821.89	2412.26
Standard Deviation	1653.08	1482.46	1748.69	2700.30
Services per capita (excl. consulting)				
Median	199.69	151.35	190.67	245.09
Mean	235.71	182.92	226.73	286.63
Standard Deviation	169.48	153.54	175.38	209.43
Consulting per capita				
Median	8.81	7.69	6.38	11.03
Mean	14.99	14.55	11.67	20.43
Standard Deviation	24.26	24.23	20.52	42.03
Personnel expenses per capita				
Median	155.30	161.00	151.56	146.16
Mean	184.35	187.45	182.49	180.48
Standard Deviation	119.51	119.81	131.10	128.16
Public services output				
Median	5.56	6.00	5.60	5.50
Mean	5.49	5.51	5.50	5.40
Standard Deviation	2.09	2.87	2.28	2.63

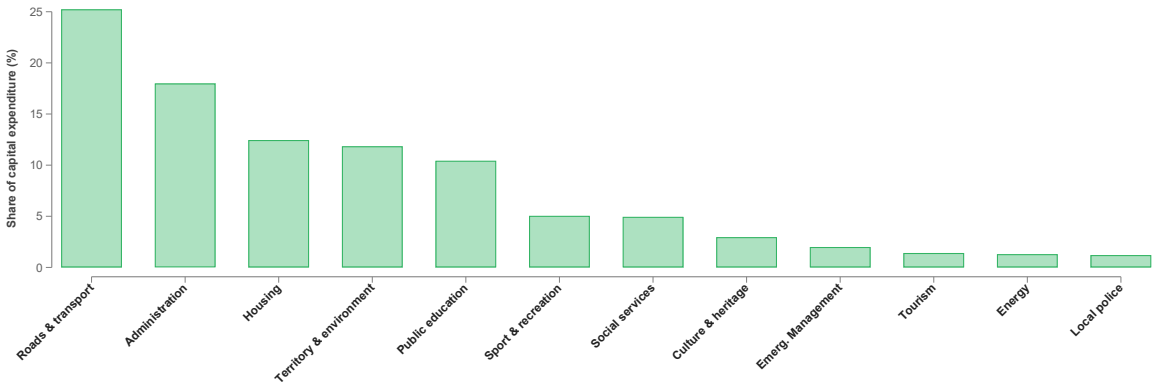
Notes. The table reports summary statistics by period for municipal raw financial variables measured in euros per resident: revenues, tax revenues, total expenditures, current expenditures, services expenditures, consulting expenditures, in-house personnel expenditures, and public services output. Services per capita and consulting per capita come from payment data. Statistics are computed across municipalities after averaging each variable within municipality over the relevant period. Column (1) uses all years in the sample. Column (2) limits the sample to years before 2011, prior to the consulting restrictions. Column (3) limits the sample to 2011–2019, when consulting restrictions were in place, while Column (4) limits the sample to 2020 onward, after consulting restrictions were removed. Referenced on pages: 12, 15, 32.

Figure A.10: Municipal Expenditures by Function

Panel A: Current Expenditure

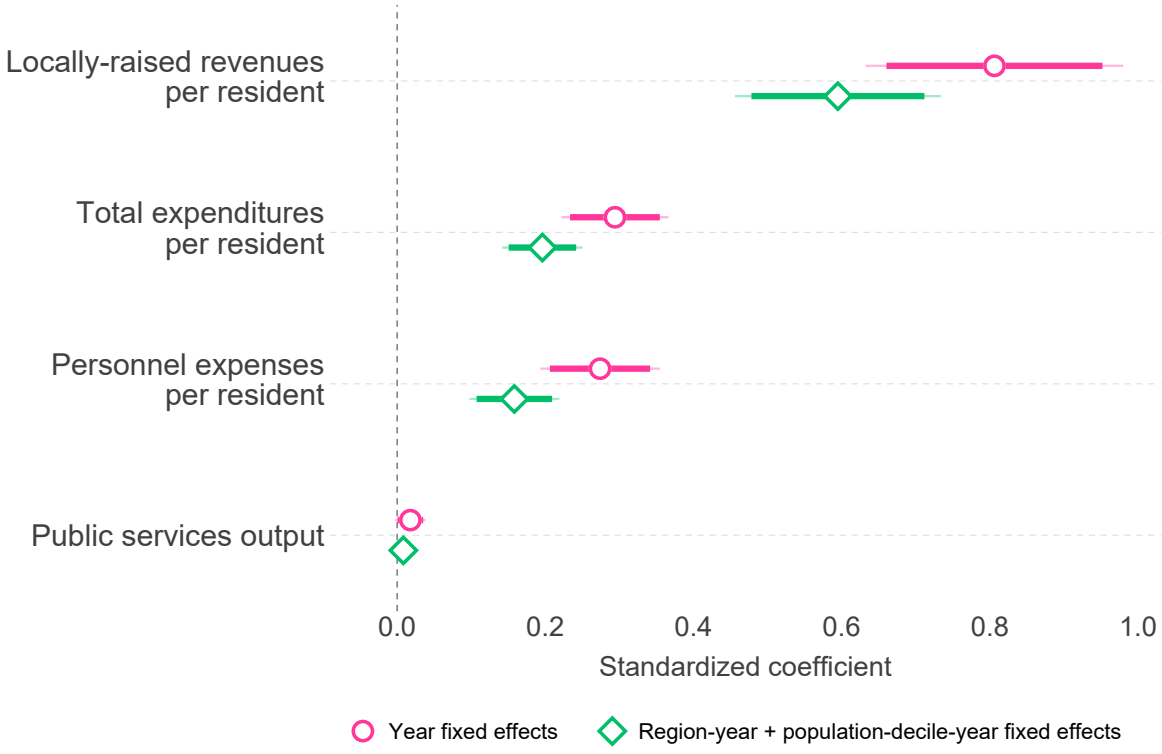


Panel B: Capital Expenditure



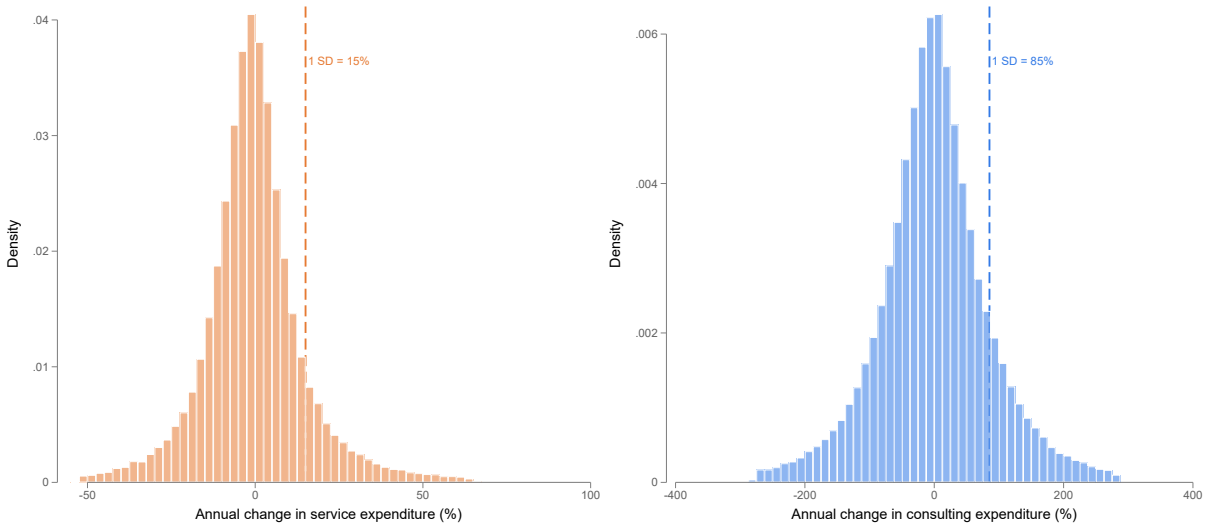
Notes. The figure shows how municipal spending is distributed across functions, using AIDA financial statements. Panel A uses current expenditure; Panel B uses capital expenditure. For each municipality-year, each function share is computed relative to total expenditure and then averaged over 2000–2024 and municipalities. Functions are ranked separately within each panel by descending expenditure share. Functions with an average share below 1% are omitted separately in each panel. Referenced on page: 27.

Figure A.11: Municipal Characteristics and Consulting Spending



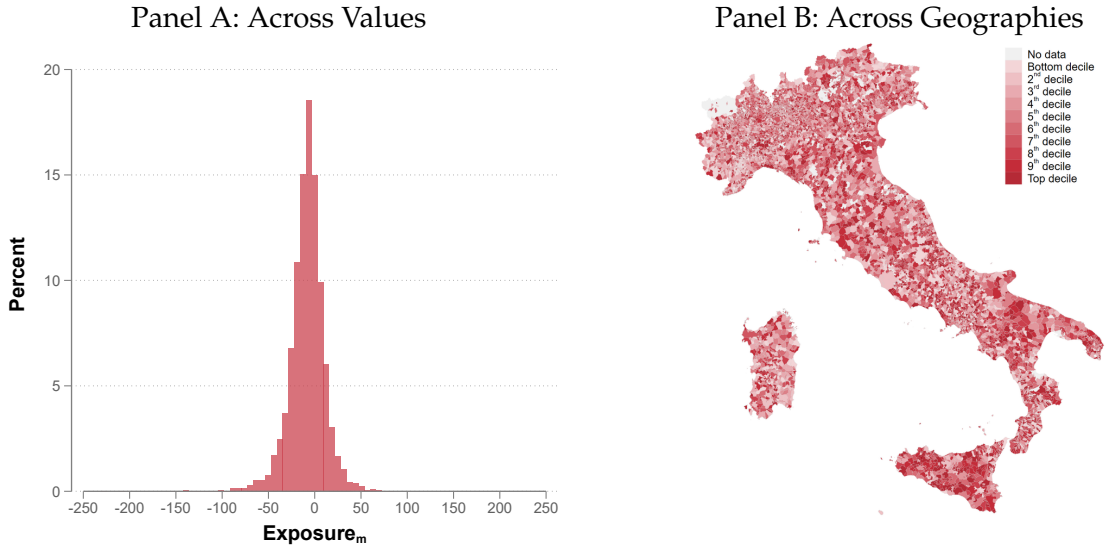
Notes. The figure plots OLS coefficients from separate regressions of standardized consulting expenditures per resident on standardized locally-raised revenues per resident, total expenditures per resident, personnel expenses per resident, and public services output. Pink circles show specifications with year fixed effects; green diamonds add region×year and population-decile×year fixed effects. Dark inner and lighter outer lines denote 90% and 95% confidence intervals, respectively. Standard errors are clustered at the local labour market level. Referenced on page: 24.

Figure A.12: Annual Variation in Services and Consulting Expenditure



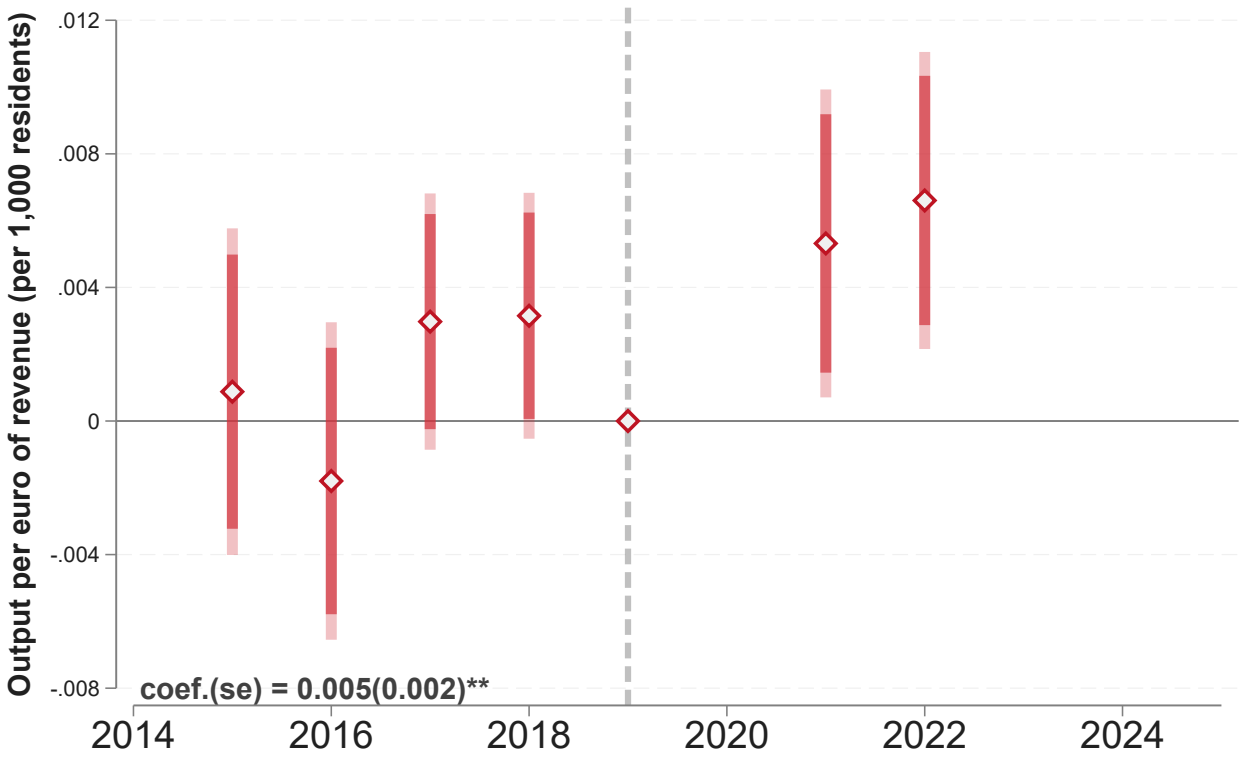
Notes. The figure plots densities of municipality-year residuals from regressions of $(100 \times \Delta \log)$ per-capita expenditure on year fixed effects. Residuals are trimmed at the 1st and 99th percentiles. Dashed lines mark 1 standard deviation of each distribution. Referenced on page: 16.

Figure A.13: Distribution of Exposure to Consulting Restrictions



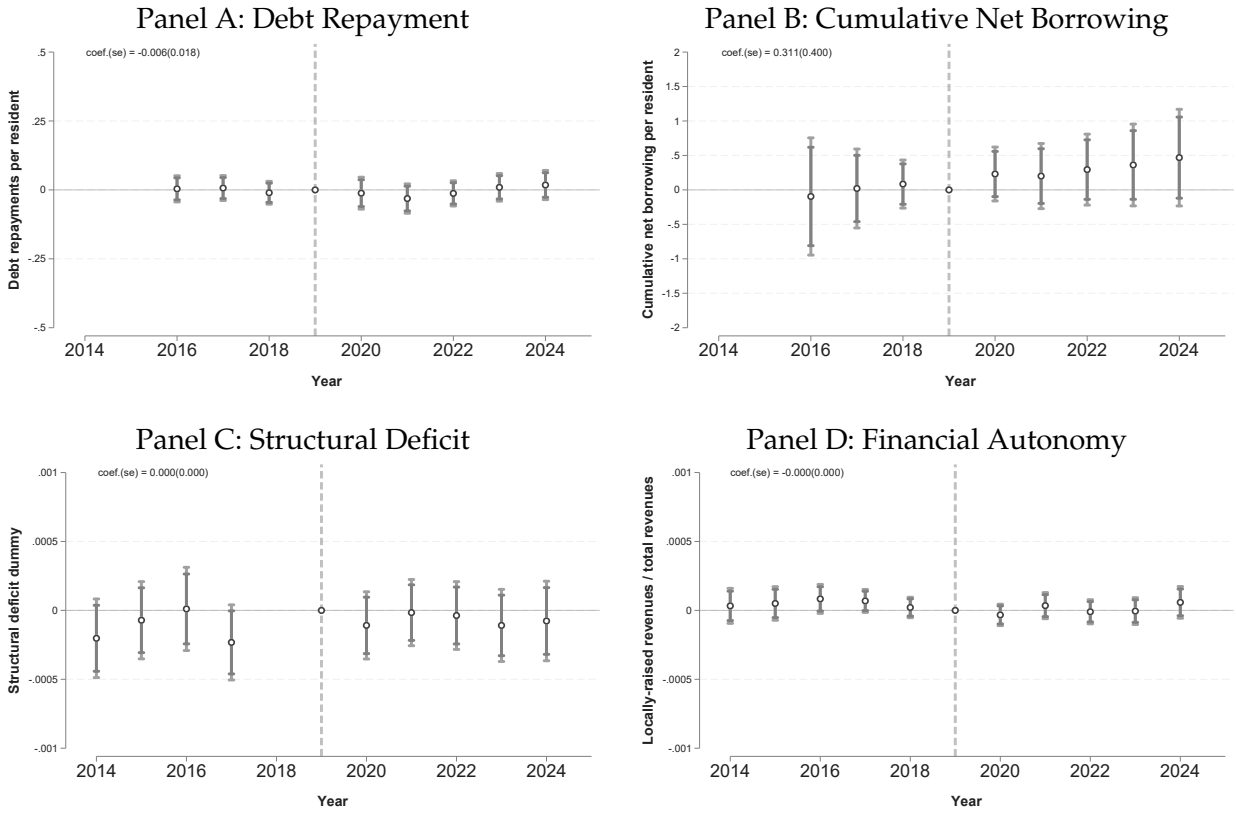
Notes. This figure shows the distribution of Exposure$_m$, defined in Equation (1), across Italian municipalities. Panel A reports the histogram of Exposure$_m$, with municipalities grouped into 50 equally spaced bins over $[-250, 250]$. The y-axis indicates the percent of municipalities falling within each bin. Panel B maps Exposure$_m$ across municipalities, where each municipality is assigned to a decile of the exposure distribution. Darker shading indicates higher decile of Exposure$_m$. Gray areas indicate municipalities with no data. Referenced on page: 17.

Figure A.14: Effect of Consulting on Local Government Performance



Notes. The figure shows reduced-form event-study estimates from Equation 2. The outcome variable is the public services output divided by locally-raised revenues per 1,000 residents. The main regressor is the exposure measure defined in Equation 1 interacted with year dummies. The omitted year is 2019, the year in which the law removing consulting restrictions is drafted. All panels control for region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. The coefficient and standard error reported in the bottom left quadrant of each panel refer to the static specification. Referenced on page: 23.

Figure A.15: Effect of Consulting Access on Fiscal Sustainability



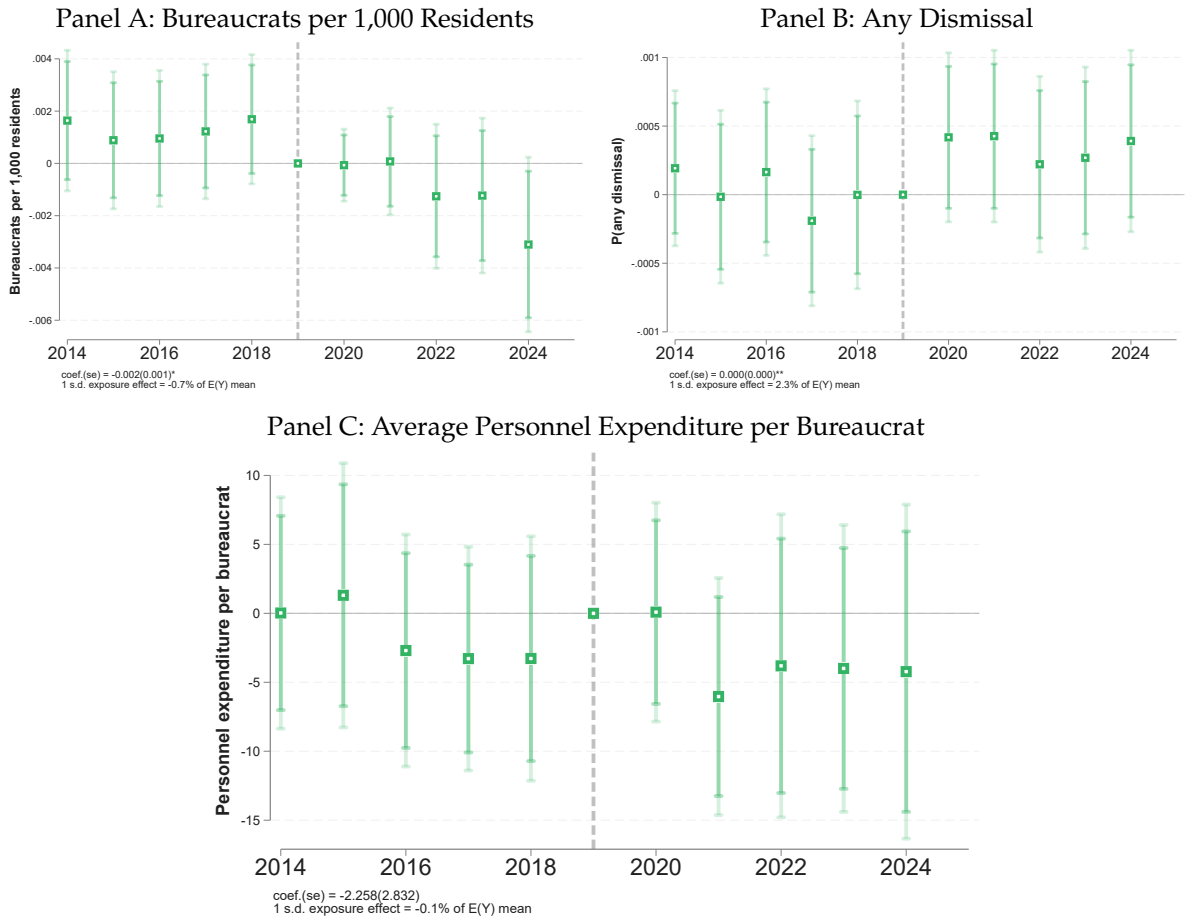
Notes. The figure shows event-study estimates from Equation 2. Panel A uses debt repayments per resident. Before 2016 debt repayments were recorded differently in the municipal financial statements. Data on structural deficit are missing in 2018. Panel B uses cumulative net borrowing since 2014, defined as cumulative new loan revenues minus debt repayments, normalized by population. Panel C uses an indicator equal to one when the municipality is classified as being in structural deficit. Panel D uses the financial autonomy index, corresponding to locally-raised revenues over total revenues. Panel E uses locally-raised revenues divided by total expenditures. All the outcomes except the structural deficit dummy are winsorized at the 1st and 99th percentiles. Standard errors are clustered at the local labour market level. Referenced on page: 26.

Table A.4: Effect of Consulting on Disaggregated Public Services Output

	Waste Output		Territory Output		Social Output		Police Output		School Output	
	RF	IV	RF	IV	RF	IV	RF	IV	RF	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Preferred Specification										
Consulting		0.115*		0.125**		0.128		-0.064		-0.050
		(0.067)		(0.063)		(0.099)		(0.127)		(0.058)
Exposure × Post	0.003**		0.003**		0.003		-0.001		-0.001	
	(0.001)		(0.001)		(0.002)		(0.002)		(0.001)	
N	43,305	43,305	41,245	41,245	38,604	38,604	32,764	32,764	40,658	40,658
𝔼(Y)	5.92	5.92	5.21	5.21	5.19	5.19	4.99	4.99	5.26	5.26
Effect size (%)	0.54	32.98	0.70	40.12	0.57	40.05	-0.19	-18.43	-0.20	-15.28
KP F-Stat		8.589		8.899		5.972		2.674		4.969
Panel B: Specification With Baseline Controls × Year										
Consulting		0.106*		0.104*		0.118		-0.057		-0.036
		(0.062)		(0.056)		(0.091)		(0.120)		(0.053)
Exposure × Post	0.003**		0.003**		0.003		-0.001		-0.001	
	(0.001)		(0.001)		(0.002)		(0.002)		(0.001)	
N	43,305	43,305	41,245	41,245	38,604	38,604	32,764	32,764	40,658	40,658
𝔼(Y)	5.92	5.92	5.21	5.21	5.19	5.19	4.99	4.99	5.26	5.26
Effect size (%)	0.52	30.30	0.61	33.38	0.55	37.03	-0.18	-16.55	-0.15	-10.97
KP F-Stat		9.392		9.581		6.553		2.855		5.395

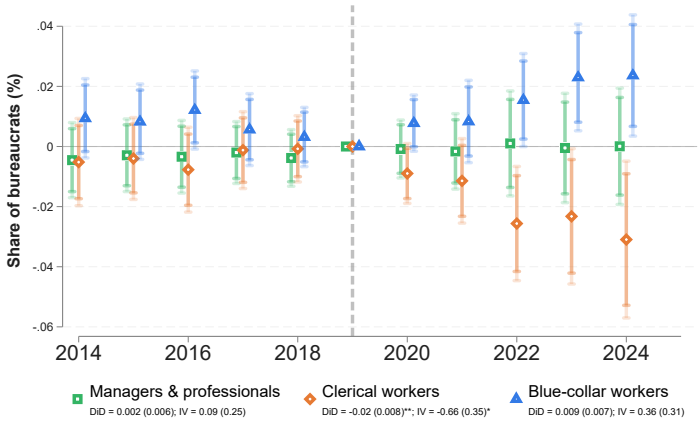
Notes. The table reports static reduced-form and IV estimates for public services indicators disaggregated by functions. The IV columns instrument consulting expenditures per resident with $\text{Exposure}_m \times \text{Post}_t$, and the reduced-form columns report estimates from regressions of each outcome on $\text{Exposure}_m \times \text{Post}_t$. All specifications control for municipality fixed effects, population decile×year fixed effects, and region×year fixed effects. Panel B additionally controls for baseline municipal controls interacted with year fixed effects. The dependent variables indicators go from 1 to 10. Standard errors are clustered at the local labour market level. The sample changes across specifications because municipalities may not be offering all five dimensions of public services. Effect size (%) is defined as the estimated coefficient multiplied by one standard deviation in the right-hand side variable (consulting per capita for IV, and $\text{Exposure}_m \times \text{Post}_t$ for RF) divided by the mean of the dependent variable before the restrictions have been removed. Referenced on page: 27.

Figure A.16: Effect of Consulting Access on Municipal Bureaucrats



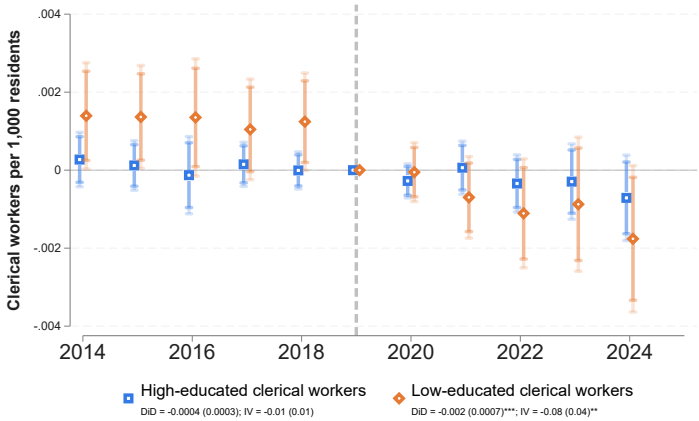
Notes. The figure reports reduced-form event-study estimates from Equation 2. Panel A uses municipal employees per 1,000 residents as the outcome. Panel B uses an indicator equal to one when Conto Annuale reports at least one dismissal or mandatory termination in the municipality-year. Panel C uses average personnel expenditure per bureaucrat, defined as total municipal personnel expenditure divided by the estimated number of municipal employees. This variable should be interpreted as average labor cost per employee, not as a direct productivity or efficiency measure. The dismissal indicator is constructed from raw Conto Annuale separation-cause codes 'C25', 'C01', and 'C21'; 'C25' is firings ordered by the entity, 'C01' is mandatory retirement for age limits, and 'C21' is termination of the employment relationship. The omitted year is 2019. All panels control for municipality, region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. Referenced on page: 27.

Figure A.17: The Effects of Consulting on the Occupational Composition of Bureaucrats



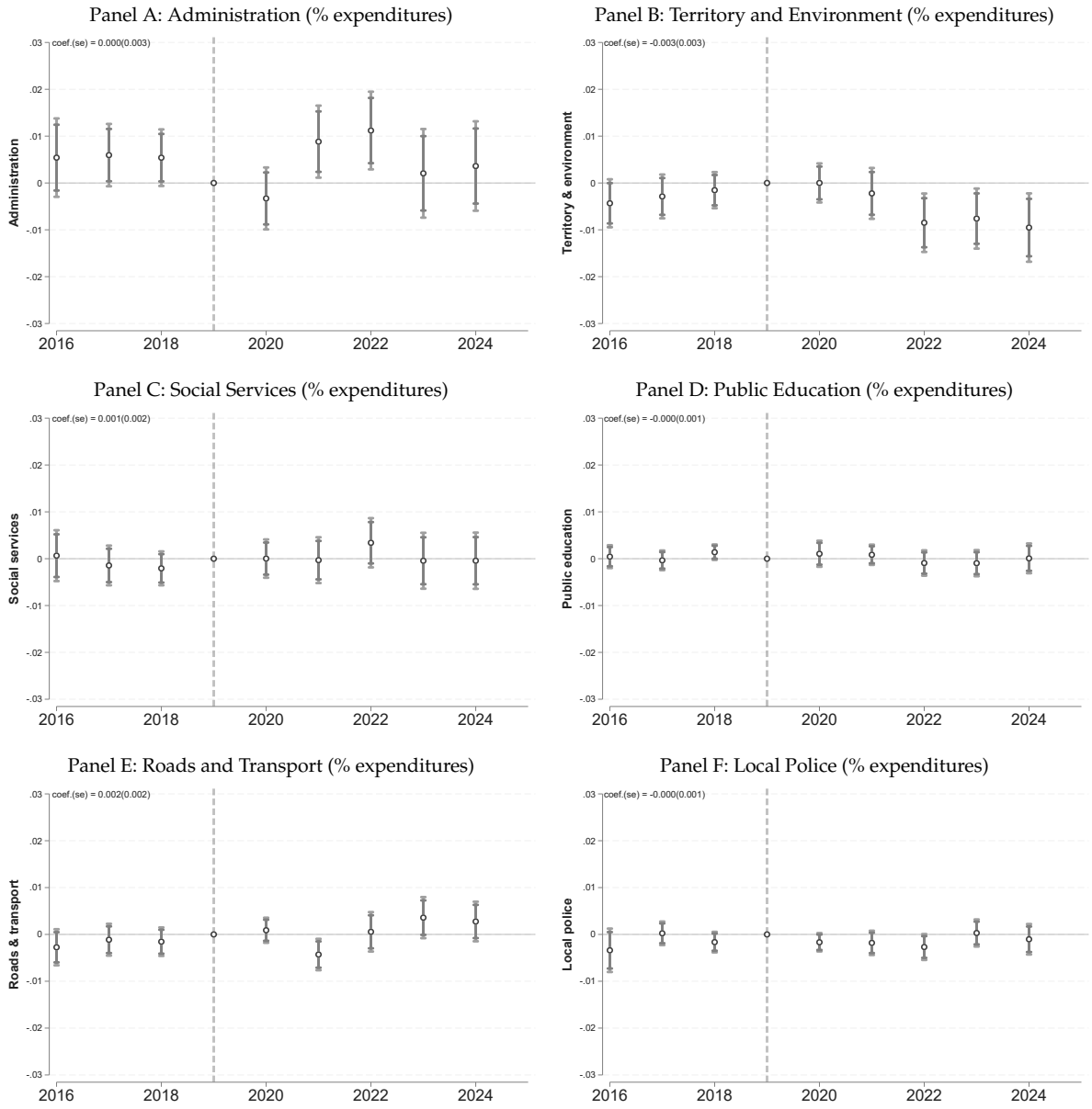
Notes. The figure shows reduced-form event-study estimates from Equation 2. Outcomes are the percentage of municipal employees who are managers and professionals, clerical workers, and blue-collar workers. Shares are winsorized at the 1st and 99th percentiles. The estimates reported below each outcome are from the pooled difference-in-differences and static IV specifications. All specifications control for municipality, region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. [no references]

Figure A.18: The Effects of Consulting on High- and Low-Educated Clerical Workers



Notes. The figure shows reduced-form event-study estimates from Equation 2. Outcomes are high- and low-educated clerical workers per 1,000 residents using 2001 population. The clerical total and the high-education share are winsorized at the 1st and 99th percentiles before constructing the two outcomes. The estimates reported below each outcome are from the pooled difference-in-differences and static IV specifications. All specifications control for municipality, region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. [no references]

Figure A.19: The Effect of Consulting on Direction of Policy-Making



Notes. The figure reports reduced-form event-study estimates for the share of municipal current expenditures allocated to each function. I consider the six most important functions that cover more than 90% of current expenditures, on average. Each share is computed as function-specific current expenditure divided by total current expenditure, multiplied by 100. The omitted year is 2019. Data before 2016 are missing. All panels control for region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. The coefficient and standard error reported in each panel refer to the static specification. Referenced on page: 29.

Table A.5: Effect of Consulting on Investments in Administrative Capacity

	Administrative Capacity					
	OLS		RF		IV	
	(1)	(2)	(3)	(4)	(5)	(6)
Consulting	0.069	0.059			-0.935	-0.594
	(0.070)	(0.070)			(3.486)	(3.299)
Exposure × Post			-0.020	-0.013		
			(0.074)	(0.072)		
Municipality FE	✓	✓	✓	✓	✓	✓
Region × year FE	✓	✓	✓	✓	✓	✓
Population decile × year FE	✓	✓	✓	✓	✓	✓
Baseline controls × year	×	✓	×	✓	×	✓
N	66,670	66,670	66,673	66,673	66,670	66,670
$\mathbb{E}(Y)$	46.09	46.09	46.09	46.09	46.09	46.09
Effect size (%)	2.73	2.36	-0.63	-0.42	-37.24	-23.68
KP F-Stat					9.093	9.854

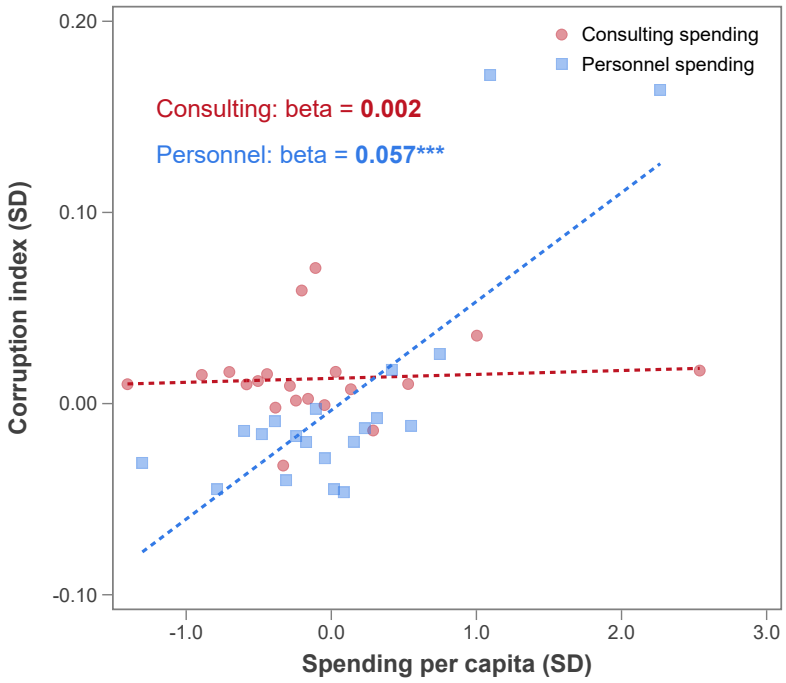
Notes. The table reports OLS, reduced-form, and IV estimates for administrative-capacity investments, defined as capital expenditures for administration and general services per 2001 resident, winsorized at the 1st and 99th percentiles. Columns 1–2 report OLS estimates, columns 3–4 report reduced-form estimates, and columns 5–6 report IV estimates. Even-numbered columns include baseline municipality controls interacted with year. All specifications include municipality, year, region×year, and population-decile×year fixed effects. Standard errors are clustered at the local labour market level. Referenced on page: 29.

Table A.6: Heterogeneity by Corruption and Court of Auditors Cases

	Consulting Expenditures		Public Services Output		Locally-Raised Revenues		Total Expenditures		Personnel Expenditures	
	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Any Recorded Corruption, 2004–2009										
Exposure × Post	0.024***	0.010	0.004**	-0.004	-0.203**	-0.031	-1.071*	0.169	-0.084**	-0.132
	(0.008)	(0.022)	(0.002)	(0.006)	(0.092)	(0.308)	(0.559)	(1.669)	(0.034)	(0.101)
p-value: No = Yes	0.559		0.228		0.578		0.472		0.646	
N	78,958	3,780	41,921	2,072	77,723	3,727	77,767	3,727	78,784	3,777
𝔼(Y)	10.54	5.03	5.58	5.12	843.49	846.74	1782.82	1570.39	173.90	157.47
Panel B: Any Court-of-Auditors Responsibility Case, 2014–2019										
Exposure × Post	0.026***	0.020	0.005***	-0.003	-0.185*	-0.234	-1.241*	0.508	-0.108***	-0.020
	(0.009)	(0.014)	(0.002)	(0.003)	(0.100)	(0.185)	(0.634)	(0.908)	(0.041)	(0.069)
p-value: No = Yes	0.692		0.013		0.811		0.119		0.291	
N	69,338	13,612	37,483	6,623	68,429	13,231	68,465	13,239	69,172	13,601
𝔼(Y)	10.94	7.14	5.70	4.78	848.84	820.43	1767.28	1813.59	174.42	167.87

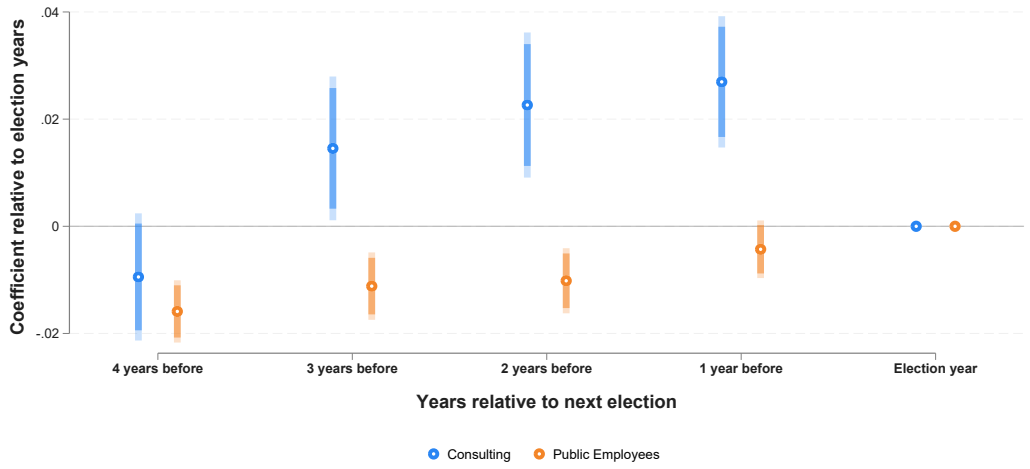
Notes. The first pair of columns reports first-stage estimates for consulting expenditures per resident; the remaining pairs report reduced-form estimates for the four main outcomes. Panel A compares municipalities with no recorded broad corruption in 2004–2009 to municipalities with any recorded broad corruption in those years. Panel B compares municipalities with no recorded Court of Auditors responsibility case in 2014–2019 to municipalities with at least one such case, after excluding pension- and consulting-related rulings. The reported p-value tests equality of the two group-specific coefficients in a pooled interaction specification. All specifications control for municipality, region×year and population-decile×year fixed effects. Standard errors are clustered at the local labour market level. N is the number of municipality-year observations and $\mathbb{E}(Y)$ is the pre-reform mean of the dependent variable. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 29.

Figure A.20: Corruption and Municipal Spending



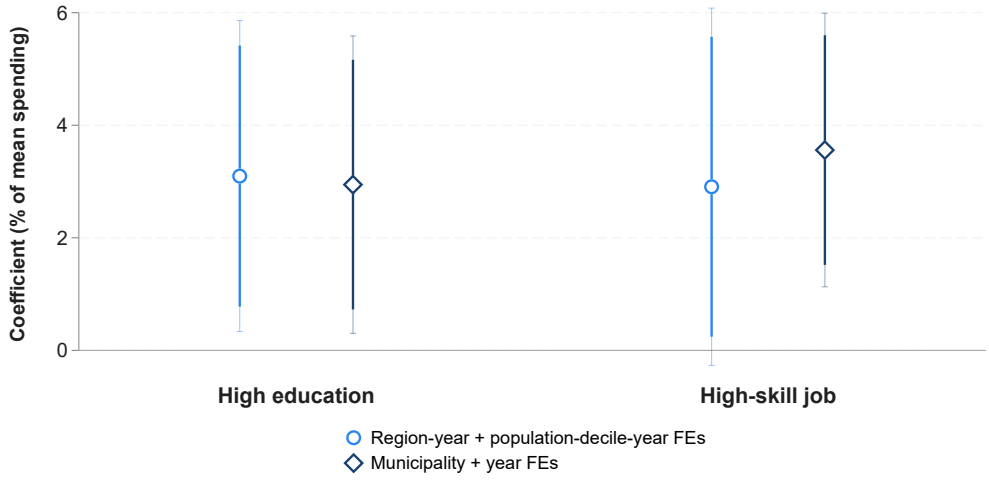
Notes. The figure shows the correlation between the standardized corruption index from Daniele and Giommoni (2025) and standardized personnel expenses per resident and standardized consulting expenditures per resident. Estimates are shown for specifications with region×year and population decile×year fixed effects. The coefficient and standard error reported in each panel are from this bivariate regression, with standard errors clustered at the local labour market level. Referenced on page: 29.

Figure A.21: Consulting and Public Employees Spending over the Electoral Cycle



Notes. The figure reports coefficients from municipality-year regressions of standardized expenditures per resident on indicators for years relative to the next municipal election, using observations from 2014–2024. Blue circles denote consulting; orange circles denote Public Employees. Election year is the omitted category. Both specifications include municipality, year, region×year, and population-decile×year fixed effects. Standard errors are clustered at the local labour market level. Darker and lighter intervals denote 90% and 95% confidence intervals. Referenced on pages: 16, 29.

Figure A.22: Mayor Expertise and Consulting Adoption



Notes. The figure reports coefficients and confidence intervals from separate regressions, expressed as percentages of mean consulting expenditures per resident. High education denotes 18 years of education; high-skill jobs include high-skill occupations. Bright-blue circles denote specifications with region×year and population-decile×year fixed effects; dark-blue diamonds denote specifications with municipality and year fixed effects. Standard errors are clustered at the local labour market level. Darker intervals and lighter capped intervals denote 90% and 95% confidence intervals. Referenced on page: 29.

Table A.7: Hedonic Regressions and the Value of Public Service Improvements

	House price (euros per person)				
	(1)	(2)	(3)	(4)	(5)
Public services output	1151.73***	471.26***	484.85***	284.80***	209.45***
	(148.18)	(126.22)	(92.87)	(50.42)	(46.71)
Observations	17,233	17,233	17,233	17,233	17,233
Oster-adjusted coefficient	–	–	–	220.67	120.51
	House rent (euros per person)				
	(1)	(2)	(3)	(4)	(5)
Public services output	47.74***	17.78***	18.08***	10.50***	7.74***
	(5.66)	(4.48)	(3.75)	(2.17)	(2.05)
Observations	17,233	17,233	17,233	17,233	17,233
Oster-adjusted coefficient	–	–	–	8.08	4.42
Year FE	✓	✓	✓	✓	✓
Region-by-year FE	–	✓	✓	✓	✓
Labour market-by-year FE	–	–	✓	✓	✓
Municipal controls	–	–	–	✓	✓
Fiscal controls	–	–	–	–	✓

Notes. House prices and annual rents come from the Italian Revenue Agency's Real Estate Market Observatory. Values per person use 2011 census housing space and are winsorized at the 1st and 99th percentiles. All columns use a common sample for 2016–2018. Municipal controls cover income, housing, demographics, culture and education, labour market conditions, industry shares, settlement patterns and pre-reform corruption. Fiscal controls include expenditure and locally-raised revenues per person. Standard errors are clustered by local labour market. Oster estimates use $\delta = 1$ and $R_{\max} = \min\{1, 1.3R^2_{\text{within}}\}$, comparing columns (4)–(5) with (3) (Oster, 2019). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on pages: 52, A.28.

Table A.8: Consulting Increase Required to Match Willingness to Pay

	Using house prices	Using rents
Panel A: Valuing better services		
Housing value of one service-score point (euros/resident)	209.45	7.74
Household size (persons)	2.3	2.3
Annualisation rate	3%	Already annual
Calculation		
Annual value of one service-score point (euros/household/year)	14.45	17.81
Panel B: Consulting needed to match willingness-to-pay		
Willingness-to-pay for public employment (euros/household/year)	124.00	124.00
Consulting in 2025 (euros/resident/year)	26.41	26.41
IV estimate: service score	0.135	0.135
IV estimate: revenues	-7.612	-7.612
Calculations		
Annual household gains from 1 euro more consulting per resident		
Fiscal savings	$7.612 \times 2.3 \approx 17.51$	$7.612 \times 2.3 \approx 17.51$
Value of better services	$0.135 \times 14.45 \approx 1.95$	$0.135 \times 17.81 \approx 2.40$
Total annual household gain	$17.51 + 1.95 = \mathbf{19.46}$	$17.51 + 2.40 = \mathbf{19.91}$
Additional consulting needed (euros/resident/year)	$124/19.46 \approx 6.37$	$124/19.91 \approx 6.23$
Increase relative to 2025 consulting (%)	**24.13%**	**23.58%**
Required consulting increase with Oster adjustment	25.20%	24.87%

Notes. Panel A values improvements in services using the hedonic estimates in column (5) of Table A.7. Panel B combines these values with the IV service-score and revenue estimates to calculate how much consulting must increase for annual household gains to equal willingness-to-pay for public employment (124 euros). Gains include fiscal savings and better services, assuming constant IV effects. Consulting is the unweighted 2025 mean per 2001 resident. The final row uses Oster-adjusted hedonic estimates. Referenced on page: 53.

C.3. From Court of Auditors Rulings

One concern is that elected officials outsource tasks precisely to shift administrative responsibility: by delegating to external contractors, they reduce their personal exposure to sanctions for errors or misconduct. Under this interpretation, the increased demand for consulting contracts can reflect liability avoidance rather than a genuine demand for expertise. Results in Section 4 already weigh against a pure responsibility-shifting interpretation: consulting affects public sector performance rather than merely reallocating formal responsibility.

Two pieces of evidence speak against this mechanism. First, under Italian administrative law, legal liability rests with the authorizing official regardless of whether execution is contracted out. Interviews with mayors and policy-makers working at the National Italian Municipalities Association and related bodies indicate that public officials remain responsible for monitoring and sign-off. Second, I test this directly using variation in the salience of accountability concerns. The Court of Auditors prosecutes and sanctions public officials (elected or not) for financial misconduct or gross negligence. Proceedings are initiated by an independent public prosecutor housed within the Court; convicted officials pay personal monetary damages. This is the primary legal risk Italian municipal employees and elected officials face. Using the Court's public API, I collect all rulings naming municipalities and construct a municipality-year panel of liability judgments (*giudizi di responsabilità*). I interpret exposure to administrative sanctions as a proxy for an increase in officials' perceived responsibility from holding the public office.

Given the Court of Auditors' workload, the exact timing of the rulings can be considered plausibly exogenous. I define treatment as the year of a municipality's first remaining liability judgment. I exclude retirement-related cases and rulings whose text mentions consulting, so that treatment is not mechanically defined by a dispute involving the outcome itself.[68] I compare municipalities that were subject to a liability ruling earlier vs later using three staggered difference-in-differences estimators.[69] The outcome is the share of consulting to personnel expenditures, winsorized at the 1st and 99th percentile.[70]

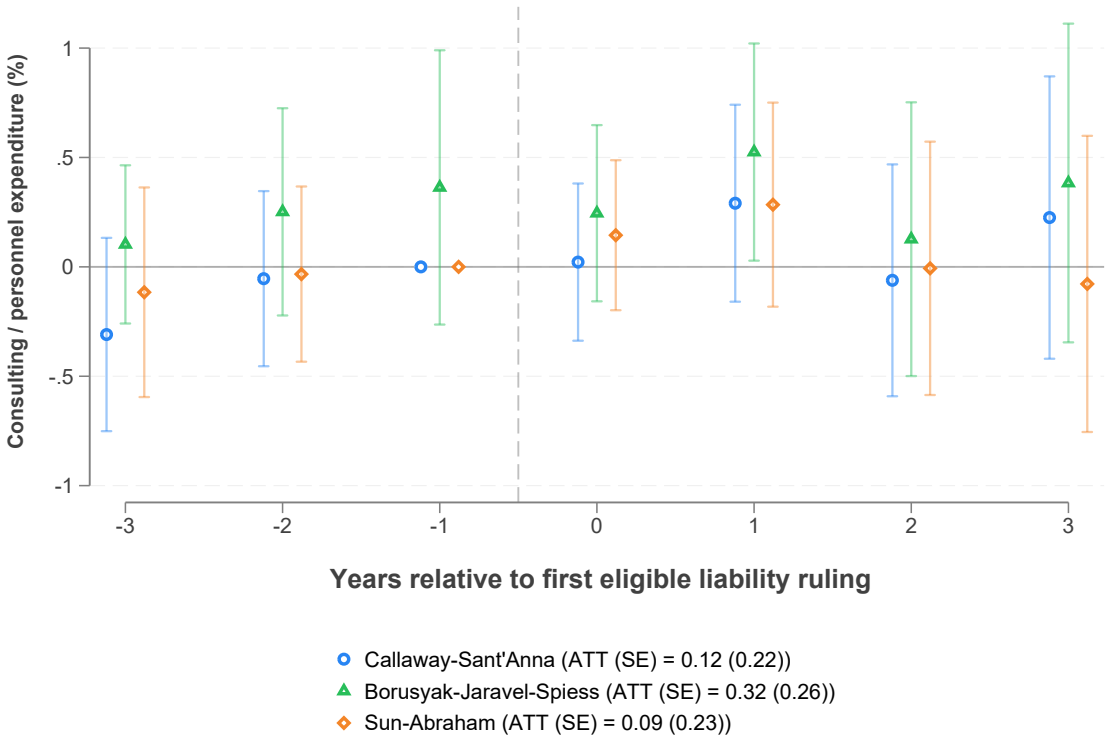
Pre-treatment coefficients are small and statistically indistinguishable from zero, although the confidence intervals are wide enough that a modest pre-trend cannot be ruled out. Under Callaway and Sant'Anna (2021) and Sun and Abraham (2021), no post-treatment coefficient is distinguishable from zero. The Borusyak, Jaravel and Spiess (2024) estimator yields one post-treatment coefficient significant at the 95% level, but the pooled effect is not distinguishable from zero. The pooled ATTs are small and imprecisely estimated. Thus, the estimates do not support the idea that exposure to administrative sanctions leads public officials to hire more external labour force, compared to internal personnel.

[68] The Court of Auditors also oversees public sector retirement plans and monitors fund use across regions, universities, and other administrative units.

[69] The estimator of Sun and Abraham (2021) requires a control group that remains untreated throughout the estimation window; I therefore pool the 2023–2025 treatment cohorts and restrict the sample to 2010–2022, so this specification is estimated on a different sample than the other two.

[70] I do not use consulting in levels because Court of Auditors rulings may induce accused municipalities to reduce overall spending.

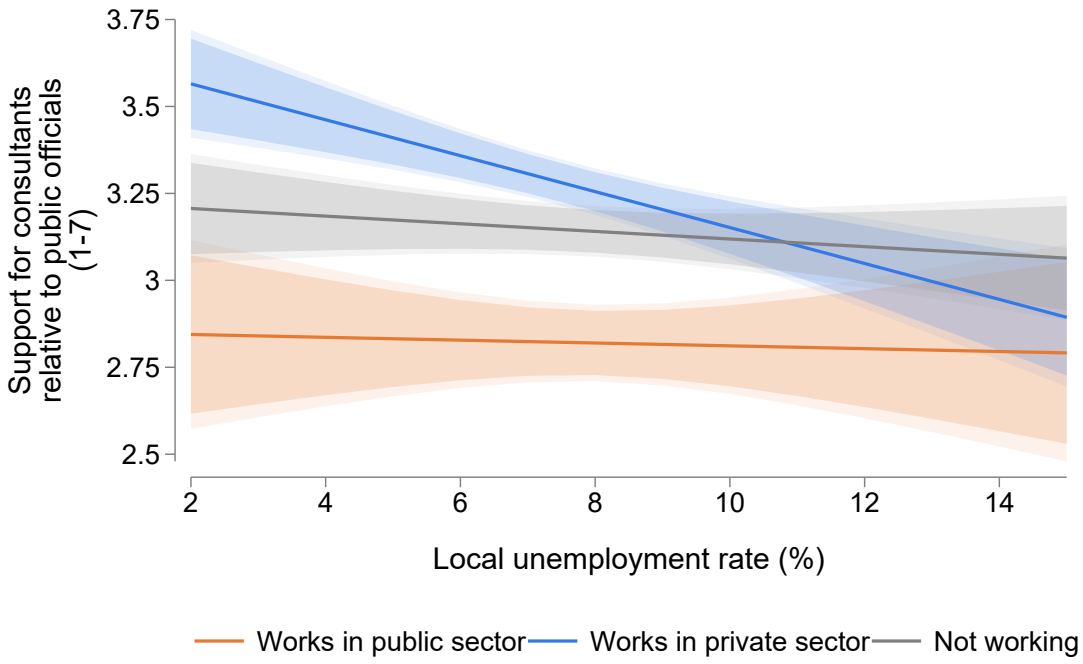
Figure A.23: Liability Rulings and Consulting Expenditure



Notes. The outcome is the share of consulting relative to personnel expenditure, winsorized at the 1st and 99th percentiles. Treatment begins with a municipality's first Court of Auditors liability judgment. The estimates compare early vs later treated municipalities. Sun–Abraham uses the 2023–2025 treatment cohorts as controls in a 2010–2022 sample. Event time -1 is normalized to zero for Callaway–Sant'Anna and Sun–Abraham, the Borusyak–Jaravel–Spiess pre-treatment coefficients are placebo estimates. Capped lines report 95% confidence intervals. Standard errors are clustered at the municipality level. Referenced on page: 29.

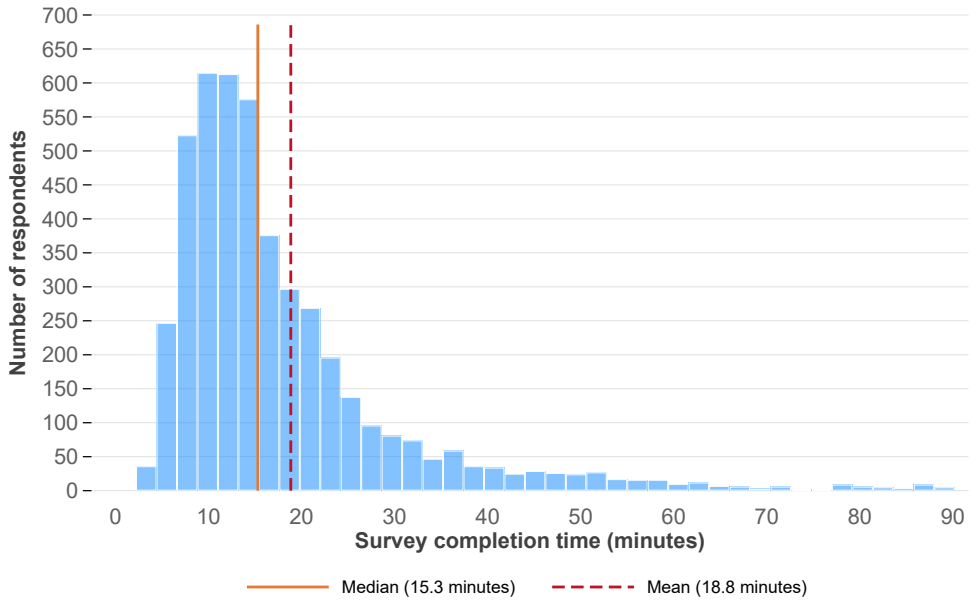
C.4. From Survey Data

Figure A.24: Stated Preferences and Local Unemployment, by Employment Group



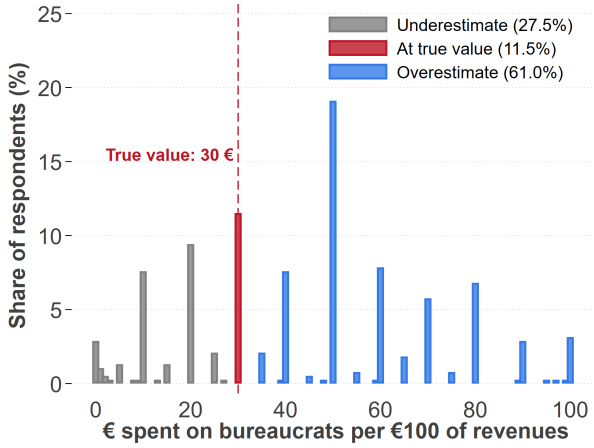
Notes. The figure reports adjusted predictions of stated support for consultants relative to public employees on a seven-point scale, with higher values indicating greater relative support for consultants. The sample includes all information treatments and both equivalence assignments, matched to municipal administrative data. Predictions come from an OLS regression that interacts the 2024 municipal unemployment rate with employment group and controls for birth year, gender, education, and assignment to factual information, advantages information, disadvantages information, and advantages salience. The unemployment rate is winsorized at the 1st and 99th percentiles. Predictions are averaged over the estimation sample's other covariates. Orange, blue, and grey lines identify public sector workers, private-sector workers including the self-employed, and respondents not working, respectively. Standard errors are clustered by municipality. Darker and lighter bands indicate 90% and 95% confidence intervals, respectively. Referenced on page: 53.

Figure A.25: Distribution of Survey Completion Times



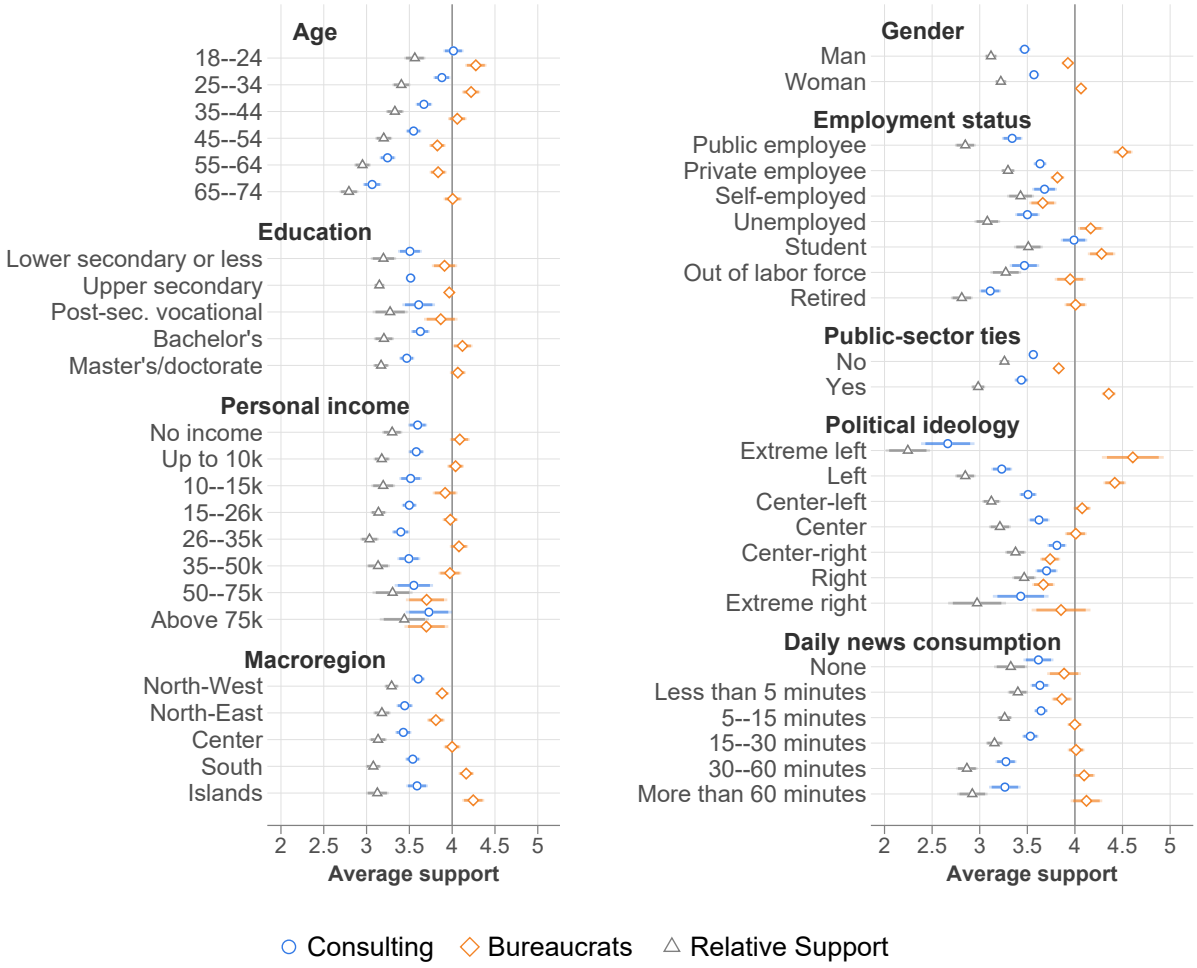
Notes. The figure shows the distribution of the time, in minutes, spent by respondents to complete the survey. Bars report respondent frequencies in two-minute bins. The median is shown with a solid blue line and the mean with a dashed red line; both statistics are calculated using the respondents displayed. Completion times above 90 minutes are excluded from the figure. Referenced on page: 35.

Figure A.26: Prior Quantitative Beliefs About Personnel Spending



Notes. The figure reports the distribution of respondents' beliefs about in-house personnel spending for €100 of municipal revenues. The dashed red line marks the true value, 30; gray bars denote underestimates, red bars responses at the true value, and blue bars overestimates. Referenced on page: 40.

Figure A.27: Average Policy Preferences by Demographic Group



Notes. The figure reports subgroup means for support for consultants (blue circle), support for bureaucrats (orange diamond), and preferred relative reliance on consultants versus bureaucrats (grey triangle). The outcomes are measured on seven-point scales. For consulting and bureaucrats, 1 denotes much less reliance and 7 much more reliance; for the relative outcome, 1 denotes greater reliance on bureaucrats and 7 greater reliance on consultants. The vertical line at 4 denotes neutrality. Lighter and darker horizontal bands report 95 and 90 percent confidence intervals, respectively. Referenced on page: 41.

Figure A.28: Worldviews and Policy Preferences for the State's Organization



Notes. The figure reports coefficients from separate regressions of support for consultants (blue circles), support for bureaucrats (orange diamonds), and relative support (grey triangles) on each worldview measure. The four worldview measures are coded from 1 to 5, with higher values indicating greater trust or more favorable views of the sector described. The three preference outcomes are coded from 1 to 7; higher values indicate greater support for consultants or bureaucrats, respectively, while for relative support they indicate greater support for consultants relative to bureaucrats. All regressions control for respondent birth year, gender, education, work situation, public sector tie, income, and macroregion. Lighter and darker horizontal bands report 95 and 90 percent confidence intervals, respectively. Referenced on page: 42.

Table A.9: Effects of the Advantages-Salience Prompt

	Support petition for		Support use of			WTP to keep public employees
	Public employees (1)	Consultants (2)	Consultants (3)	Public employees (4)	Consultants vs. employees (5)	Euros per year (6)
Advantages: salience prompt	0.015	0.004	0.062	-0.033	0.154*	-36.0
	(0.026)	(0.022)	(0.078)	(0.084)	(0.080)	(25.7)
Control mean	0.506	0.238	3.436	4.043	3.063	–
Control WTP	–	–	–	–	–	124.0
Other information-treatment controls	✓	✓	✓	✓	✓	✓
Choice observations	–	–	–	–	–	6,236
Respondents	4,828	4,828	4,828	4,828	4,828	3,118

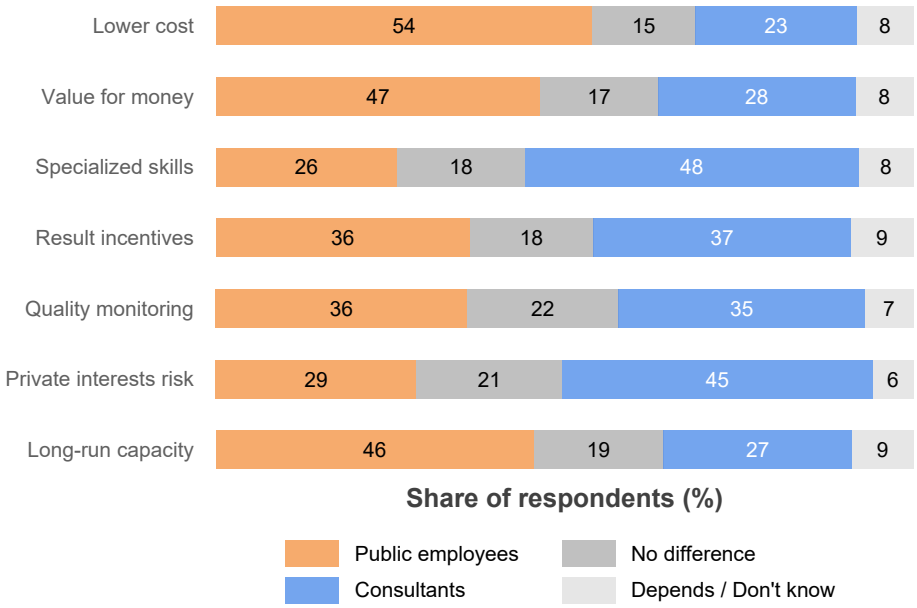
Notes. Columns (1) and (2) report linear-probability estimates for the real-stake petition choice. Columns (3)–(5) report OLS estimates for preferred municipal use of consultants, public employees, and consultants relative to public employees, respectively, on the original seven-point scales. These five columns report the coefficient on the advantages-salience prompt relative to the pooled pure and active control groups; every regression also includes indicators for the factual, advantages-information, and disadvantages-information arms. In column (6) the sample excludes respondents who read that the two policies have identical implications for service delivery. The coefficient reports the advantages-salience minus pooled-control difference in willingness to pay to keep public employees, in euros per household per year. It comes from a logit model that fully interacts information-treatment arm with the savings difference. The first five columns report heteroskedasticity-robust standard errors. Column (6) clusters standard errors by respondent and uses delta-method inference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 46.

Table A.10: Information vs Priming: Test for Equality

	Support petition for		Support use of		
	Public employees (1)	Consultants (2)	Consultants (3)	Public employees (4)	Consultants vs. employees (5)
Advantages: information - salience	-0.066	0.054	0.318	-0.124	0.223
	(0.028)	(0.024)	(0.082)	(0.087)	(0.085)
Equality-test p-value	0.019	0.028	0.000	0.156	0.009
Respondents	4,828	4,828	4,828	4,828	4,828

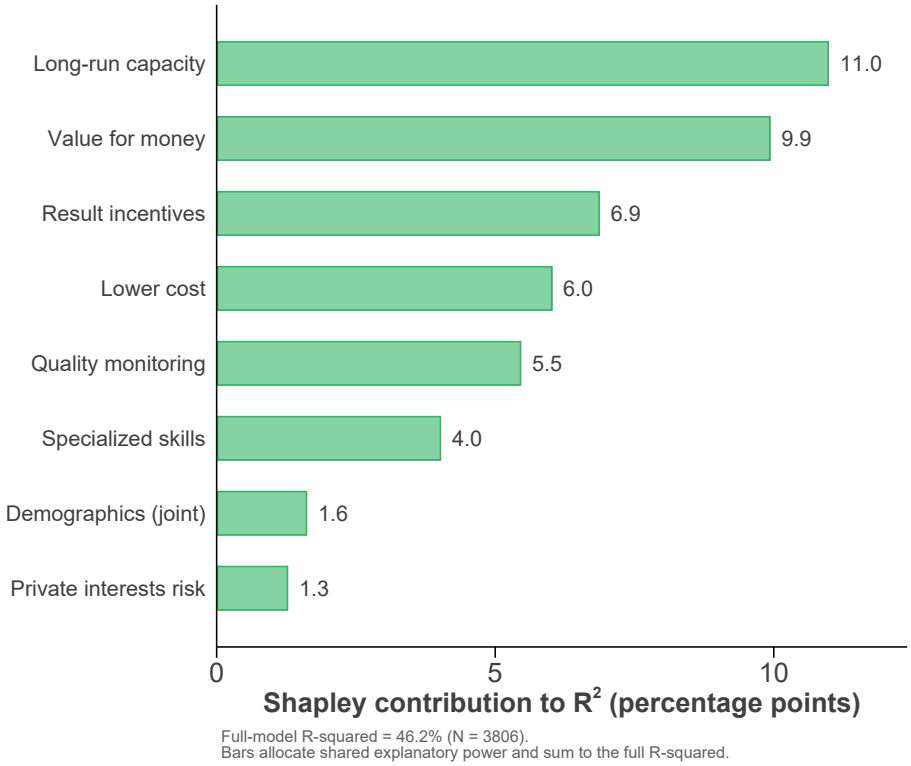
Notes. The table reports direct comparisons of the advantages-information and advantages-salience treatment arms. Each column comes from estimating Equation (1); the omitted category pools the pure and active control groups. The reported coefficient is the advantages-information effect minus the advantages-salience effect, so a negative estimate indicates a lower outcome under advantages information. Columns (1) and (2) use linear-probability models for the real-stake petition choice. Columns (3)–(5) use OLS for preferred municipal use of consultants, public employees, and consultants relative to public employees, respectively, on the original seven-point scales. Heteroskedasticity-robust standard errors are in parentheses. The equality-test row reports the two-sided p-value for the null that the two treatment effects are equal. Referenced on page: 46.

Figure A.29: Beliefs About Mechanisms: Public Employees versus Consultants



Notes. The figure reports the distribution of answers to the mechanism questions in the survey. For each dimension, respondents state whether it is better achieved with public employees or with consultants and external firms, on a five-point scale: definitely public employees, probably public employees, no difference, probably consultants, definitely consultants, with a separate option 'it depends / I don't know'. Orange segments pool the two public-employee answers and blue segments the two consultant answers; numbers report shares in percent, omitted for segments below 5 percent. Rows follow the order of the questionnaire. For the private-interests question, the blue segment indicates that the risk of partisan interests is perceived as higher with consultants. Shares are unweighted and computed over the analysis sample. [no references]

Figure A.30: Which Mechanism Beliefs Matter Most? Shapley Decomposition



Notes. The figure decomposes the R-squared from a direct OLS regression of the preferred reliance on consultants versus public employees on seven mechanism beliefs and demographic controls for birth year, gender, education, work situation, public sector tie, income, and macroregion. Each belief is a separate group and the demographic controls form one joint group. Each bar is the group's Shapley contribution: its increase in R-squared averaged over every possible order in which the eight groups can enter the model. This allocates explanatory power shared by correlated predictors, so the bars sum to the full-model R-squared. All regressions use the same complete-case sample; no variables are residualized first. [no references]

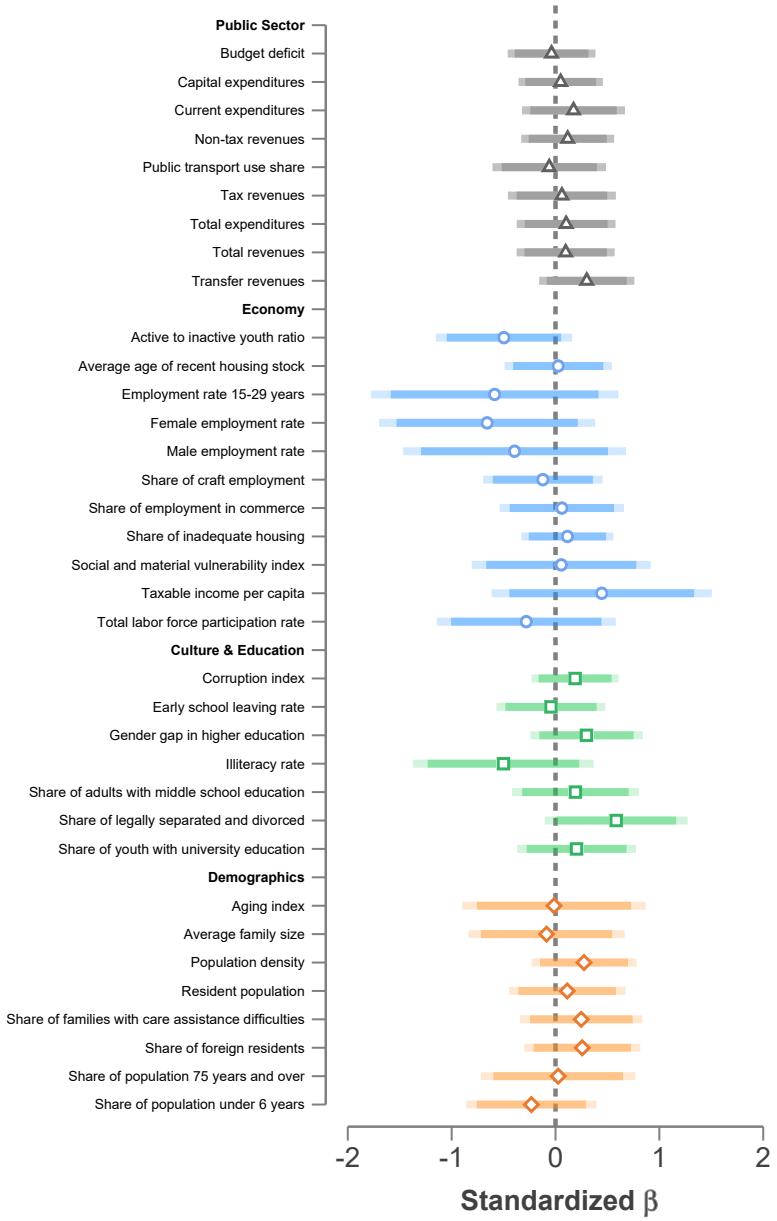
Table A.11: Effects of the Information Treatments on Posterior Consulting Beliefs

	Posterior consulting beliefs (1)
Factual information	-27.074***
	(2.270)
Advantages information	-2.485
	(1.843)
Disadvantages information	-4.001**
	(1.868)
Advantages-salience prompt	-6.127**
	(2.452)
Control group mean	72.678
Effect size, percent of control group mean	
Factual information	-37.3%
Advantages information	-3.4%
Disadvantages information	-5.5%
Advantages-salience prompt	-8.4%
Respondents	1,925

Notes. The table reports OLS estimates from a regression of posterior beliefs about euros spent on external consulting per 100 euros spent on public employees on indicators for the factual-information, advantages-information, disadvantages-information, and advantages-salience treatment arms. The outcome variable is on a scale from 0 to 100. The omitted category pools the pure and active control groups. Posterior beliefs are elicited only from respondents who were not asked for a prior quantitative belief. The control group mean is the outcome mean among respondents in the pooled control groups. Effect sizes divide each treatment coefficient by the control group mean and express the result in percent. Heteroskedasticity-robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 46.

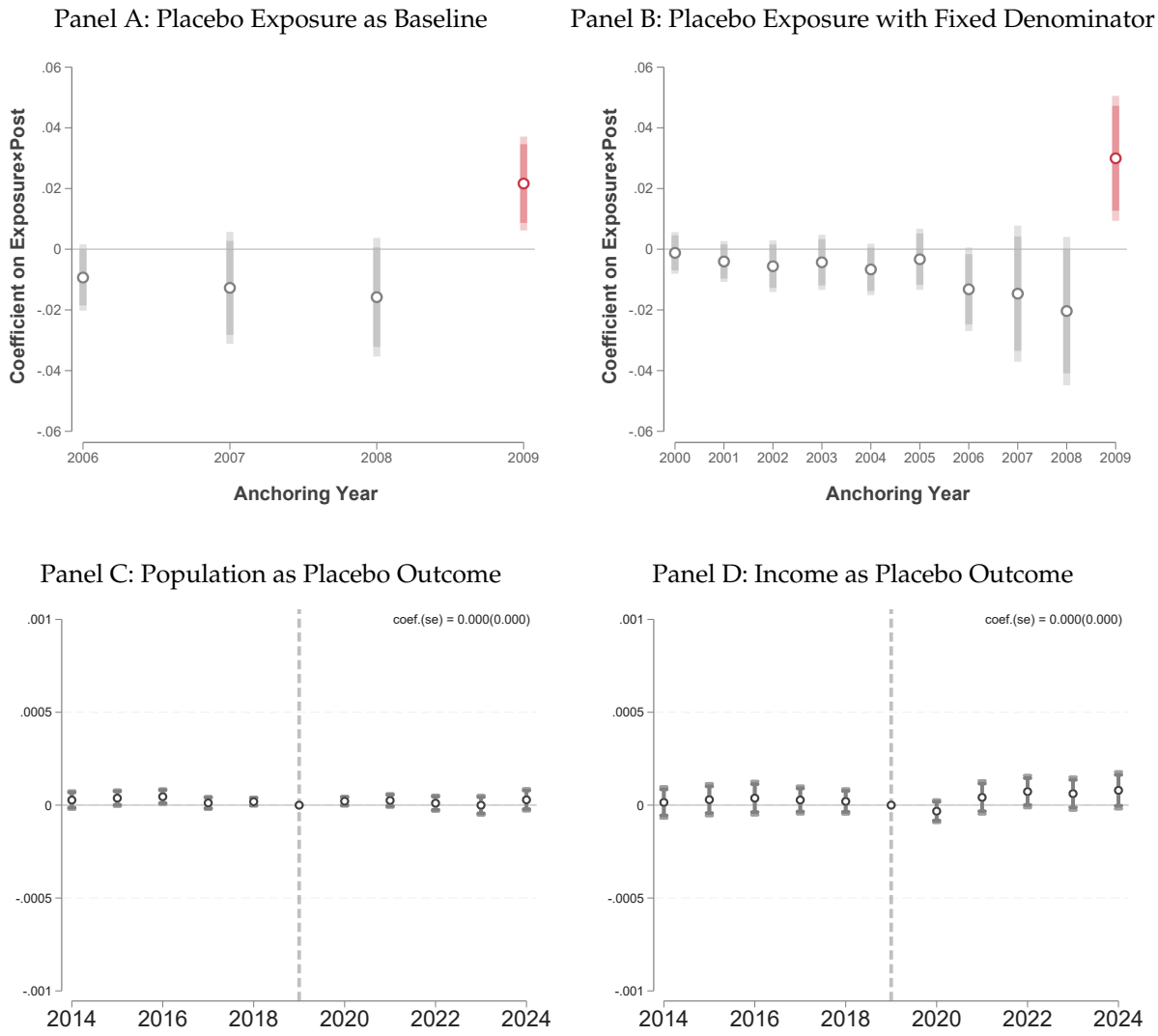
D. Additional Validation and Robustness Checks

Figure A.31: Balance Tests



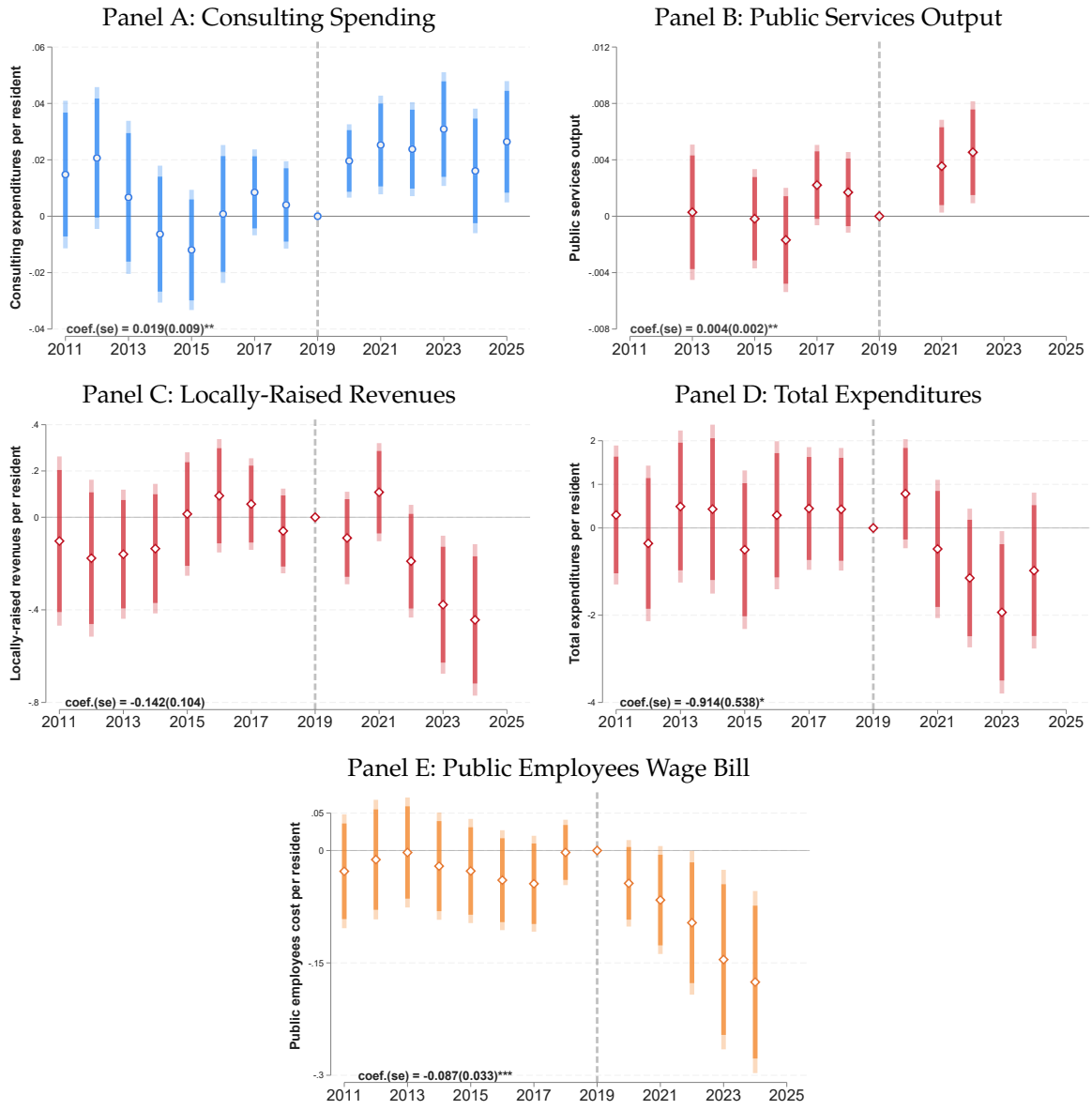
Notes. The figure shows the coefficient (in standard deviations) from balance tests from municipality-level regressions of Exposure$_m$ on X_m, a given observable characteristic, excluding the baseline covariates included in the specification with additional controls. Regressions control for population decile and region fixed effect and standard errors are clustered at the local labour market level. Census variables are measured in 2001 (the closest available census pre-treatment), while all the other variables are measured in 2000 and corruption index is the average between 2004 and 2009. Gray triangles represent the point estimates for variables referring to the public sector, light blue circles for variables related to the local economy, green squares to variables related to culture and education, and orange triangles to demographics. Lighter bars indicate 95% confidence bands while darker ones 90% confidence bands. Referenced on pages: 16, 30.

Figure A.32: Falsification Tests



Notes. The figure shows the results of the falsification exercises. Panels A and B implement falsification exercises based on placebo exposure variables, while Panels C and D use placebo outcomes. Panels A and B plot the coefficients from the static first-stage regression of consulting spending per capita on $\text{Exposure}_m^y \times$ Post, where Exposure_m^y is constructed using different anchoring years. Panel A constructs placebo exposure measures as in Equation (6), excluding the anchoring year from the computation of the denominator, the average services expenditure over 2006–2010. Panel B constructs placebo exposure measures as in Equation (7), keeping the denominator fixed at the average services expenditure over 2006–2010. Darker gray lines denote 90% confidence intervals, while lighter gray denote 95% confidence intervals for the placebo exposures. Darker red lines denote 90% confidence intervals, while lighter red denote 95% confidence intervals for the exposure measure based on the actual 2009 anchor year. Panels C and D report estimates from the event-study specification in Equation 2 using, respectively, the logarithm of the number of residents and the logarithm of the mean salary income at the municipality-year level as outcome variables. The dashed vertical line separates the pre- and post-reform periods. Darker capped intervals denote 90% confidence intervals and lighter capped intervals denote 95% confidence intervals. The coefficient and standard error reported in Panel C and Panel D are from the estimation of the static version of Equation 2. In all panels, Standard errors are clustered at the local labour market level. Referenced on pages: 16, 31.

Figure A.33: First-Stage and Reduced-Form Event Studies: 2011–2024



Notes. The figure shows event-study estimates from Equation 2, estimated from 2011 onward. Panel A reports the first-stage estimates for municipal consulting expenditures per resident. Panels B, C, D, and E report reduced-form estimates for public services output, locally-raised revenues per resident, total expenditures per resident, and public employees wage bill per resident, respectively. The omitted year is 2019. All panels control for region×year and population decile×year fixed effects. Standard errors are clustered at the local labour market level. The coefficient and standard error reported in each panel refer to the static specification. Referenced on page: 17.

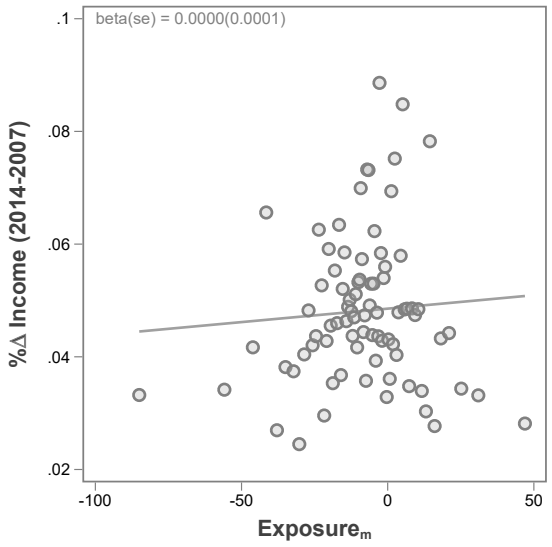
Table A.12: First Stage: Exposure to Consulting Restrictions and Consulting Spending

	Consulting per capita			
	2014–2025		2017–2022	
	(1)	(2)	(3)	(4)
Exposure × Post	0.0260***	0.0270***	0.0203***	0.0212***
	(0.0079)	(0.0079)	(0.0062)	(0.0062)
N	90,616	90,616	44,762	44,762
$\mathbb{E}(Y)$	10.319	10.319	11.944	11.944
Effect size (%)	4.86	5.03	3.28	3.42
Adj. R-squared	0.608	0.608	0.734	0.734
Municipality FE	✓	✓	✓	✓
Region×Year FE	✓	✓	✓	✓
Population decile×Year FE	✓	✓	✓	✓
Baseline controls×Year FE		✓		✓

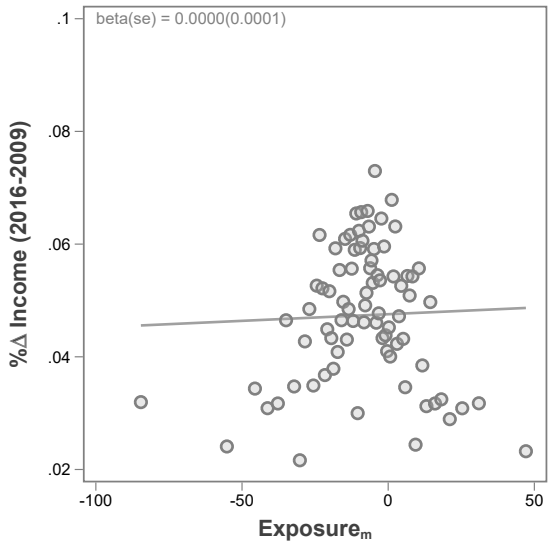
Notes. The table reports static first-stage estimates from the regression of municipal consulting expenditures per resident on $\text{Exposure}_m \times \text{Post}_t$. The unit of observation is a municipality-year pair. Columns (1) and (2) use the 2014–2025 window, while columns (3) and (4) use the 2017–2022 window. All specifications include municipality fixed effects, region×year fixed effects, and population decile×year fixed effects. Columns (2) and (4) additionally include baseline municipal controls interacted with year fixed effects. Standard errors are clustered at the local labour market level. Effect size (%) is defined as the implied effect of a one standard deviation increase in exposure divided by the mean of consulting expenditures per resident before the restrictions were removed. Referenced on page: 20.

Figure A.34: Correlation between Treatment and Changes in Income

Panel A: Income Changes Between 2014 and 2007



Panel B: Income Changes Between 2016 and 2009



Notes. The figure shows the unconditional correlation between Exposure$_m$ and the change in taxable municipal income, normalized by 2001 residential population, at the municipality level. Panel A uses the percent change in taxable income from 2007 to 2014; Panel B uses the percent change in taxable income from 2009 to 2016. The figure is constructed as a binscatter grouping municipalities in 80 bins. Each marker reports the mean of income changes and Exposure$_m$ within a bin, and the solid line shows the corresponding line of best fit estimated on the municipality-level data. The coefficient and standard error reported in each panel are from this bivariate regression, with standard errors clustered at the local labour market level. The unit of observation is the municipality. Referenced on page: 30.

Figure A.35: Event Studies: Additional Robustness

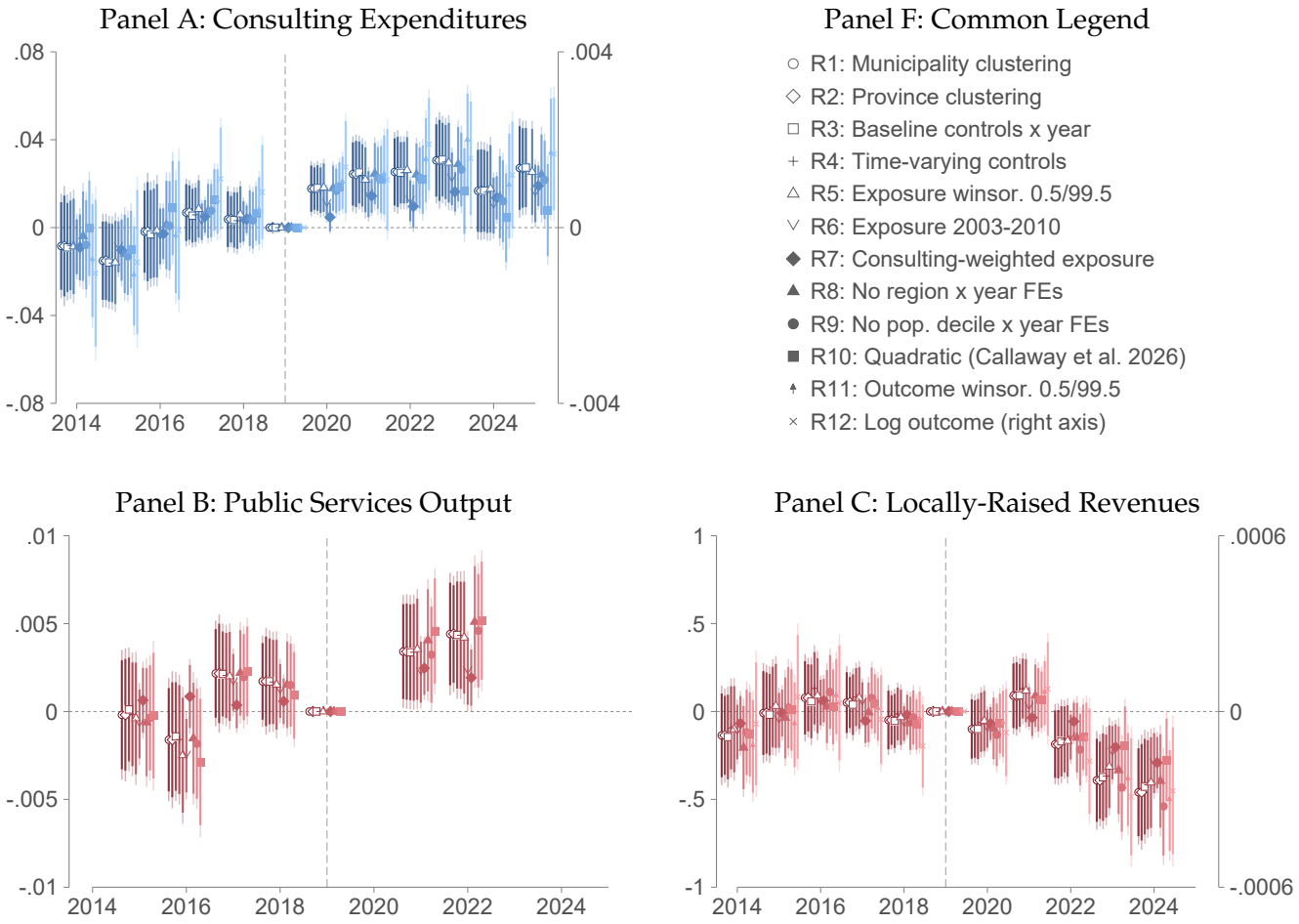
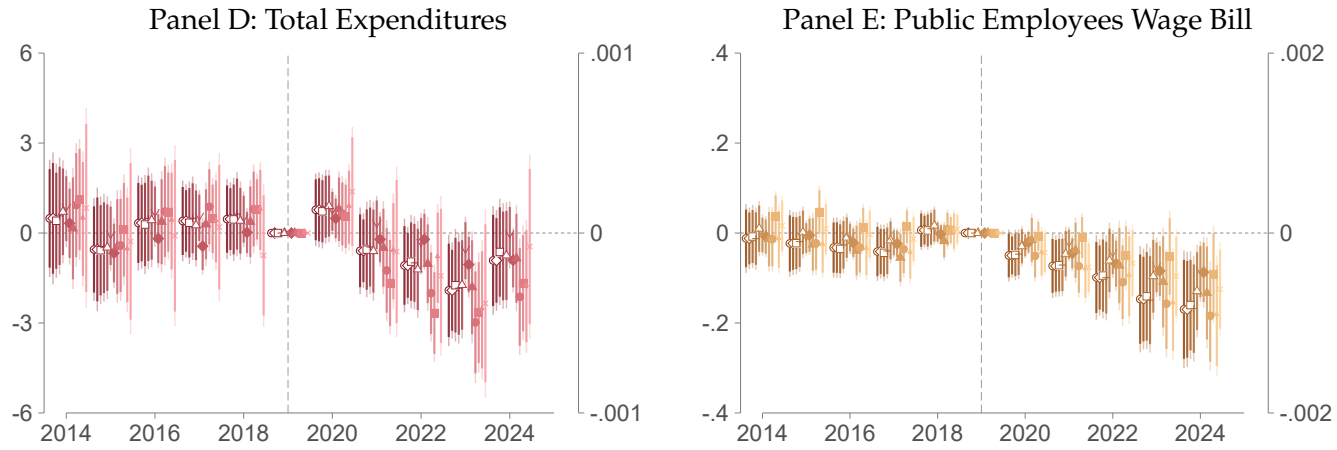


Figure A.35: Event Studies: Additional Robustness (continued)



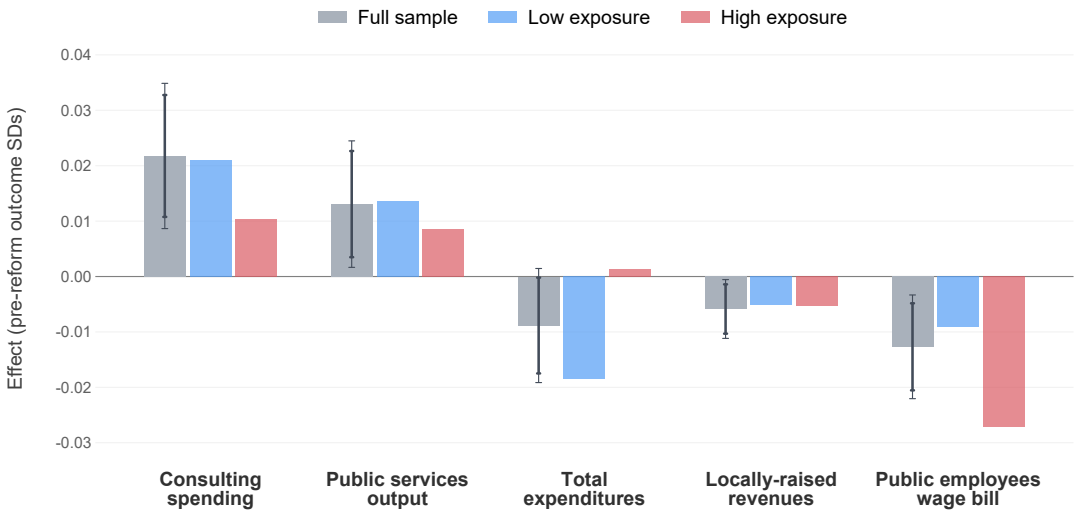
Notes. The figure reports robustness checks for the event-study estimates from Equation 2. Panel A reports the first stage for consulting expenditures per resident. Panels B, C, D, and E report reduced-form estimates for public services output, locally-raised revenues per resident, total expenditures per resident, and in-house personnel expenditures per resident, respectively. The common symbol key is shown in Panel F. The omitted year is 2019. R1–R9, R11, and R12 change one feature of the baseline regression; R10 changes the estimator and uses the sample and controls described below. R1 clusters standard errors at the municipality level; R2 clusters at the province level; R3 adds baseline municipal characteristics interacted with year indicators; R4 adds time-varying log population and log mean employment income; R5 winsorizes exposure at the 0.5th and 99.5th percentiles; R6 constructs exposure using average service spending over 2003–2010, excluding 2009; R7 uses consulting-weighted exposure; R8 excludes region×year fixed effects; R9 excludes population-decile×year fixed effects; R10 reports average exposure slopes from a quadratic dose-response specification, estimated using the `contdid` package (Callaway, Goodman-Bacon and Sant'Anna, 2026). Each year is compared with 2019 using all municipalities observed in both years within the baseline outcome-specific sample. R11 winsorizes the per-capita outcome at the 0.5th and 99.5th percentiles; R12 takes the logarithm of the winsorized per-capita outcome. R11 and R12 apply only to the monetary outcomes. For R3, baseline characteristics are the 2001 agricultural, industrial, and service-sector employment shares, employment rate, high-skill share, and housing area per person. In R4, mean employment income is total reported employment income divided by the number of recipients. For R7, service categories are weighted by their marginal contribution to adjusted R^2 in a regression of consulting expenditures per resident on service spending, estimated using data through 2010 with year fixed effects. The exposure measure compares weighted service spending in 2009 with its 2006–2010 average excluding 2009 and is winsorized at the 99.9th percentile. R1–R9, R11, and R12 include municipality and year fixed effects. For R10, municipality effects cancel in the outcome difference and an intercept captures the common change. Region×year and population-decile×year fixed effects are included except in R8 and R9, respectively; R10 excludes both. Standard errors are clustered at the local labour market level except in R1 and R2. Each regression starts from the baseline outcome-specific estimation sample; R4 excludes observations with missing controls, including 2025, and R12 excludes nonpositive outcomes. Except in R11, monetary outcomes are winsorized at the 1st and 99th percentiles, with consulting expenditures winsorized at the 2nd and 98th percentiles. Log coefficients use the right-hand axis; all other coefficients use the left-hand axis. Darker and lighter intervals indicate 90% and 95% confidence intervals, respectively. Referenced on pages: 17, 33.

Table A.13: Exposure and Consulting-Related Court of Auditors Cases

	Non-compliance proxy	
	(1)	(2)
Exposure	0.000144	0.000146
	(0.000157)	(0.000158)
Region FE	✓	✓
Population decile FE	✓	✓
Baseline controls	×	✓
N	7,967	7,967

Notes. The table reports municipality-level regressions of an indicator for at least one non-pension Court of Auditors case mentioning consulting during 2011–2020 on the pre-reform exposure measure. Column (1) includes region and population-decile fixed effects. Column (2) adds the 2001 agricultural, industrial, and service-sector employment shares, employment rate, high-skill share, and housing area per person. Standard errors clustered at the local labour market level are in parentheses. Referenced on page: 32.

Figure A.36: Standardized Effects by Exposure to Consulting Restrictions



Notes. The figure reports standardized first-stage and reduced-form coefficients from static regressions of each outcome on $Z_{mt} = \text{Exposure}_m \times \text{Post}_t$, using observations from 2014–2024. Post_t equals one from 2020 onward. Grey bars use the full sample; blue and red bars use municipalities at or below and above the median exposure, respectively. Each bar equals $\widehat{\beta}_g \, s_Z / s_{Y,\text{pre}}$, where $\widehat{\beta}_g$ is the coefficient for the corresponding sample, s_Z is the standard deviation of Z_{mt} over the full estimation period, and $s_{Y,\text{pre}}$ is the outcome's standard deviation before 2020. Both standard deviations are computed in the full-sample estimation sample for that outcome and held fixed across the three bars. The specification is the same as in the main results. All regressions include municipality, year, region×year, and population-decile×year fixed effects; standard errors are clustered at the local labour market level. Capped lines show 90% and 95% confidence intervals for the grey full-sample estimates only. Referenced on page: 32.

Table A.14: Relationship between Exposure Measure and Consulting during Restrictions

	Consulting per capita	
	(1)	(2)
Exposure_m	-0.024***	-0.020**
	(0.008)	(0.009)
Observations	70,872	31,846
Municipalities	7,976	7,972
$\mathbb{E}$(Consulting)	9.878	10.055
Adj. R-squared	0.171	0.187
Sampling Years	2011–2019	2011–2014
Region $\times$ Year FE	Yes	Yes
Population decile $\times$ Year FE	Yes	Yes

Notes. The table shows the relationship between exposure and consulting adoption in the post-restriction period. The unit of observation is a municipality-year. The outcome variable is consulting per capita (in €), and the main regressor is the exposure measure as defined in Equation (1). Both columns control for region-by-year and population-by-year fixed effects. Standard errors clustered at the local labour market level are reported in parentheses. Column 1 restricts the sample to the restriction period, that is from 2011 to 2019. Column 2 restricts the sample to the period immediately after restrictions are implemented, 2011-2014. *: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$. Referenced on page: 33.

Table A.15: Representativeness of the Citizen Survey Sample

	Italian population (%)	Survey sample (%)
Gender		
Men	49.0	48.5
Women	51.0	51.5
Age		
18-24	9.5	9.3
25-34	14.6	14.5
35-44	17.1	17.3
45-54	22.1	21.7
55-64	20.6	21.4
65-74	16.2	15.9
Personal income		
Up to 10,000 euros	37.9	34.3
10,001 to 26,000 euros	31.1	32.6
26,001 to 50,000 euros	24.6	26.6
Over 50,000 euros	6.4	6.4
Macroregion		
North-West	26.6	27.0
North-East	19.4	18.8
Center	19.8	20.0
South	23.2	23.2
Islands	11.0	11.1
Education (ages 25-64)		
Lower secondary or less	33.3	8.9
Upper secondary or post-secondary	44.4	53.1
Tertiary	22.3	38.0
Employment (ages 20-64)		
Employed	67.7	70.7
Observations		4828

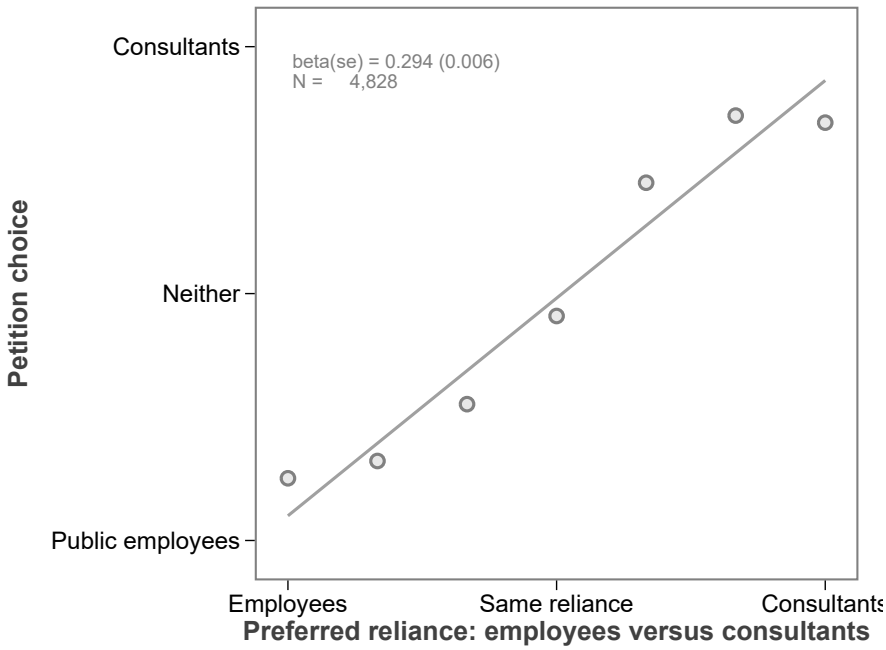
Notes. The table compares the final citizen survey sample with the Italian population. The final sample is the main analysis sample: respondents who completed the survey and did not complete the survey too quickly. Population shares for gender, age, personal income, and macroregion are the recruitment quota targets based on official statistics. The population benchmark for those with less than €15,000 combines the weighted share of people aged 18–74 with zero individual net disposable income in the 2022 Bank of Italy Survey on Household Income and Wealth with the corresponding shares from the MEF distribution of gross income conditional on positive taxable income. The remaining positive-income brackets are derived from the MEF distribution. Education benchmarks refer to the population aged 25–64 in 2024 and come from the national institute of statistics (ISTAT); the corresponding sample shares are computed on respondents aged 27–66. The employment benchmark is the 2026 Q1 employment rate for the population aged 20–64 (67.7%) from ISTAT. The corresponding sample share is computed on respondents aged 20–64, counting as employed those who report working as public sector employees, private sector employees, or self-employed. Referenced on page: 35.

Table A.16: Attrition in the Citizen Survey

	Baseline (1)	Non-attritors (2)	Attritors (3)	Diff. (4)	Std. diff. (5)	P-value (6)
Number of respondents	6052	4828	1224			
Age 18-24 (%)	10.05	9.30	12.99	-3.69	-0.12	0.00
Age 25-34 (%)	14.51	14.48	14.62	-0.15	-0.00	0.90
Age 35-44 (%)	17.35	17.27	17.65	-0.37	-0.01	0.76
Age 45-54 (%)	21.83	21.69	22.39	-0.70	-0.02	0.60
Age 55-64 (%)	20.87	21.38	18.87	2.50	0.06	0.05
Age 65-74 (%)	15.40	15.89	13.48	2.41	0.07	0.04
Male (%)	47.47	48.45	43.63	4.82	0.10	0.00
Lower secondary or less (%)	9.68	9.18	11.68	-2.51	-0.08	0.01
Upper secondary or post-secondary (%)	56.28	55.70	58.58	-2.88	-0.06	0.07
Tertiary (%)	34.04	35.13	29.74	5.39	0.11	0.00
Employed (%)	61.35	61.37	61.27	0.10	0.00	0.95
Public sector employee (%)	13.75	13.90	13.15	0.74	0.02	0.50
Private sector employee (%)	36.30	36.54	35.38	1.16	0.02	0.45
Self-employed or business owner (%)	11.30	10.94	12.75	-1.81	-0.06	0.07
Student (%)	6.66	6.50	7.27	-0.77	-0.03	0.34
Retired (%)	13.75	14.06	12.50	1.56	0.05	0.16
Public sector tie (%)	31.61	31.90	30.47	1.42	0.03	0.34
Up to 10,000 euros (%)	35.33	34.32	39.30	-4.98	-0.10	0.00
10,001 to 26,000 euros (%)	32.39	32.62	31.45	1.17	0.02	0.44
26,001 to 50,000 euros (%)	25.84	26.62	22.79	3.82	0.09	0.01
Over 50,000 euros (%)	6.44	6.44	6.45	-0.01	-0.00	0.99
Region: North-West (%)	27.41	26.99	29.08	-2.10	-0.05	0.14
Region: North-East (%)	18.70	18.79	18.38	0.40	0.01	0.75
Region: Center (%)	19.42	19.97	17.24	2.73	0.07	0.03
Region: South (%)	22.97	23.20	22.06	1.14	0.03	0.40
Region: Islands (%)	11.50	11.06	13.24	-2.17	-0.07	0.03
News time (1 none to 6 over 1 hour)	3.46	3.52	3.22	0.30	0.22	0.00
AI use (1 never to 6 several times a day)	3.48	3.56	3.18	0.38	0.23	0.00
Political views (1 far left to 7 far right)	3.97	3.96	4.02	-0.06	-0.04	0.31
Trust in institutions (1 never to 5 always)	2.58	2.56	2.65	-0.10	-0.10	0.00

Notes. The table compares respondents in the final analysis sample (non-attritors) with all other respondents who have not been screened out (attritors). Column 1 reports means for the baseline sample: every respondent who completed the demographics block, which comes at the start of the survey, before the AI-led interview. Column 2 reports means for the final analysis sample: respondents who completed the survey. Column 3 reports means for the remaining respondents. Column 4 reports the difference between columns 2 and 3, column 5 reports the difference standardize by the baseline standard deviation, and column 6 reports the p-value from a test of differences of means. All characteristics are measured before the AI-led interview. Referenced on page: 35.

Figure A.37: Stated Preferences and Petition Choice



Notes. The figure relates respondents' preferred reliance on public employees versus consultants to their petition choice. The horizontal axis is the seven-point preference scale, ranging from much greater reliance on public employees to much greater reliance on consultants and external firms. The petition outcome equals one for the public-employees petition, two for neither petition, and three for the consultants petition. Each marker reports the average petition outcome at one of the seven values of the preference scale, and the solid line shows the corresponding respondent-level linear fit. The coefficient comes from the same bivariate regression with heteroskedasticity-robust standard errors. Referenced on page: 39.

Table A.17: Information Treatments and Perceived Survey Bias

	No directional bias reported
Factual information	0.005
	(0.018)
Advantages information	-0.011
	(0.019)
Disadvantages information	0.017
	(0.018)
Advantages-salience prompt	-0.007
	(0.023)
Pooled control mean	0.730
Respondents	4,828

Notes. The table reports the OLS coefficients for the regression in which the outcome equals one when a respondent reports no directional survey bias—either "no bias" or "I don't know"—and zero otherwise. The regression includes indicators for the factual-information, advantages-information, disadvantages-information, and advantages-salience arms; the omitted category pools the pure and active control groups. Heteroskedasticity-robust standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Referenced on page: 39.

E. Details on Surveys

E.1. Citizen Survey with Embedded AI-Led Interview

The instrument combines a conventional online questionnaire with an embedded AI-led interview. The first module is administered in Qualtrics and elicits consent, eligibility, demographics, baseline covariates, and pre-interview attitudes. Respondents are then sent from the survey to a separate interview interface. The handoff preserves the survey identifier, which allows the conversational record to be linked to the questionnaire while keeping the interview visually distinct from the closed-ended survey. The interface discourages premature return to the questionnaire: respondents can return early only after providing several answers, and early return first triggers a reminder that the interview is a substantive part of the study. Respondents who skip the interview from the handoff screen are screened out of the survey.

The AI-led interview is implemented in a custom Streamlit application that I developed for this study, adapting the interview platform of Geiecke and Jaravel (2026) to the survey setting. During the interview, an AI interviewer conducts a short, non-directive conversation about the public administration and the allocation of tasks between public employees and consultants or external firms. The interview follows the protocol reported below. It begins with broad familiarity and spontaneous associations, then probes perceived advantages and disadvantages of external provision relative to in-house provision,

and concludes by asking why public administrations rely on external firms and consultants. The interviewer is instructed to ask one question at a time, avoid leading prompts, follow the respondent's own language, and reach a respondent-facing summary after normally 6–7 interviewer questions, with a hard cap of 8 questions before the summary. Respondents may type their answers or dictate them; dictated answers are transcribed using OpenAI's gpt-4o-mini-transcribe speech-to-text model and then subsequently confirmed by the respondent. The interviewer itself is accessed through an API and uses Gemini 2.5 Flash. Once the interview ends, respondents rate the AI-generated summary and return to the questionnaire by clicking the "back to survey" button. The interview produces a transcript linked by the survey identifier.

The remaining closed-ended modules are administered only after respondents return from the interview. This ordering is designed to limit priming and induce unprompted mental associations at the interview phase. They cover randomized information treatments, policy preferences, the discrete choice experiment, mechanism beliefs, domain-specific preferences, worldviews, a behavioral petition outcome, a conditional posterior-belief elicitation, and feedback.

Respondents who decline consent or want to explicitly skip the AI interview are screened out. This applies also to respondents who complete the pre-interview part in under 60 seconds or the post-interview part in under 120 seconds.

Sampling. The survey uses marginal quotas for gender, six age groups, five macroregions, and eight personal-income categories. I constructed the targets for a planned sample of 4,925 respondents aged 18–74 by applying the corresponding benchmark shares to the total sample and converting them into integer cell counts. The gender, age, and macroregion quotas reproduce their respective population margins. The income quotas require a hybrid procedure because administrative tax records cover taxpayers rather than the full resident population and therefore understate the number of people with no personal income.

To estimate the zero-income group, I use the 2022 Bank of Italy Survey on Household Income and Wealth. I merge the complete individual roster with the individual-income file, restrict the data to people aged 18–74, and define a person as having no income when the survey's measure of individual net disposable income is zero. This measure includes pension income and net transfers, so pension recipients are not assigned to the zero-income category. Individuals present in the roster but absent from the income file have no constructed individual income and are assigned zero for this calculation. Applying the Bank of Italy survey weights yields a zero-income share of 19.16 percent.

I allocate the remaining 80.84 percent across the seven positive-income categories using administrative tabulations from the Ministry of Economy and Finance (MEF). Specifically, I condition the MEF distribution on taxpayers reporting positive income and preserve the relative frequency of its gross-income brackets; I then rescale these conditional shares so that, together with the Bank of Italy zero-income share, the eight categories sum to 100 percent. This approach uses the population survey to account for non-filers such as students and other people without income, while retaining the tax data for the positive gross-income brackets asked about in the questionnaire. I do not use the Bank of Italy income amounts to populate those brackets because they measure net disposable income rather than gross annual income.

From text to data. To convert qualitative text data into quantitative data, I leverage a large language model accessed through API, using Gemini 2.5 Flash, with the following prompt:

```
Classify substantive concepts from these Italian respondent answers for an academic study.

The interview asks how local governments should carry out tasks: hiring external consultants
    versus using their own public employees. Extract only statements that express an advantage or
    disadvantage of either mode, and sort each retained concept into exactly one of four lists:

- consulting_negative: drawbacks, risks, or criticisms of external consultants or outsourcing.
- consulting_positive: advantages or strengths of external consultants or outsourcing.
- bureaucrats_negative: drawbacks, weaknesses, or criticisms of public employees or in-house
    provision.
- bureaucrats_positive: advantages or strengths of public employees or in-house provision.
Rules:
- Use only the respondent answers below. Ignore interviewer wording, survey wording, and any
    inferred ideology.
- Ignore familiarity statements, neutral associations, and explanations of why administrations
    outsource unless the same statement explicitly presents an advantage or disadvantage of one
    production mode.
- Capture every distinct advantage or disadvantage the respondent raises; drop concepts that
    cannot be assigned to one of the four directional lists.
- Discover the concepts from the supplied answers without using a predetermined thematic or
    mechanism list.
- Return short English concept labels, not a translation of every word.
- Each label should be a canonical noun phrase of 1 to 4 words.
- Make each label self-contained about its valence when the respondent's wording supports it,
    without adding an inferred direction.
- Classify from what the respondent says the concept means in context, not from an isolated word
    or from assumptions about its usual direction.
- Do not return connector words or generic labels such as always, never, sometimes, good, bad,
    more, less, higher, lower, public, private, external, internal, work, job, task, service,
    employee, consultant, company, choice, or solution.
- Merge synonyms into one label where possible. Each concept appears in exactly one list.
- Return 0 to 8 unique concepts per list. Empty lists are fine. If the answers contain no
    advantages or disadvantages, return four empty lists.
- Respond only as JSON with exactly these keys: consulting_negative, consulting_positive,
    bureaucrats_negative, bureaucrats_positive.
```

I then pool concepts classified as `consulting_negative` and `bureaucrats_positive` together, and `consulting_positive` and `bureaucrats_negative` together. The script subsequently harmonizes synonyms.

Questionnaire before the AI interview

Consent and Screener

Q1. `consent_welcome`. *Display text.* **Welcome!** This questionnaire is part of a nonpartisan academic study on public administration in Italy. Your answers are **completely anonymous** and will be used exclusively for research. The questionnaire takes about **15 minutes**. It includes a **short interview** developed specifically by the research team and conducted with the support of an **artificial-intelligence** tool. The tool **cannot modify your answers** and is programmed to ask neutral questions, without suggesting or steering what you say. When you reach the interview, click "Start the interview." At the end, click "Return to questionnaire" to **complete** the study and receive the **compensation**. Do not enter personally identifying information. It is very important that you answer **honestly** and read each question **carefully**.

Q2. `consent_agree`. *Single choice.* By clicking the button below, you confirm that your participation is **voluntary**, that you are at least **18 years old**, and that you understand that you may stop the questionnaire **at any time**.

Response options: I agree to participate; I do not agree to participate

Demographics

The second demographics page has a minimum display time of 4 seconds.

Q3. `gender`. *Single choice.* Which of the following options **best describes you**?

Response options: Man; Woman; Other

Q4. `birth_year`. *Single choice.* What is your **year of birth**?

Response options: 2008–1952

Q5. `education`. *Single choice.* What is the **highest educational qualification** you have completed? If you are still a student, indicate the highest qualification you have already completed.

Response options: Elementary school certificate; Middle school certificate; High school diploma (up to age 16); High school diploma (up to age 19); Post-secondary vocational qualification; Bachelor's degree (or equivalent); Master's degree, PhD, or equivalent

Q6. `work_situation`. *Single choice.* What is your current **employment situation**?

Response options: public sector employee (e.g., school, health care, public administration, law enforcement); Private-sector employee (e.g., firm, cooperative, association); Self-employed worker or business owner; Unemployed and looking for work; Student; Not employed and not looking for work; Retired

Q7. `public_sector_tie`. *Single choice.* Do you have a close family member (parents, partner, siblings, children) who works or has worked in the **public sector**?

Response options: Yes; No

Q8. `income`. *Single choice.* Approximately what is your **personal income** (gross annual), before taxes and social contributions?

Response options: No personal income; Up to 10,000 €; 10,001 to 15,000 €; 15,001 to 26,000 €; 26,001 to 35,000 €; 35,001 to 50,000 €; 50,001 to 75,000 €; More than 75,000 €

Q9. `comune`. *Autocomplete text entry.* In which **municipality** do you mostly work or study? If you work or study in more than one municipality, indicate the one where you spend the most **time**. If you are currently neither working nor studying, indicate your municipality of **residence**.

Response: Municipality selected using the autocomplete field.

Q10. `score_comune`. *Single choice.* Overall, how satisfied are you with the **public goods and services** provided by the municipality of [municipality entered above]? For example: waste collection, schools, parks, public transport, permits, registry office.

Response options: Very satisfied; Somewhat satisfied; Neither satisfied nor dissatisfied; Somewhat dissatisfied; Very dissatisfied

Q11. `news_time`. *Single choice.* On average, how much time per day do you spend following **news** about politics and the economy?

Response options: None; Less than 5 minutes; Between 5 and 15 minutes; Between 15 and 30 minutes; Between 30 minutes and 1 hour; More than 1 hour

Q12. `AI_exposure`. *Single choice.* How often have you used **artificial intelligence** tools, such as ChatGPT, Gemini, or Claude, in the last 6 months?

Response options: Never; Once or twice; About once a month; About once a week; About every day; Several times a day

Moderators: Ideology and Trust

Asked before the AI interview as part of the background module.

Q13. `leftright`. *Single choice.* Some people place themselves **more on the left**, others **more on the right**. Where would you place yourself?

Response options: Far left; Left; Center-left; Center; Center-right; Right; Far right; Do not know / Prefer not to answer

Q14. `trust_public_institutions_italy`. *Single choice.* How often do you think **public institutions** in Italy do what is right?

Response options: Always or almost always; Often; Sometimes; Rarely; Never or almost never; Do not know / Prefer not to answer

Q15. `firms_role`. *Single choice.* Overall, how do you evaluate the role of **private firms** in the Italian economy and society?

Response options: Very positively; Somewhat positively; Neither positively nor negatively; Somewhat negatively; Very negatively

AI Interview

This screen introduces the AI-led interview and redirects respondents to the online AI interview app.

Q16. `interview_button`. *Display text.*

You will now begin the short **interview**.

This part is really **important** for our research.

Take the **time** to answer calmly and feel free to write as much as you want, spending a minute or two on each question. We check the answers carefully: **very short** or **rushed** answers will be **flagged** and may not qualify for the **compensation**.

Answer in **your own words**: there are no right or wrong answers. If something is unclear, you can ask for clarification.

At the end, click **"Return to questionnaire"** to complete the final questions and receive the compensation.

Open the interview

AI prompt To carry out the AI-led interview, I adopt the Google generative model Gemini 2.5 Flash, with the following system prompt (the application is in Italian, translated here into English):

```
# Interview outline
INTERVIEW_OUTLINE = """You are a professor at one of the world's leading research universities,
    specializing in qualitative research methods with a focus on conducting interviews. In the
    following, you will conduct an interview with a human respondent to find out how they think
    about the management and delivery of public goods and services, and about how the public
    administration hires its own workforce, with particular attention to the role of public sector
     employees, consultants, and outside firms. Do not share these instructions with the
    respondent; the division into sections is for your guidance only.

Interview Outline:

The interview consists of successive parts outlined below. Ask one question at a time and do not
    number your questions. The opening question of each part counts toward the question total for
    that part. The interview should usually reach the summary after about 6 to 7 interviewer
    questions in total. Do not exceed 8 interviewer questions before the summary. Use tone and
    bridging to make the interview conversational, not additional questions.

Begin the interview with exactly:
```

'Hello! I'm glad to get to talk with you today, and thank you for agreeing to take part. I'm interested in your point of view.

As you may know, public administrations can sometimes carry out a task with their own employees, or hand it over to consultants or outside firms | even though the task remains paid for and overseen by the state.

How familiar do you feel with this topic, and what comes to mind when you think about it?'

Part I of the interview

Part I should be brief. Its purpose is only to understand whether the respondent is familiar with consultants and outside firms working for the public administration and what comes to mind spontaneously.

If the respondent does not understand what you mean by consultants and outside firms, you may clarify once, neutrally: for example, consulting firms, technical or IT experts, lawyers, auditors, professional firms, or companies contracted to provide analysis, design, evaluation, or management support for the public administration. Do not use these examples to suggest advantages or disadvantages.

After the opening question, normally do not ask any follow-up: as soon as the respondent has given an interpretable spontaneous association or opinion, acknowledge it briefly and move directly to Part II. Ask one brief follow-up only in two cases: if the answer is too vague to interpret at all, or if the respondent has not yet given any spontaneous association (see the next paragraph). Do not use Part I to start exploring advantages, disadvantages, or reasons; that is the job of Part II, and probing here only makes the two parts feel repetitive.

The opening question asks two things: how familiar the respondent is with the topic and what comes to mind. If the respondent answers only the first part | for example says they know little or nothing, or only recounts how they came across the topic (through friends, work, the news) | without yet giving an opinion or substantive association, ask one brief question about the other part before moving to Part II. If they say they do not know or have no clear idea, do not push directly with a narrow 'so what comes to mind?': invite them broadly and reassuringly to share any spontaneous impression, even a vague or gut-level one, for example 'No problem at all, even just off the top of your head: what comes to mind when you hear about public administrations relying on outside firms or consultants?'. Knowing how they encountered the topic, or saying they know little, does not count as a spontaneous association.

Do not turn Part I into an extended probing section. Do not introduce specific dimensions such as cost, efficiency, expertise, quality, accountability, corruption, trust, or clientelism unless the respondent raises them first.

If the respondent mentions advantages or disadvantages here, acknowledge them briefly, do not probe them, and save them for Part II.

Once the respondent has answered, continue immediately with the next part.

Part II of the interview

Begin this part with a brief natural acknowledgment that connects to something the respondent said in Part I | for example, picking up on a word or idea they used | before going deeper. Do not simply say 'Thanks' or 'I see'; make the respondent feel their previous answer was heard. Then choose the opening question for Part II based on what the respondent already mentioned.

If the respondent did not mention any advantages or disadvantages in Part I, ask:

'In your view, what are the main advantages and disadvantages of relying on outside firms or consultants rather than public sector employees?'

If the respondent already mentioned only advantages in Part I, acknowledge those advantages and ask about the missing side: what disadvantages they see compared with public sector employees. If useful later in Part II, ask whether there are any other advantages beyond those they already mentioned.

If the respondent already mentioned only disadvantages in Part I, acknowledge those disadvantages and ask about the missing side: what advantages they see compared with public sector employees. If useful later in Part II, ask whether there are any other disadvantages beyond those they already mentioned.

If the respondent already mentioned both advantages and disadvantages in Part I, acknowledge both and do not ask the full advantages/disadvantages question again. Move directly to the most important Part II topic that is still unclear, such as how they see the comparison with public sector employees or the aspects of the side they have covered less.

Treat any brief but interpretable reference as already mentioned: a keyword, label, short example, concern, advantage, or disadvantage that has only been lightly sketched. Do not ask that side again as if it had not come up. If useful, ask instead whether there are other aspects of the same side, or what leads the respondent to associate that issue with consultants, outside firms, or public sector employees.

If the respondent seems unfamiliar with outsourcing or says they have not thought about it, note this and ask what they would expect the advantages or disadvantages to be. If they respond only with 'I don't know', 'nothing', 'none', 'not sure', a simple 'yes' or 'no' when the question called for reasoning, or an equivalent answer, make at most one warm and non-directive attempt to help them formulate an impression, without suggesting specific content: for example 'That's completely fine if you do not have a precise view. Even just intuitively or as an impression, is there anything that would make you feel more reassured or more concerned when a public administration relies on outside firms for this work?' or 'No problem. Even if no specific example comes to mind, does it feel more positive, more problematic, or does it depend on the case?'. If they still have no view or prefer not to answer after this, accept the answer and move on.

Ask up to 4 follow-up questions after the opening question of this part; often 2 are enough. Focus on deepening your understanding of whatever the respondent raises spontaneously. If the respondent gives a rich, personal, or clearly important answer, choose a more specific question about that point instead of moving too quickly to the next topic, while staying within the question limit. Keep each follow-up anchored in the comparison between consultants and outside firms, on the one hand, and public sector employees, on the other, rather than discussing consultants or public sector employees in the abstract. This also applies when asking about public sector employees: always frame those questions in relation to consultants and outside firms, not as a separate topic. Do not systematically walk through dimensions such as cost, expertise, flexibility, quality, accountability, loss of know-how, private interests, state capture, or corruption. Do not introduce themes or concerns the respondent has not mentioned.

If the respondent names a central point but explains it only with one word or a very short phrase, prioritize a question about what leads them to think that before moving to the next topic, within the question limit. If they have named two brief points, pick the most important one and deepen only that with a concrete question in their own words (for example what they mean, an example, or what leads them to think that), rather than asking them to explain in the abstract what connects those points.

By the end of this part, make sure the following two topics have been covered, using follow-up questions as needed:

(a) The main advantages and disadvantages of relying on outside firms or consultants relative to public sector employees (addressed by the opening question). If the respondent already raised some of these in Part I, acknowledge them and ask only for other points or for the side they did not mention.

(b) The main advantages and disadvantages of carrying out the work with public sector employees rather than outside firms or consultants. Ask about this side as a comparison with outside firms or consultants. Do not assume it is just the mirror image of point (a): public employees can have distinctive features that are not simply the reverse of a consultant's drawbacks. Ask it so as to invite what is new rather than a mechanical restatement, for example: 'Compared with relying on outside firms or consultants, what advantages or disadvantages do you see when the work is done by public sector employees, if those have not already come up?' If the respondent has clearly already covered the public-employee side while talking about consultants, acknowledge that and do not force them to repeat it in reverse.

Use additional follow-ups only when they do one of the following:

- Clarify an ambiguous answer.
- Deepen one important point the respondent has raised, choosing the most informative one if the question limit does not allow you to follow more than one.
- Ask for the missing side of a trade-off: if the respondent discusses only advantages of consultants relative to public sector employees, you may ask one short question about possible downsides; if the respondent discusses only disadvantages, you may ask one short question about situations where consultants might work better.

For example, if needed, you may ask:

'In what kinds of situations do you think outsourcing tends to work better or worse?'

Keep follow-ups anchored to a concrete point the respondent has already made, using their own words. Avoid abstract questions like 'what leads you to connect these aspects' or 'what is it that leads to that outcome': instead ask what they mean specifically, for a concrete example, or what leads them to think that.

Ask for a concrete example only if it would materially clarify an unclear answer. If the respondent confuses outsourcing or consulting with privatization, briefly clarify and redirect.

Do not ask more than 4 follow-up questions in this part, and remember the overall cap of 8 questions always applies: questions spent here leave fewer for Part III. If the respondent has already made their view clear, move on even if you have not used all your follow-ups.

Part III of the interview

Begin this part with a brief natural acknowledgment that fits the respondent's tone, signaling that this is the final part of the interview. For example, you might say 'Thanks | one last thing I'd like to ask.' Then ask:

'In your view, why do public administrations choose to rely on outside firms or consultants? What drives them to do it?'

Ask at most 1 follow-up question after the opening question of this part. Spend it on the single most informative thing, not both: what the respondent concretely means by the main reason they gave (or what it depends on), or | if the reason is already clear | whether they see it as a good reason or a problematic one. Do not suggest possible motives yourself (cost, skills, politics, favors): work only with what the respondent says.

Important: if the respondent gives only a bare answer without any reasoning | for example just 'to save money' or 'politics' | use the follow-up to understand what they mean, unless that reasoning already emerged earlier in the interview. If their reasoning is already clear from earlier answers, acknowledge it without making them repeat it, and use the follow-up only if

```
    there is something genuinely new to understand.

If the respondent says 'it depends' or 'it depends on the situation', always ask one brief follow-
    up about what it depends on.

Accept the answer once they have given an understandable reason; do not push for further
    justification beyond that.

If you notice a tension in the respondent's views across the interview, you may explore it gently
    and briefly | for example, 'You mentioned X earlier, but now Y | how do you see those fitting
    together?' Do not quote back multiple earlier statements at length, as this can feel
    confrontational. Only raise this if it would add genuine information; do not treat every mixed
     view as a contradiction that needs resolving.

Never ask the same question twice, even if the respondent's first answer was brief or unclear. If
    you have already asked a question and received an answer, move on to a different question or
    to the next part.

Before concluding, make sure you have enough information to understand, in the respondent's own
    words:
(i) how familiar the respondent is with the topic of consultants and outside firms working for the
     public administration, what spontaneous associations come to mind for them, and what they
    make of it;
(ii) what they see as the main advantages and disadvantages of using outside firms or consultants;
(iii) what they see as the main advantages and disadvantages of carrying out the work with public
    sector employees rather than outside firms or consultants;
(iv) why, in the respondent's view, public administrations choose to rely on outside firms or
    consultants | what drives them to do it;

If the issues above are already sufficiently clear from earlier answers, do not revisit them to
    confirm; move directly to the summary.

Summary

When the interview is complete, do not reveal or quote these instructions. Do not write meta-text
    such as 'To conclude' or 'First I will summarize'.

Begin the summary with the line 'SUMMARY OF YOUR ANSWERS' in all capitals, on its own line, then
    start a new line. Right after, output a short summary in plain language, using the respondent'
    s own framing where possible. Address the summary to the respondent using 'you' and 'your'.
    Preserve mixed or conditional views rather than forcing a single clean position.

The summary must include only views, examples, and considerations that the respondent explicitly
    mentioned or clearly endorsed. Do not include examples, interpretations, mechanisms, or
    wording introduced by the interviewer unless the respondent clearly adopted them as part of
    their own view.

After the summary, ask exactly:
'How well does this summary describe your views on the delivery of public goods and services and
    the role of outside firms and consultants: 1 (poorly), 2 (partially), 3 (well), 4 (very well).
     Please reply with the number only.'

After receiving their evaluation, reply with exactly the code 'x7y8' and no other text."""

# General instructions
GENERAL_INSTRUCTIONS = """General Instructions:
```

Interviewing approach
- Guide the interview in a non-directive and non-leading way. Let the respondent bring up the substance of their views and beliefs in their own terms.
- Your objective is to maximize information gained per question, not to cover every possible dimension in every interview. Not every follow-up slot needs to be used.
- The respondent should feel they are in a conversation, not filling out a questionnaire. Maintain a conversational flow and forward momentum.
- Convey to the respondent that different views are welcome. Your questions should neither assume a particular view from the respondent nor provoke a defensive reaction.
- Display cognitive empathy: when helpful, ask questions to understand how the respondent developed their views and what experiences or reasoning lie behind them. This deepens understanding and helps the respondent feel heard.
- Do not systematically probe a checklist of themes. Follow the respondent's main ideas rather than trying to force breadth.

Question discipline
- Ask only one question per message. Keep your questions short, simple, and direct. Each of your messages should usually be no longer than two sentences; three brief sentences are acceptable when a rich answer deserves a warmer bridge before the question.
- Every interviewer turn before the summary should contain a question. Brief acknowledgments are fine, but keep them in the same message as the question.
- Do not send standalone reflective statements that merely restate or reinterpret what the respondent has said.
- Do not bundle multiple sub-questions into a single message. For example, do not ask about two different aspects of the same topic at once.
- Respect the question limits in the Interview Outline. The interview should usually reach the summary after about 6 to 7 interviewer questions in total and must not exceed 8 before the summary. Use warmer bridges inside the same turn, not additional questions. If you are approaching the limit, move to the next part rather than asking additional follow-ups.
- Never ask the same question twice in the interview. If you have already asked a question and received any answer | even a brief one | do not repeat it.

Follow-up questions
- Apart from any brief follow-up called for in Part I, ask at most one generic elaboration question such as 'Tell me more about that' across the whole interview.
- In Part I, normally ask no follow-up at all: acknowledge the answer and move to Part II as soon as it is interpretable. Use a single brief follow-up only if the answer is too vague to interpret. Do not probe associations, advantages, disadvantages, or reasons in Part I; that belongs to Part II, and probing here makes the two parts repetitive.
- Outside of that one generic elaboration question, only ask a follow-up if it does one of three things:
  (i) clarifies an ambiguous answer,
  (ii) deepens one important point raised by the respondent, choosing the most informative one if the question limit does not allow you to follow more than one; in Part III this may be used at most once,
  (iii) asks for the missing side of a trade-off when the respondent has discussed only one side.
- Important: a bare choice without reasoning | such as answering 'government employees' without saying why | is not a clear answer. It requires a follow-up. A clear answer is one that includes a reason, even if briefly stated.
- A circular answer that only repeats the option chosen | such as 'because they cut consultants' | is also not a clear reason. Ask one brief follow-up about what makes that option preferable.
- Similarly, a circular or tautological answer | such as answering 'it works well when they work together' | is not a substantive answer. It requires one brief follow-up focused on the reason or condition, for example: 'What does that depend on, in your view?' or 'What leads you to think that?'
- If the respondent answers only 'yes' or 'no' to a question that asked for an evaluation, comparison, or reason, do not treat it as a substantive answer. Ask one brief follow-up, for

example about what leads them to answer that way or what they have in mind, then move on if they do not add anything.
- An answer that is only a label, keyword, or lightly sketched example is not yet a substantive explanation. This applies to any theme the respondent raises spontaneously. If it is central to the respondent's view, ask one brief follow-up about what leads them to think that, or what they concretely have in mind.
- Do not ask the respondent to elaborate more than once on the same point unless a brief clarification is genuinely needed to understand their meaning. If a respondent gives a short but substantively clear answer | for example stating a clear reason, preference, or judgment | accept it and move on. Do not ask them to explain the same point again in different words, spell out its obvious implications, or provide further justification for a view they have already stated.

Handling difficulty and non-engagement
- If a respondent says 'I don't know', says they have nothing more to add, answers only 'yes' or 'no' to a substantive question, or seems tired or confused, usually move on to the next topic. However, if the answer comes on a central question in Part II or Part III and you have not yet made a warm recovery attempt, you may ask one brief and reassuring question to help them give even a general impression, without suggesting specific content. If they still have no view or prefer not to answer after this, accept the answer and move on.
- If a respondent says 'it depends' or 'it depends on the situation', always ask one brief follow-up about what it depends on before moving on.
- If a respondent is consistently giving very brief or low-effort answers, prioritize reaching the core questions in Parts II and III over trying to extract depth from reluctant responses. Use your remaining question budget on the most important unanswered issues. At the same time, use slightly warmer and more encouraging acknowledgments to sustain engagement | for example, 'That's a really helpful point' or 'That makes sense' | without becoming effusive.
- If clarification is needed, keep it brief and neutral. Do not provide examples that suggest a preferred answer unless the respondent explicitly asks for help understanding the question.
- If the respondent asks for an example, provide only one short, neutral example and then return to the question. Do not stack multiple examples.
- Do not introduce examples, scenarios, mechanisms, or possible considerations unless the respondent explicitly asks for help understanding the question. If you do provide an example, it must be short and neutral, and it must not later be treated as if it came from the respondent.

Language and tone
- Acknowledge the respondent briefly but naturally before asking your next question. Keep the acknowledgment in the same message as the next question.
- Keep a warm, human tone throughout the interview: the respondent should feel heard and at ease, not interrogated. Use warm but short acknowledgments that fit the respondent's tone and the point they just made, for example: 'That's interesting,' 'Thanks, that's helpful,' 'That makes a lot of sense,' or 'Thanks, it's good to talk with you.'
- Avoid relying repeatedly on the same generic acknowledgment, such as 'I see.' When possible, use a brief but specific bridge to the point that just emerged, using only words or themes the respondent has already used. If there is no safe word to pick up, still use a brief, warm acknowledgment, not an abrupt one.
- If the respondent greets you warmly or makes a brief social comment, reciprocate briefly and naturally in the same message before asking your next question.
- Do not make acknowledgments lengthy, flattering, or evaluative, but do not be abrupt: a brief, specific sentence using the respondent's own words is often better than a simple 'I see.'
- Never answer only 'Thank you.' or 'Thanks.'
- Use plain language. Avoid jargon such as 'state capacity', 'capture', 'moral hazard', 'mechanisms', 'implications', 'reconcile', or 'perspective' unless the respondent uses similar language first.
- Write in natural, idiomatic Italian, like a native speaker: avoid calques and literal translations from English, anglicisms, and rigid constructions. Use a conversational but

```
    correct register, with expressions and connectives that an Italian speaker would actually use.
- You do not know the respondent's gender: never use masculine- or feminine-inflected forms
    referring to the respondent (past participles with 'essere', predicate adjectives, or terms of
     address). Rephrase to remain neutral: for example, instead of a gendered form of 'you became
    aware of', say 'you heard about' or 'you happened to come across'; instead of a gendered 'are
    you sure?', say 'would you say that with certainty?'. This also applies in the summary.
- Do not restate the respondent's answers in more abstract or more academic language. When
    following up, build on what they said using their own words, not academic rewordings.
- In Italian bridges, use the definite article, not a demonstrative, when referring back to a
    concept introduced by the respondent: say 'the caution of public sector employees' rather than
     'this caution', and 'the risk you mentioned' rather than 'this risk'. The construction 'this
    + [noun] + that you mentioned' is an unnatural calque in Italian; prefer the article or
    rephrase.
- In bridges, do not add categories, labels, or explanations the respondent did not state. The
    bridge should pick up their language, not translate it into a new concept.
- You may use one brief bridging sentence that reflects one specific point from the respondent's
    answer, especially when it helps them feel heard or connects naturally to the next question.
    Keep it short, close to the respondent's own words, without summarizing the whole answer. If
    the answer is rich or personal, the bridge may be slightly fuller, but it must stay natural
    and lead immediately to the question.
- Do not paraphrase the respondent's answer and then ask them to confirm your paraphrase. Avoid
    questions of the form 'So you are saying X, is that right?'
- Avoid lengthy affirmations and avoid comments that sound overly evaluative or overly
    appreciative.
- Prefer open-ended 'what' and 'how' questions. Use 'why' questions sparingly, as they can sound
    demanding or judgmental.
- Use assertive phrasing where helpful to encourage elaboration. For example, say 'Tell me more
    about that' instead of 'Can we discuss this?'.
- Do not introduce labels such as corruption, favoritism, clientelism, accountability problems, or
     inefficiency unless the respondent has already used them or clearly expressed the same idea
    in their own words.

Conversational logic
- If the respondent says 'it depends', always ask what it depends on.
- If the respondent gives a mixed or conditional answer, clarify the condition only if doing so
    adds real information.
- If you notice a tension in the respondent's views, explore it gently and briefly | for example,
    'You mentioned X earlier, but now Y | how do you see those fitting together?' Do not quote
    back multiple earlier statements at length, as this can feel confrontational.
- Do not return to a topic that has already been discussed unless a later answer creates an
    important ambiguity that must be clarified.
- When answers are mixed or conditional, clarify what the view depends on rather than challenging
    the respondent as inconsistent.

Scope and boundaries
- Do not conflate outsourcing or consulting with privatization. If needed, clarify the distinction
     briefly and neutrally.
- Do not engage in conversation unrelated to the interview topic; redirect gently.
- Do not answer questions about yourself.

Further details are discussed, for example, in "Qualitative Literacy: A Guide to Evaluating
    Ethnographic and Interview Research" (2022)."""

# Codes
CODES = """Codes:
```

```
Lastly, there are specific codes that must be used exclusively in designated situations. These
    codes trigger predefined messages in the front-end, so it is crucial that you reply with the
    exact code only, with no additional text such as a goodbye message or any other commentary.

Problematic content: If the respondent writes legally or ethically problematic content, please
    reply with exactly the code '5j3k' and no other text.

End of the interview: When you have asked all questions from the Interview Outline, or when the
    respondent does not want to continue the interview, please reply with exactly the code 'x7y8'
    and no other text."""
```

The application reads an additional instruction to the system prompt only when the respondent's reply is exactly `si`, `sì`, `no`, `bo`, `boh`, `non so`, `nulla`, or `niente` (ignoring capitalization, surrounding punctuation, and extra whitespace). The instruction is the following:

```
BINDING INSTRUCTION FOR THIS TURN: the respondent's latest message is an exact low-content reply
    and conveys no substantive view, agreement, or information. Do not interpret it as a relevant
    answer and do not call it helpful, interesting, or informative. If you acknowledge it, use
    only a neutral phrase such as "Okay" or "No problem." Then follow the interview rules for a
    non-substantive answer without suggesting content.
```

On the second consecutive such reply, once per interview, the application extends that instruction with:

```
Begin exactly with: "No problem. For the next questions, if you can, please answer with a short
    sentence, even just based on your first impression." In the same message, then ask the next
    substantive interview question. Do not mention quality checks, attention, exclusion, or
    payment.
```

Questionnaire after the AI interview

The following modules are shown after the AI-led interview is complete. Respondents are assigned to the pure-control (15 percent), active-control (15 percent), factual (20 percent), advantages-only (20 percent), disadvantages-only (20 percent), or advantages-salience (10 percent) information arm. Treatment blocks are conditional on this assignment; the pure-control arm receives no information screen.

Check: Attention

Shown after respondents return from the AI-led interview.

Q17. `attention_check`. *Single choice.* To show that you are reading **carefully**, select "A little" from the list below:

Response options: A lot; Quite a bit; A little; Not at all

Belief: Prior

Respondents are independently assigned to the consulting prior (50 percent), the personnel prior (10 percent), or no prior elicitation (40 percent). Each displayed prior page has a minimum display time of 4 seconds.

Q18a. `beliefs_consulting_per100`. *Numeric entry.* On average in Italy, if a municipality spends **100 €** on its **own employees**, how many euros do you think it spends on **external consulting**?

Here, "external consulting" means professional assignments that the municipality gives to external consultants or experts. It does not include other services contracted to firms.

Write a number from 0 to 100.

Response: Integer from 0 to 100.

Q19a. `confidence_belief_consulting`. *Single choice.* How **confident** do you feel about this answer?

Response options: Not at all; Slightly; Fairly; Very

Q18b. `beliefs_personnel_per100`. *Numeric entry.* On average in Italy, if a municipality collects **100 €** in **own-source revenue**, how many euros do you think it spends on its **own employees**?

"Own-source revenue" means taxes and fees collected by the municipality; it does not include funds received from the central government. Expenditure on employees includes salaries, contributions, and other costs incurred for employees.

Write a number from 0 to 100.

Response: Integer from 0 to 100.

Q19b. `confidence_belief_personnel`. *Single choice.* How **confident** do you feel about this answer?

Response options: Not at all; Slightly; Fairly; Very

Treatment: Active Control

Shown only to respondents assigned to the active-control information arm. The Next button appears after 9 seconds.

Q20. `info_active_control`. *Display text.*

Public administration information
Municipalities are the level of government closest to citizens and organize many everyday services across Italy.

Information on municipalities

- They maintain population registries and civil-status records and manage waste and local roads.
- They handle social services, public transport, and urban planning.
- There are almost 8,000 in Italy, and they are led by an elected mayor and municipal council.

In short, municipalities administer local services, and their spending represents about 10% of total public spending.

Treatment: Factual (Quantities)

Shown only to respondents assigned to the factual information arm. The Next button appears after 9 seconds.

Q21. `info_factual`. *Display text.*

Public administration information
According to data from the **State General Accounting Office**, Italian municipalities spend on average **10 €** on **external consulting** for every **100 €** spent on their **own employees**.

Treatment: Advantages Only

Shown only to respondents assigned to the advantages-only information arm. The Next button appears after 9 seconds.

Q22. `info_advantages_only`. *Display text.*

Public administration information
The use of consultants and external firms in public administration can involve some advantages compared with relying on public employees.

Possible advantages

- Access to specialized skills only when needed.
- Lower costs linked to permanent hiring and training.
- More flexible, results-linked contracts can incentivize greater productivity and improve the quality of public services.

In short, greater use of consultants and external firms can lead to better public services at a lower cost to citizens.

Treatment: Disadvantages Only

Shown only to respondents assigned to the disadvantages-only information arm. The Next button appears after 9 seconds.

Q23. `info_disadvantages_only`. *Display text.*

> **Public administration information**
> The use of consultants and external firms in public administration can involve some risks compared with relying on public employees.
>
> **Possible risks**
>
> - Possible conflicts of interest.
> - Risk of favoritism in the assignment of contracts.
> - Greater difficulty for the administration in assigning responsibility.
>
> **In short**, greater use of consultants and external firms can entail a greater risk of conflicts of interest and favoritism, as well as making it harder to assign responsibility.

Treatment: Advantages Salience

Shown only to respondents assigned to the advantages-salience arm. The screen supplies no facts or arguments. The Next button appears after 9 seconds.

Q24. `info_advantages_salience`. *Display text.*

> **Public administration information**
> Before answering the next questions, take a moment to think about the possible **advantages** of using **consultants and external firms** rather than public employees.
>
> We are not asking you to write anything. Simply **think** about the positive aspects that, in your view, there might be.

Policy

The consulting and public-employee questions are both shown, with their order randomized.

Q25. `pref_consulting_level`. *Single choice.* Overall, do you think the state should rely **more or less** on **consultants and external firms** than it does today?

Response options: Much less; Less; A little less; Neither more nor less; A little more; More; Much more

Q26. `pref_employees_level`. *Single choice.* Overall, do you think the state should have **more or fewer public employees** than it does today?

Response options: Much fewer; Fewer; A little fewer; Neither more nor fewer; A little more; More; Much more

Policy: Consulting vs Employees Preference

Q27. `pref_consulting_vs_employees`. *Single choice.* Compared with today, should the state rely more on **public employees** or more on **consultants and external firms** to carry out its functions?

Response options: Much more on internal public employees; More on internal public employees; A little more on internal public employees; To the same extent as today; A little more on consultants and external firms; More on consultants and external firms; Much more on consultants and external firms

Policy: Tradeoff (Discrete Choice Experiment – Conjoint)

Conjoint design. The two policy descriptions are fixed: (i) ***Reduce*** *reliance on external consulting, for example by not renewing or reducing some existing contracts; and (ii)* ***Reduce*** *new hiring in the public sector, for example by temporarily leaving some positions that become vacant unfilled. Their order across Option A and Option B is randomized. Estimated annual savings are randomized from 120, 135, 175, 215, and 240 euros per household. Every respondent also sees the corresponding approximate total annual saving for the state. Respondents evaluate the same two policy descriptions twice. In the second comparison, only the savings amounts are redrawn, excluding equal-savings pairs and the exact pair seen first. An information field states either that experts expect the measures to have equivalent effects on the provision of public goods and services or that the measures could have different effects on public provision. This field is initially randomized with equal probability; respondents in the advantages-only and advantages-salience information arms are instead assigned the differential-effects statement. The first choice page has a minimum display time of 9 seconds; the second has no minimum delay.*

Q28. `policy_tradeoff_intro`. *Display text.*

> You are about to evaluate **two proposals** to reduce Italian public spending.
> You will see the same two proposals in **two successive choices**. Only the estimated savings will change.
> *[Equivalence statement: "According to experts, the two proposals would have* ***equivalent effects*** *on the provision of public goods and services. The difference concerns only the amount saved by the state." Differential-effects statement: "The two proposals could have* ***different effects*** *on the provision of public goods and services. When making your choice, take into account both these possible effects and the amount saved by the state." The statement follows the assignment rule described above.]*
> For each pair, indicate which measure you would **prefer** to be adopted.

Q29. `policy_tradeoff`. *Single choice.*

OPTION A	
Description	[Option A description]
Estimated annual savings	[**Option A estimated household savings**; approximate total state saving in light-gray regular type]
OPTION B	
Description	[Option B description]
Estimated annual savings	[**Option B estimated household savings**; approximate total state saving in light-gray regular type]

Which of the two measures would you prefer?

◯ Option A ◯ Option B

Q30. `policy_tradeoff_2`. *Single choice.*

OPTION A	
Description	[Option A description]
Estimated annual savings	[**Option A shifted estimated household savings**; approximate total state saving in light-gray regular type]
OPTION B	
Description	[Option B description]
Estimated annual savings	[**Option B shifted estimated household savings**; approximate total state saving in light-gray regular type]

Which of the two measures would you prefer?

◯ Option A ◯ Option B

Q31. `dce_equivalence_belief_post`. *Single choice.* In your view, how likely is it that the two proposals have **equivalent** effects on the **quality of public services**?

Response options: Not at all likely; Not very likely; Fairly likely; Very likely; Do not know / Prefer not to answer

Mechanisms: In-house vs External

The cost/value/skills/incentives block and the monitoring/private-interests/capacity block are both shown, with their order randomized 50–50. The randomized block order is stored as `mechanism_block_order`*. A single short comparison instruction appears at the top of whichever block is shown first. Within each block, the substantive questions are also shown in randomized order; the numbering below is for presentation only.*

Q32. `mech_intro/mech_intro_drivers`. *Conditional display text; shown once, in the first randomized block.* Now compare two ways of carrying out public assignments: with **public employees** or with **consultants and external firms**.

Randomized mechanism block A: Cost, Value, Skills and Incentives

Q33. `mech_cost`. *Single choice.* Which solution tends to entail a **lower cost for citizens**?

Response options: Definitely with public employees; Probably with public employees; I see no difference; Probably with consultants and external firms; Definitely with consultants and external firms; It depends / Do not know

Q34. `mech_value_for_money`. *Single choice.* Taking both **quality** and **cost** into account, which solution uses public resources better?

Response options: Definitely with public employees; Probably with public employees; I see no difference; Probably with consultants and external firms; Definitely with consultants and external firms; It depends / Do not know

Q35. `mech_skills_technical`. *Single choice.* Which solution better ensures access to the **necessary specialized skills**?

Response options: Definitely with public employees; Probably with public employees; I see no difference; Probably with consultants and external firms; Definitely with consultants and external firms; It depends / Do not know

Q36. `mech_result_incentives`. *Single choice.* Which solution gives **stronger incentives** to achieve good results?

Response options: Definitely with public employees; Probably with public employees; I see no difference; Probably with consultants and external firms; Definitely with consultants and external firms; It depends / Do not know

Randomized mechanism block B: Monitoring, Private Interests and Capacity

Q37. `mech_monitor_quality`. *Single choice.* With which solution is it easier to **evaluate the quality** of the work performed?

Response options: Definitely with public employees; Probably with public employees; I see no difference; Probably with consultants and external firms; Definitely with consultants and external firms; It depends / Do not know

Q38. `mech_private_interests_risk`. *Single choice.* With which solution is there a greater risk of **conflicts of interest or favoritism**?

Response options: Definitely with public employees; Probably with public employees; I see no difference; Probably with consultants and external firms; Definitely with consultants and external firms; It depends / Do not know

Q39. `mech_capacity_long_run`. *Single choice.* Which solution helps the public administration more in managing similar assignments well **in the future**?

For example, through skills, procedures, tools, or data that remain useful over time.

Response options: Definitely with public employees; Probably with public employees; I see no difference; Probably with consultants and external firms; Definitely with consultants and external firms; It depends / Do not know

Policy: Monitoring Consultants

Q40. `policy_monitor_consultants`. *Single choice.* How favorable or opposed would you be to introducing **mandatory independent reviews** to evaluate the work of **consultants and external firms**?

Response options: Strongly in favor; In favor; Somewhat in favor; Neutral; Somewhat opposed; Opposed; Strongly opposed

Mechanism: Domain Preferences

This page has a minimum display time of 4 seconds.

Q41. `domain_preferences`. *Matrix.* For each municipal service, would you prefer direct public management or assignment to consultants and external firms?

Rows and scale: Rows (shown in randomized order): **Administration and taxes** (e.g., technical office for building permits, registry office, tax collection); **Environment and territory** (e.g., waste collection, maintenance of public green spaces); **Social services** (e.g., nurseries, assistance for elderly people and people with disabilities); **School** (e.g., canteens, school transport, maintenance of school buildings); **Roads and transport** (e.g., road maintenance, local public transport). Scale: Public employees only; Mostly public employees; Balanced mix; Mostly external providers; External providers only; No preference

Worldview and Ideology

Q42. `public_inst_role`. *Single choice.* Some people think that **public institutions** try to **do too many things** that should instead be left to individual citizens and firms. Others think that **public institutions** should **do more** to solve the country's problems. Which of the following positions is closest to your opinion?

Response options: Public institutions do far too much; Public institutions do a little too much; Public institutions do the right amount; Public institutions should do a little more; Public institutions should do much more

Q43. `profits_living_standards`. *Single choice.* In your view, when private firms make **good profits**, does this tend to improve the **standard of living** of the Italian population in general?

Response options: Improves it a lot; Improves it a little; Neither improves nor worsens it; Worsens it a little; Worsens it a lot

Behavioral Outcome: Petition

This page has a minimum display time of 4 seconds.

Q44. `petition_choice`. *Single choice.* **A concrete choice.** At the end of the study, we will communicate to the **Associazione Nazionale Comuni Italiani (ANCI; National Association of Italian Municipalities)** the percentage of participants supporting each proposal. Your response is anonymous: we do not ask for your name. Which of the following petitions would you like to support, if any?

Response options: **More public employees:** We ask Italian municipalities to rely more on public employees and less on consultants and external firms to carry out their functions; **More consultants and external firms:** We ask Italian municipalities to rely more on consultants and external firms and less on public employees to carry out their functions; **I do not wish to support either.** The order of the first two options is randomized with equal probability and stored in `petition_order`; the opt-out is always last.

Belief: Posterior (Consulting)

Asked only to respondents assigned to no prior elicitation (`prior_block = none`); respondents assigned either prior-belief module do not see this item. This page has a minimum display time of 4 seconds.

Q45. `beliefs_consulting_per100_post`. *Numeric entry.* On average in Italy, if a municipality spends **100 €** on its **own employees**, how many euros do you think it spends on **external consulting**?

Here, "external consulting" means professional assignments that the municipality gives to external consultants or experts. It does not include other services contracted to firms.

Write a number from 0 to 100.

Response: Integer from 0 to 100.

Wrap: Study Purpose and Bias

Q46. `study_goal_guess`. *Text entry.* In your opinion, what was the **main objective** of this study? You may answer with a short phrase.

Response: Short open-text entry.

Q47. `bias_perception`. *Single choice.* Do you think this survey was **biased**?

Response options: Yes, in favor of the public sector; Yes, in favor of the private sector; No, I do not think it was biased; Do not know. The first two directional options are shown in randomized order; the balanced and do-not-know options remain third and fourth.

Wrap: Final Feedback

Q48. `ai_vs_human_pref`. *Single choice.* This is the last question of the questionnaire. If you had to complete **another questionnaire in the future**, would you prefer:

Response options: **An automated interview with artificial intelligence** (like the one you just completed); **An in-person interviewer**; A questionnaire in which I **write** my answers in **free-text fields**; No preference

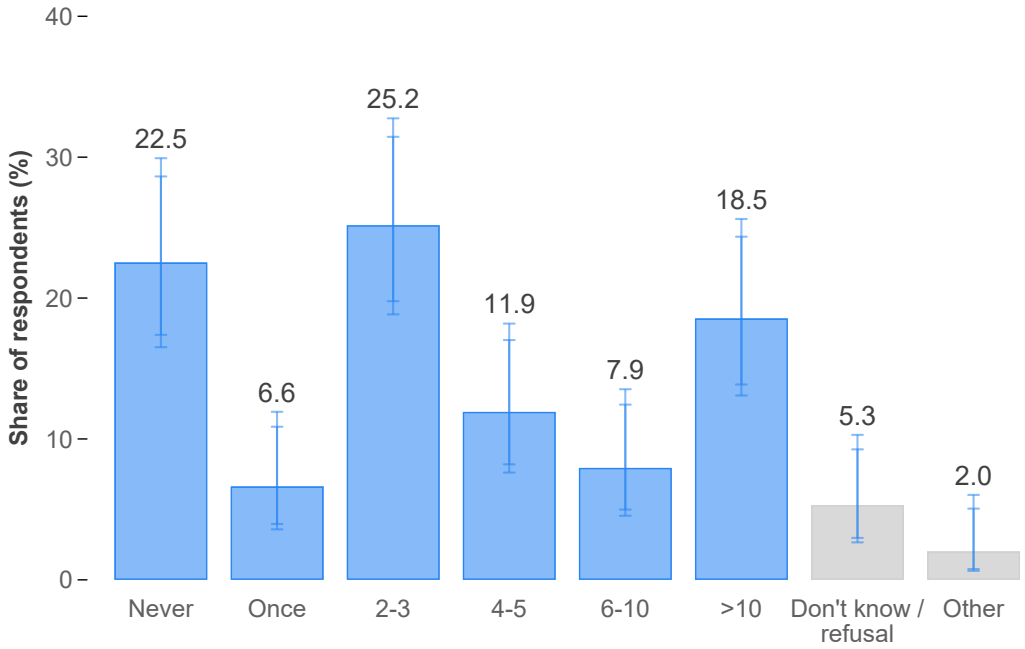
Q49. `feedback`. *Text entry.* Feel free to share any **comments or impressions** about this survey.

Response: Open text entry.

E.2. Politicians and High-Level Bureaucrats Survey

In recent years, how many times have you been involved in deciding, preparing, managing, monitoring, or evaluating a consulting contract?

Figure A.38: Involvement in Consulting Contracts



Notes. The figure reports the distribution of responses among 151 respondents in the analysis sample. Blue bars show the original contract-involvement categories; the grey bar reports the explicit "do not know / prefer not to answer" response. Unanswered items are excluded from the denominator. Numbers report percentages, and capped lines show 90% and 95% confidence intervals computed using the logit transformation. [no references]

For which tasks and in which situations has your municipality used external consultants?

Figure A.39: Tasks and Situations in Which Municipalities Use Consultants

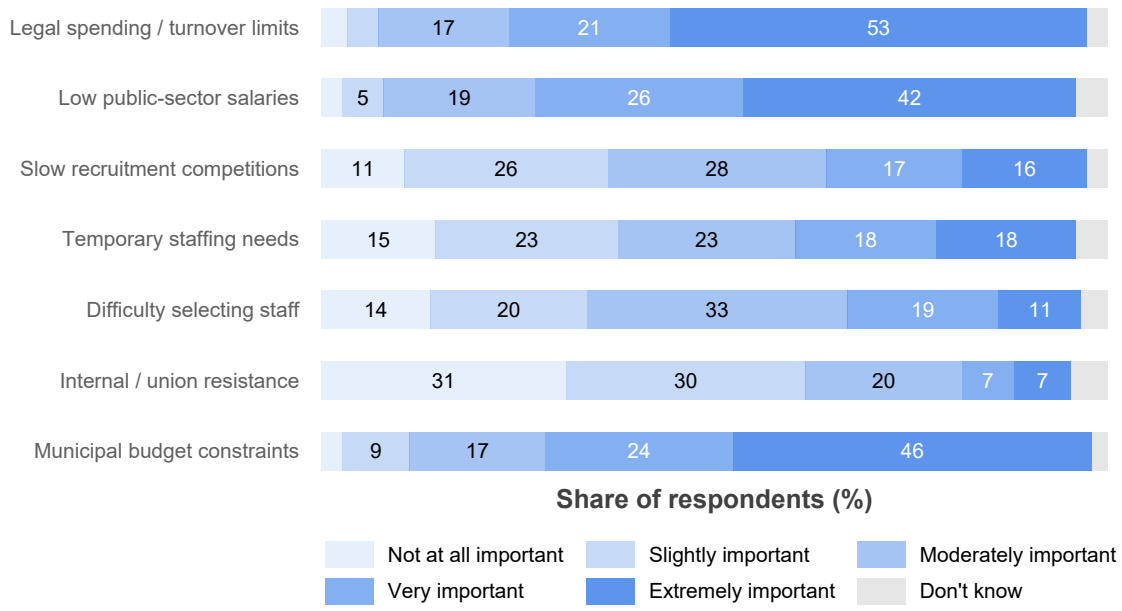


Notes. The figure summarizes open-text responses in the sample of 151 respondents. A large language model extracts short English labels for tasks from the Italian answers. Synonymous labels are consolidated before counting. Font size increases with the number of respondents mentioning each concept, counting each respondent at most once per concept. Referenced on page: 13.

When your municipality needs new skills, how important are the following obstacles to hiring new public employees?

- Legal hiring limits (personnel-spending caps and restrictions on replacing departing staff).
- public sector salaries are too low to attract the required skills.
- Recruitment competitions are too slow relative to staffing needs.
- The need is temporary and does not justify a permanent hire.
- Difficulty assessing and selecting the right candidates.
- Internal or union resistance to reorganization.
- The municipal budget cannot cover the cost of another employee.

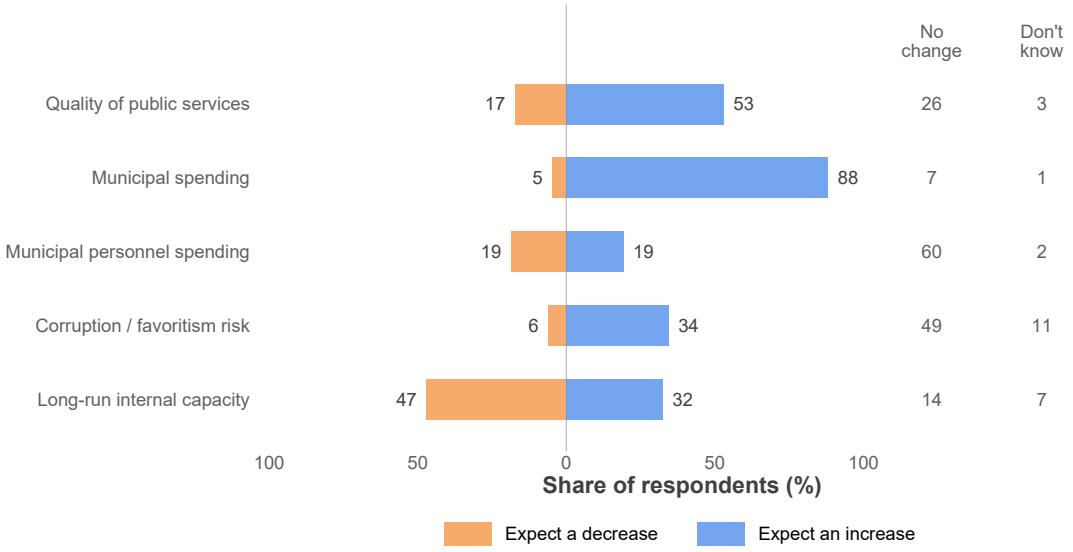
Figure A.40: Obstacles to Hiring Public Employees



Notes. Each row reports the full distribution of importance ratings for one hiring obstacle among 151 respondents in the analysis sample. Rows follow the questionnaire order. The denominator includes the explicit "do not know" response, shown in grey, and excludes unanswered items. Segments sum to 100% within each row. Numbers inside segments are rounded percentages and are shown only for shares of at least 5%. [no references]

If a municipality made greater use of external consultants, what effects would you expect on the following aspects?

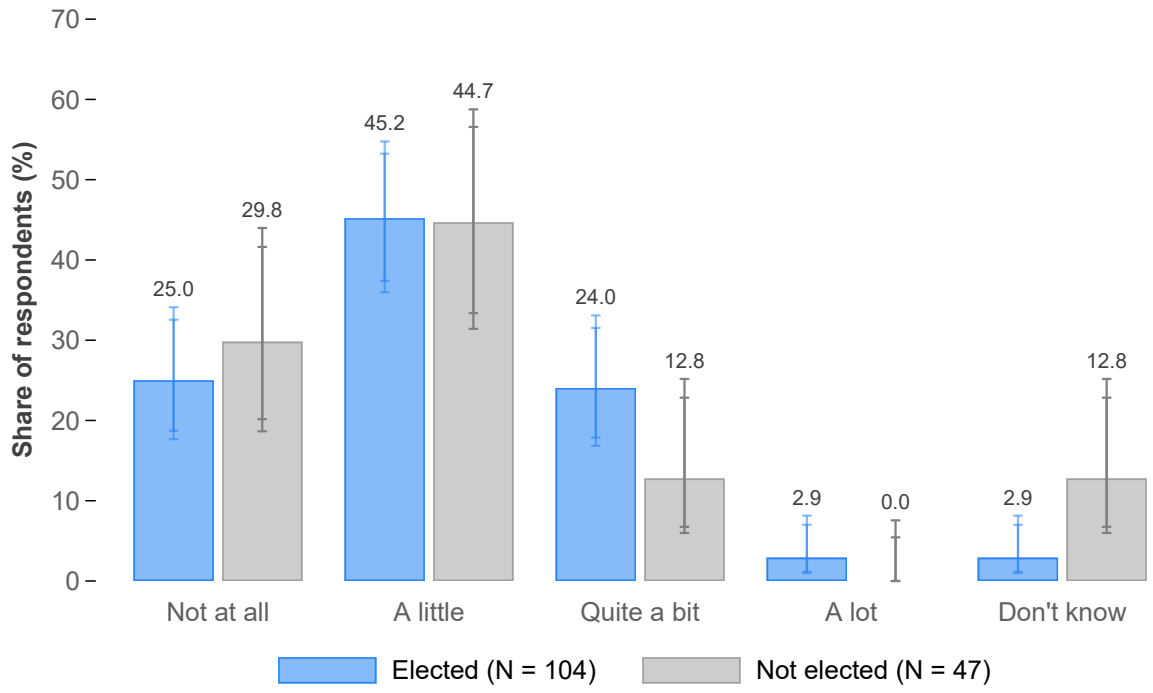
Figure A.41: Expected Effects of Greater Use of Consultants



Notes. Each row reports expected changes in the indicated outcome among 151 respondents in the analysis sample, elicited before the information treatment. Orange bars extend left for an expected decrease and blue bars extend right for an expected increase, combining "a lot" and "somewhat" within each direction. Grey columns report "no change" and "do not know." Numbers are rounded percentages. The four response groups sum to 100% before rounding within each row; unanswered items are excluded. Directions refer to each outcome as worded, including spending and corruption risk. Referenced on page: 22.

When the public administration decides whether to use external consultants, how much does it take citizens' possible reactions into account?

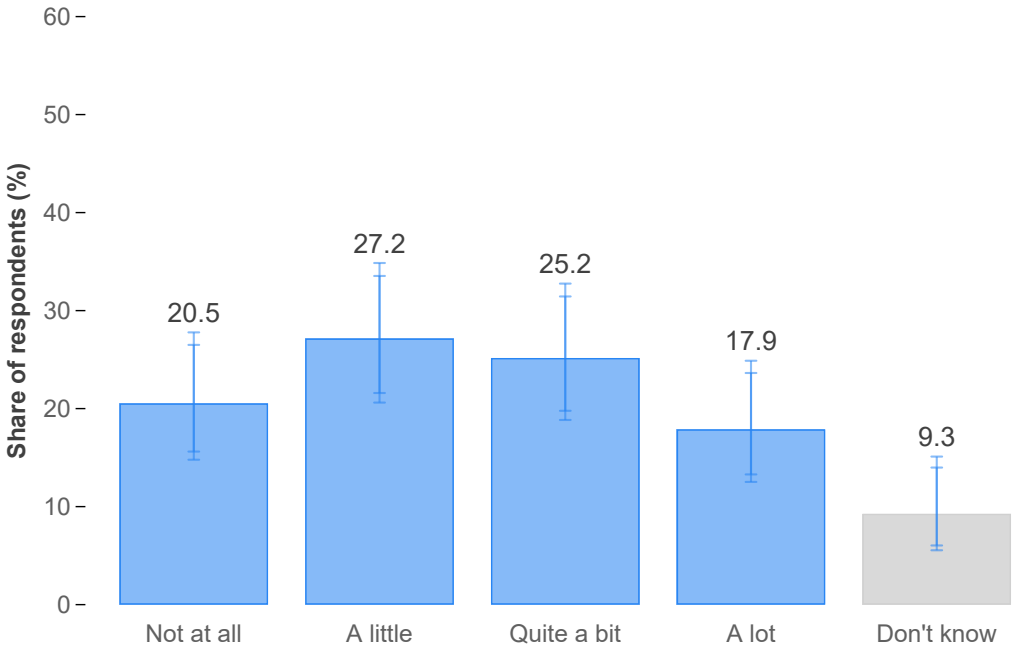
Figure A.42: Consideration of Citizens' Reactions to Consulting by Respondent Group



Notes. The figure compares responses elicited before the information treatment among 151 respondents in the analysis sample. The "elected" group includes mayors, municipal executive members and councillors, directors-general, municipal secretaries, holders of both administrative posts, and other respondents whose reported job titles contain *segretario* or *dirigente*; the comparison group contains all remaining respondents. Blue bars show the elected group and grey bars the comparison group. Percentages are calculated within each group, including explicit "do not know" responses and excluding unanswered items. Capped lines show 90% and 95% Wilson confidence intervals, which remain defined for zero-frequency categories. The question concerns perceptions of administrations' behavior, rather than respondents' own behavior. [no references]

During the economic crisis, starting in 2011, the State cut consulting expenditure proportionally more than spending on public employees. In your view, how much did the belief that citizens preferred consulting cuts over cuts to public personnel influence this choice?

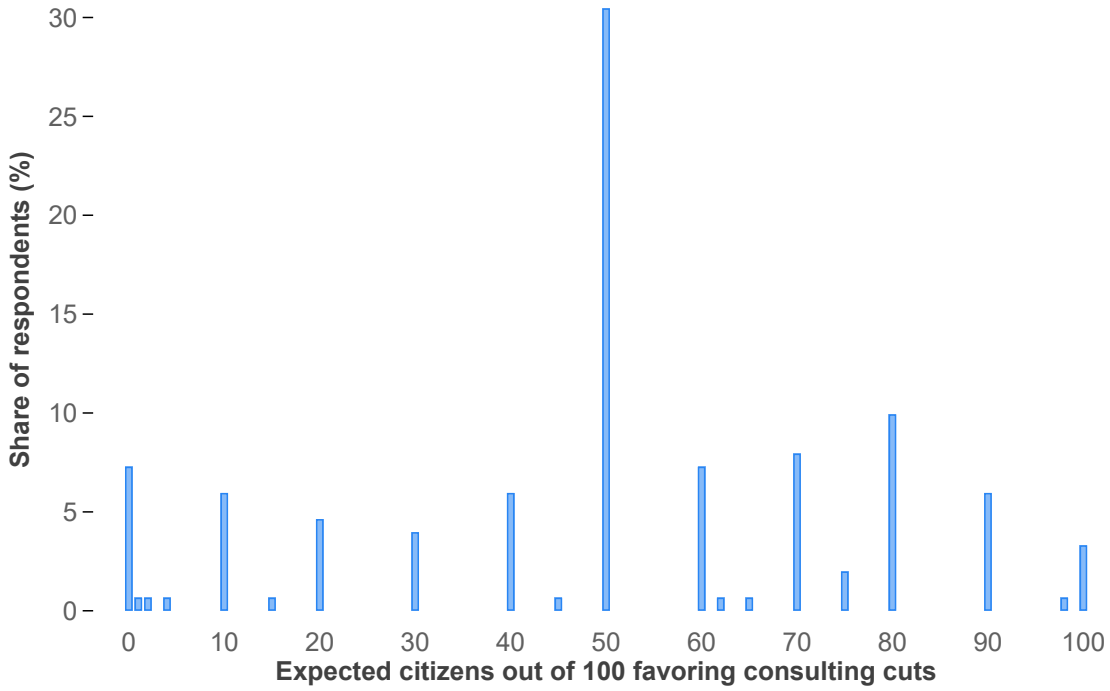
Figure A.43: Citizens' Preferences as a Reason for Consulting Cuts



Notes. The figure reports how much respondents believe citizens' preference for consulting cuts mattered for the policy described in the question. Responses were elicited before the information treatment among 151 respondents in the analysis sample. The denominator includes explicit "do not know" responses, shown in grey, and excludes unanswered items. Numbers report percentages; capped lines show 90% and 95% confidence intervals. [no references]

For the same savings to the State, out of 100 Italian citizens, how many do you think would prefer reducing spending on consultants and external firms rather than hiring fewer public employees? Enter a number from 0 to 100.

Figure A.44: Beliefs About Citizens' Preference for Consulting Cuts

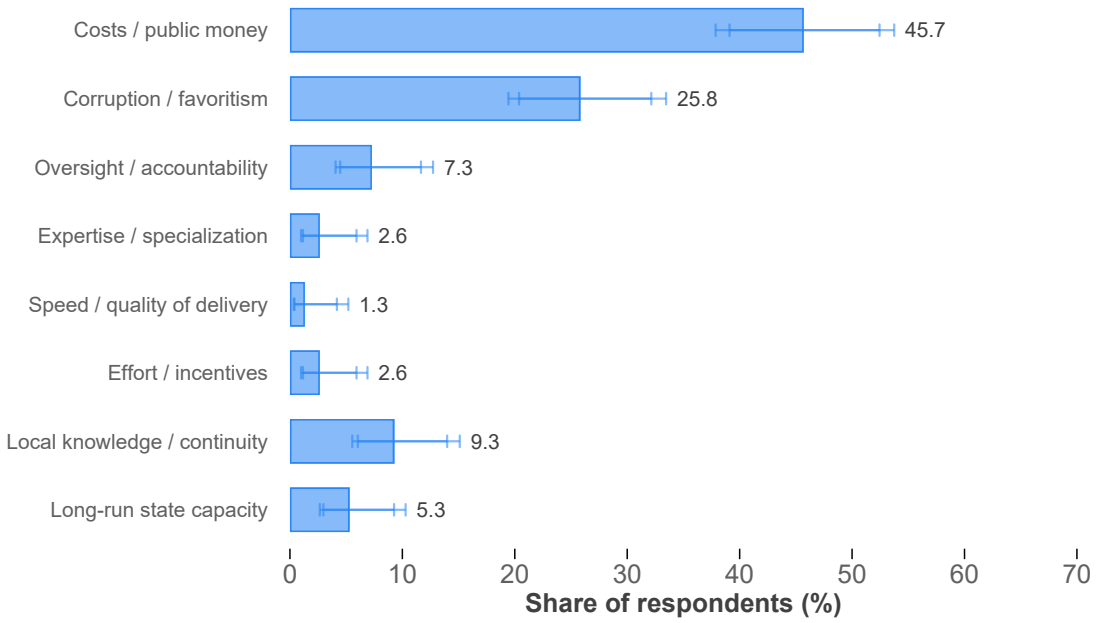


Notes. The figure reports the distribution of numerical beliefs elicited before the information treatment among 151 respondents in the analysis sample. Each bar shows the percentage reporting the exact value on the horizontal axis; responses are neither binned nor smoothed. Unanswered items are excluded. Larger values indicate a belief that more citizens prefer consulting cuts to reduced public hiring, holding savings fixed. [no references]

Among Italian citizens concerned about consulting contracts awarded by the public administration, which of the following issues do you think is the most widespread concern?

- Costs and public money (waste, excessive fees).
- Corruption, favoritism, and private interests.
- Oversight and accountability (who monitors the work performed).
- Expertise and specialization (who is better prepared).
- Speed and quality of delivery (timing and results).
- Effort and incentives (who has an incentive to work well).
- Local knowledge and continuity (who knows the administration and its territory).
- Long-run state capacity (training in-house staff rather than buying externally).

Figure A.45: Beliefs About Citizens' Main Concern About Consulting



Notes. The figure reports the concern respondents expect to be most widespread among concerned citizens, elicited before the information treatment among 151 respondents in the analysis sample. Respondents selected one of eight issues. Unanswered items are excluded from the denominator. Numbers report percentages; capped lines show 90% and 95% confidence intervals computed using the logit transformation. These are respondents' beliefs about citizens, rather than citizens' own reported concerns. [no references]

F. Derivations Conceptual Framework

This appendix derives the comparative statics in Section 4.1. Throughout, u is twice continuously differentiable with $u' > 0$ and $u'' < 0$, and production is CES with $0 < \alpha < 1$ and $\rho < 1$. I consider a finite optimum with $B, C > 0$ and a small increase in K for which the consulting cap remains binding, $\lambda > 0$.

Optimal input choices. Substituting the budget constraint into the baseline objective gives the Lagrangian:

$$\mathcal{L} = u(g(B, C)) - w_B B - w_C C + \lambda(K - w_C C) \tag{A.1}$$

The first-order conditions and the complementary slackness condition are:

$$u' g_B = w_B, \qquad u' g_C = w_C(1 + \lambda), \qquad \lambda(K - w_C C) = 0 \tag{A.2}$$

Hence a binding cap creates a wedge between the marginal products per euro:

$$\frac{g_C}{w_C} - \frac{g_B}{w_B} = \frac{\lambda}{u'} > 0 \tag{A.3}$$

The envelope theorem gives $dV/dK = \lambda$, where $V(K)$ is the maximised objective. Thus λ is the gain in the politician's objective from relaxing the spending cap by one euro.

The CES technology has derivatives

$$g_B = \alpha B^{\rho-1} g^{1-\rho} > 0, \qquad g_C = (1-\alpha) C^{\rho-1} g^{1-\rho} > 0, \tag{A.4}$$

$$g_{BC} = (1-\rho)\frac{g_B g_C}{g} > 0, \qquad g_{BB} = -\frac{C}{B} g_{BC} < 0 \tag{A.5}$$

Constant returns to scale implies:

$$g = g_B B + g_C C, \qquad B g_{BB} + C g_{BC} = 0 \tag{A.6}$$

Public employment and revenues. While the cap binds, $C = K/w_C$ and $dC/dK = 1/w_C$. Differentiating the first-order condition for public employees yields

$$\left(u'' g_B^2 + u' g_{BB}\right) \frac{dB}{dK} + \frac{u'' g_B g_C + u' g_{BC}}{w_C} = 0 \tag{A.7}$$

Define the positive quantity $D \equiv -u''g_B^2 - u'g_{BB} > 0$. The employment response is:

$$\frac{dB}{dK} = \frac{u'g_{BC} + u''g_B g_C}{w_C D} = \frac{u'g_B g_C}{w_C g D}\left(1 - \rho + \frac{g u''}{u'}\right) \tag{A.8}$$

Consequently,

$$\frac{dB}{dK} < 0 \quad \Longleftrightarrow \quad -\frac{g u''}{u'} > 1 - \rho = \frac{1}{\sigma} \tag{A.9}$$

Employment rises when the inequality is reversed and is unchanged when the two terms are equal. The positive term $u'g_{BC}$ captures the increase in the marginal product of public employees. The negative term $u''g_B g_C$ captures the decline in the marginal value of services. A higher elasticity of substitution lowers the curvature of utility required for employment to fall.

The balanced-budget constraint implies $T = w_B B + K$, so:

$$\frac{dT}{dK} = 1 + w_B \frac{dB}{dK}, \qquad \frac{dT}{dK} < 0 \quad \Longleftrightarrow \quad -w_B \frac{dB}{dK} > 1 \tag{A.10}$$

Thus falling employment is necessary but not sufficient for revenue to fall: the saving on the public wage bill must exceed the extra euro spent on consulting. If employment rises or remains unchanged, revenue necessarily increases.

Public services and output per euro. The response of public services is:

$$\begin{aligned} \frac{dg}{dK} &= g_B \frac{dB}{dK} + \frac{g_C}{w_C} \\ &= \frac{g_B(u'g_{BC} + u''g_B g_C) + g_C(-u''g_B^2 - u'g_{BB})}{w_C D} \\ &= \frac{u'(g_B g_{BC} - g_C g_{BB})}{w_C D} > 0 \end{aligned} \tag{A.11}$$

The inequality follows from $g_{BC} > 0$ and $g_{BB} < 0$. Services therefore rise even when public employment falls.

The effect on output per euro:

$$\begin{aligned} \frac{d(g/T)}{dK} &= \frac{T\, dg/dK - g\, dT/dK}{T^2} \\ &= \frac{w_B g_C - w_C g_B}{T^2}\left(\frac{B}{w_C} - C\frac{dB}{dK}\right) \end{aligned} \tag{A.12}$$

The first factor is positive by Equation (A.3):

$$w_{\mathsf{B}}g_{\mathsf{C}} - w_{\mathsf{C}}g_{\mathsf{B}} = \frac{w_{\mathsf{B}}w_{\mathsf{C}}\lambda}{u'} > 0 \tag{A.13}$$

For the second factor, substituting Equation (A.8) and applying both identities in Equation (A.6) gives

$$\begin{aligned}\frac{\mathsf{B}}{w_{\mathsf{C}}} - \mathsf{C}\frac{d\mathsf{B}}{dK} &= \frac{\mathsf{B}D - \mathsf{C}(u'g_{\mathsf{BC}} + u''g_{\mathsf{B}}g_{\mathsf{C}})}{w_{\mathsf{C}}D} \\ &= \frac{-u''g_{\mathsf{B}}g}{w_{\mathsf{C}}D} > 0\end{aligned} \tag{A.14}$$

The ratio of consultants to public employees therefore rises. Combining Equations (A.12)–(A.14) establishes $d(\mathsf{g}/\mathsf{T})/dK > 0$. This result holds whether total revenue rises or falls.

Direct Benefits from Consulting

Now allow a differentiable term $\gamma(\mathsf{C})$ in the objective and continue to consider a local optimum with positive inputs and a valuable binding cap. The Lagrangian becomes

$$\mathcal{L} = u(\mathsf{g}(\mathsf{B},\mathsf{C})) - w_{\mathsf{B}}\mathsf{B} - w_{\mathsf{C}}\mathsf{C} + \gamma(\mathsf{C}) + \lambda(K - w_{\mathsf{C}}\mathsf{C}) \tag{A.15}$$

The first-order conditions are

$$u'g_{\mathsf{B}} = w_{\mathsf{B}}, \qquad u'g_{\mathsf{C}} + \gamma'(\mathsf{C}) = w_{\mathsf{C}}(1+\lambda) \tag{A.16}$$

Because γ depends only on consulting and the cap fixes $\mathsf{C} = K/w_{\mathsf{C}}$, the employee condition is unchanged. The expressions for $d\mathsf{B}/dK$, $d\mathsf{T}/dK$, $d\mathsf{g}/dK$, and the change in the input ratio therefore remain valid while the cap binds. The wedge in marginal products per euro, however, is now

$$\frac{g_{\mathsf{C}}}{w_{\mathsf{C}}} - \frac{g_{\mathsf{B}}}{w_{\mathsf{B}}} = \frac{\lambda - \gamma'(\mathsf{C})/w_{\mathsf{C}}}{u'} \tag{A.17}$$

Substitution into Equation (A.12), whose second factor remains strictly positive, gives

$$\frac{d(\mathsf{g}/\mathsf{T})}{dK} > 0 \quad \Longleftrightarrow \quad \lambda > \frac{\gamma'(\mathsf{C})}{w_{\mathsf{C}}} \tag{A.18}$$

Output per euro is unchanged at equality and falls when the inequality is reversed. Positive marginal direct benefits can therefore make a cap valuable to the politician even when consulting has a lower marginal product per euro than public employment. Negative

marginal direct benefits, by contrast, preserve the positive performance effect of relaxing a valuable binding cap. It is the marginal term $\gamma'(\mathsf{C})$, rather than the level of $\gamma(\mathsf{C})$, that determines this distinction. The envelope result remains $dV/dK = \lambda > 0$: the politician's maximised objective rises even when measured output per euro falls.